

Control Related Metrics for Multidisciplinary Design Optimization

by

Bernardo Bahia Monteiro

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Doctoral Committee:

Professor Carlos Eduardo Stolf Cesnik, Co-Chair
Professor Ilya Vladimir Kolmanovsky, Co-Chair
Professor Joaquim Rafael Rost Ávila Martins
Professor Peter Joseph Seiler
Dr. Fabio Vetrano, Airbus S.A.S.

I like to think of Bode's integrals as conservation laws. They state precisely that a certain quantity—the integrated value of the log of the magnitude of the sensitivity function—is conserved under the action of feedback. ...

... This applies to every controller, no matter how it was designed. ...

It is curious, somehow, that our field has not adopted a name for this quantity being conserved ..., to put it on a par with some of the great quantities of physics such as mass, momentum, or energy. ... Let me propose that we simply call it dirt.

Gunter Stein, "Respect the Unstable", 2003
based on his Bode Lecture (1989)

After more than 13 years I still remember Gunter's superb lecture. ... The lecture is like really good wine; it ages superbly.

Karl J Åström, 2003

Bernardo Bahia Monteiro
bbahia@umich.edu
ORCID iD: 0000-0002-8501-2056

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Abstract

The push for cleaner aviation to meet the net-zero carbon emissions by 2050 goal will require introducing innovative technologies in future aircraft. Active control is a key enabler of these technologies. For example, actively reducing gust and maneuver loads allows the design of lighter and higher-aspect-ratio wings that reduce fuel burn.

Traditionally, airframe and controller design are done sequentially. However, due to the couplings between these disciplines, this approach cannot guarantee that the final design is optimal on a system level. Multidisciplinary design optimization (MDO) offers a solution by simultaneously considering all disciplines and their cross-couplings. However, doing a detailed controller design early in the design cycle may still be undesirable due to the additional complexity brought into the MDO process, the fact that many state-of-the-art control design techniques are not suitable for inclusion in gradient-based optimization, and the presence of uncertainties about the final control architecture. Thus, approaches that introduce meaningful control-related constraints for airframe design without controller co-design are of interest.

This work develops a method to estimate the performance of the closed-loop system, without controller design, based on parameterizing and designing the sensitivity transfer function of the closed-loop system. This approach efficiently approximates the best achievable closed-loop performance under stochastic disturbances, quantified by a class of risk metrics based on the power spectral density of signals in the system. It also enforces a technological limit on the bandwidth of the closed-loop system via a constraint on the Bode sensitivity integral relation. Analytical formulas for the derivatives of these metrics are derived in direct and adjoint modes and verified against finite difference approximations. These formulas allow fast and accurate derivative calculation, enabling the inclusion of the proposed metrics in gradient-based optimization problems.

This approach is demonstrated in the MDO design of the wingbox of an aircraft with a gust load alleviation (GLA) system including a fatigue constraint, and in the analysis of the glideslope tracking performance of a supersonic configuration aircraft on the approach to land phase of flight, considering constraints on the risk of glideslope deviation, stall, and saturation of the elevator.

The inclusion of maneuver load alleviation (MLA) in the design is considered through a strategy that is minimally intrusive to a pre-existing MDO problem. This strategy relies on a nested (bi-level) optimization approach and includes a model order reduction process to improve computational performance. It is demonstrated in the aerostructural design of a cantilevered wing, which includes a flutter constraint.

The implementation of these methods in aeroelastic and MDO software is discussed, emphasizing the efficient use of algorithmic differentiation (AD). This enables the definition of new metrics without direct interaction with the AD tool, saving programming effort.

CHAPTER 1

Introduction

The use of active control is essential to maximizing the effectiveness of innovative technology in new aircraft designs. Active control technology has intrinsic couplings with the other disciplines involved in aircraft design, and these couplings can be naturally incorporated into the process of multidisciplinary design optimization (MDO). However, including a fully-fledged controller design in the MDO problem can be undesirable because the control law can be easily changed late in the design cycle, as opposed to the aircraft itself. This chapter is organized as follows. Section 1.1 introduces the motivation for developing an approach that approximates upper bounds for the aircraft's performance with active control without designing the controller. Section 1.2 presents a literature review that gives a historical perspective on the inclusion of active control into aircraft MDO, summarizes the existing approaches, and reviews some of the fundamental limitations that are unavoidable when using feedback control. In addition, the challenging and relevant problem of aircraft flying under continuous gust (i.e., stochastic turbulence) is presented in Section 1.2.3. The chapter ends with a description of the objectives of this dissertation (Section 1.3) and an outline of the remaining chapters (Section 1.4).

1.1 Motivation

The aeronautical industry has committed itself to a bold goal of net-zero CO₂ emissions by 2050 [1]. To accomplish this goal, various novel technologies will have to be matured and employed in the design and operation of aircraft, the most important of which will be the use of Sustainable Aviation Fuels (SAFs), which is predicted to account for 65% of the reduction in emissions according to the International Air Transport Association (IATA) [1]. However, the aeronautical industry also has a long history of reducing emissions by increasing the energetic efficiency of aircraft, driven not only by climate considerations but

also due to the fuel making up 21% of the operational cost of large U.S. passenger airlines, second only to labor costs (33%, as of 3Q 2023) [2]. These improvements in energy efficiency will become even more critical since SAFs are predicted to be more expensive than fossil fuels [3]. What drives these improvements can be identified by looking at Breguet’s range equation:

$$\text{range} = \frac{\overbrace{V}^{\text{aerodynamics}}}{\underbrace{g}_{\text{propulsion}}} \frac{\overbrace{L/D}^{\text{aerodynamics}}}{\underbrace{\text{SFC}}_{\text{propulsion}}} \log \left(\underbrace{\frac{W_i}{W_f}}_{\text{structures}} \right) \quad (1.1)$$

where V is the flight speed, g is the acceleration due to gravity, L/D is the lift to drag ratio, SFC is the specific fuel consumption, and W_i/W_f is the initial to final weight ratio.

Breguet’s equation shows that an airplane can fly further with the same amount of fuel and payload (i.e., $W_i - W_f$ fixed) if it has

- better aerodynamic performance, reflected in the VL/D figure of merit;
- better propulsive and thermodynamic efficiencies, reflected in the specific fuel consumption, SFC; or
- greater structural efficiency, reflected in the weight ratio, W_i/W_f .

Active control can enable performance enhancements in all of these disciplines. For example:

- The use of relaxed stability margins, enabled by the use of feedback controllers, can reduce the trim drag by up to 4% of the cruise drag [4]
- The use of Full Authority Digital Engine Controllers (FADEC) in virtually all commercial aircraft engines improves their performance and safety [5–8]
- The use of active load alleviation enables reductions in structural weight [9–16]

According to Livne [17]:

It has become quite clear in recent years, in the context of the development of new innovative configurations such as the truss-braced wing, the variable-camber continuous trailing-edge flap wing, natural laminar flow wings, or high-aspect-ratio composite wings using advanced composite layout and construction technologies that active control is essential to allowing such configurations to fully benefit from the new technologies they introduce.

Such innovative configurations are shown in Figure 1.1. Airbus predicts that the use of high aspect ratio wings with advanced control systems can “improve the airframe efficiency



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- (a) The Airbus A350, a modern airliner with high-aspect-ratio composite wings, at the Farnborough International Airshow
- (b) The Subsonic Ultra Green Aircraft Research (SUGAR) truss-braced-wing configuration undergoes wind tunnel testing at the NASA Ames Research Center



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- (c) Rendering of the Airbus ZEROe Turbopan concept aircraft
- (d) NASA/Lockheed Martins X-59 quiet supersonic aircraft

Figure 1.1 Current and future aircraft configurations that benefit from control co-design

in terms of aerodynamic and structural performance between 10 and 15% at integrated wing level compared to the current design of a civil transport aircraft,” and aims to demonstrate that with the Ultra-Performance Wing (UP Wing) project [18]. It is important to note that higher aspect ratio wings that are also lighter due to the use of active load alleviation and possibly even active flutter suppression will result in more flexible aircraft designs. The (possibly nonlinear) effects of this increased flexibility have to be accounted for when analyzing the closed-loop characteristics of these designs [19, Chapter 8].

Traditionally, complex systems such as aircraft are designed using a sequential strategy that simplifies the dependence between different engineering disciplines, allowing them to be addressed both independently and consecutively [20]. The controller design is usually done at the end of this sequential process when the plant is fully defined, and because resources are committed to its manufacture, making changes in the plant design is very costly. For instance, in the case of aircraft design, the airframe necessitates specialized tooling and jigs to be manufactured, and it is, therefore, very expensive to alter the design

once production starts. On the other hand, the control algorithms consist of software, which can be efficiently changed and adjusted as far in the design cycle as in-flight testing (e.g., [21]) as well as during the life cycle of the aircraft.

This sequencing is desirable because it streamlines the design process. However, because of the couplings between the control law and the physical system, it cannot guarantee system-level optimality [22]. This is clear, for example, when designing an aircraft with maneuver load alleviation (MLA) or gust load alleviation (GLA) systems, since the presence of those systems allows for weight savings if the structure is designed accordingly. The inclusion of the control discipline in the design process can be made naturally in the framework of multidisciplinary design optimization (MDO) [23].

When control design is considered within an MDO setting, it is usually based on using simpler control design methods than those used when designing only the controller but not the plant. For example, when considering an aircraft including a GLA system, flap deflection scheduling [24], linear quadratic (LQ) [13], proportional integral (PI) [10], and static output feedback [11] controller architectures were considered when designing the control system at the same time as the plant in the co-design setting. On the other hand, linear quadratic Gaussian (LQG) [25–27], adaptive feedforward [28], model predictive control (MPC) [29–31], $\mathcal{H}_2/\mathcal{H}_\infty$ [32–34], and μ -synthesis [35] controllers were proposed for control design in the sequential setting, where aircraft design is completed before controller design.

The difference between the control design methods used for co-design and sequential design motivates the search for an approach to calculate a bound on the performance of the closed-loop system without a fully-fledged controller design, which we refer to as a *controller-agnostic* approach. This approach starts by assuming the existence of a well-designed controller and tries to find the limitations of its performance when combined with the plant. The plant is then designed assuming closed-loop performance at the limits of practically achievable performance and the detailed control design is left to later in the design cycle.

One such approach, common in the aerospace industry, is to use heuristics based on domain-specific knowledge and on previous designs to estimate the “risk” of developing a control system that will deliver adequate closed-loop performance. Such risk metrics are based only on the open-loop design (i.e., the plant). This was formalized for use in MDO by Anderson and Mason [36].

Another controller-agnostic approach consists of incorporating an optimal control problem into the MDO problem [37]. The inputs to the effectors are designed together with the physical system, and closed-loop metrics can be evaluated. This approach provides no guarantees that the designed control inputs can be later realized by a feedback control law or



NASA, 1987

Figure 1.2 The X-29, with its forward-swept aeroelastically-tailored composite wings, was the epitome of the Control Configured Vehicle design concept

that the controller that will be later designed will achieve the required robustness margins.

The need for such guarantees becomes clear when looking at the case of the X-29 airplane design (Figure 1.2). The X-29 was a technology demonstrator with an aeroelastically-tailored forward-swept composite wing, controlled with a triplex fly-by-wire flight control system [38]. At low speeds, it had a high degree of static instability (as much as 35% negative static margin [38]), that manifested itself as an unstable real pole as large as around $+6 \text{ rad/s}$ in some flight conditions [39]. Stein [39] makes a simple argument that shows that the design requirement of a 45 deg phase margin is not achievable for this aircraft, no matter the control design architecture.

Further limitations are shown in Rogers and Collins [40], which concluded that, for a redesign of the X-29's control laws, "when actuator design limitations are considered, the expected superiority of the $\mathcal{H}_\infty$ design over the $\mathcal{H}_2$ design is not obtained. The limited performance designs also do not exhibit the precision flight-path control modes that are characteristic of the $\mathcal{H}_\infty$ designs." This result provides further evidence that the limitations of the control system arise from the physical design itself, combined with technological limitations (in this case the actuator's bandwidth) and not from the controller design.

It is thus necessary to take the fundamental limitations of control into account when designing the physical system, before performing the detailed controller design [41–43]. This dissertation proposes a method to incorporate control system limitations such as saturation and bandwidth limits into the calculation of controller-agnostic closed-loop design metrics that are suitable for large-scale multidisciplinary design optimization of flexible aircraft.

1.2 Literature review

1.2.1 Control considerations in aircraft design

1900–1985: Conventional design

Since the advent of human flight, in the turn from the XIX to the XX century, control and stability considerations have been of the utmost importance. The importance of these disciplines was already known empirically to pioneers such as Otto Lilienthal, Samuel Langley, and the Wright brothers [44]. Indeed, to increase the control authority of the Wright Flyer, the Wright brothers designed it slightly unstable but with a three-axis control system that allowed the pilot to stabilize the airplane [45]. By breaking with the concept of inherent stability of the time, the Wright brothers effectively increased their design space which enabled a better-performing design than would otherwise be possible.

Later technological advances in sensors, computers, and actuators allowed the design of control loops that are independent of the pilot, and thus were not hindered by the workload and bandwidth limitations of the pilot. In 1995, Hönlinger et al. [46] highlighted the most important functions enabled by Active Control Technology (ACT) from a structural perspective:

- Maneuver Load Control [referred to in this work as Maneuver Load Alleviation (MLA)]
- Gust Load Alleviation (GLA)
- Fatigue Life Enhancement
- Deformation Control and Elastic Mode Control
- Flutter Suppression and Flutter Margin Improvement
- Ride Comfort Improvement

The list of these functions remained largely unchanged through the following decades [17].

The initial development of these functions happened in the 1960s and 1970s in what became known as the Control Configure Vehicle (CCV) concept, which was initially applied to the XB-70, B-52, F-4, and F-8 aircraft [47], and later to the L-1011, F-117, B-2, and X-29 [36]. “This concept included improvements of handling qualities, reduction of elastic aircraft vibrations through elastic mode feedback, and possible improvement of flutter margins” [46]. It was enabled by the simultaneous and complementary developments in the understanding of aircraft dynamics, control theory, and the application of feedback controllers to aircraft [48]. However, as of 1995, “the full potential of this technology has

not been exploited enough” [46]. The missing link is due to the interdisciplinary nature of the problem. According to Hönlinger et al. [46]:

To take full advantage of the EFCS [Electronic Flight Control System], including all possible Active Control functions, a multidisciplinary design approach is the necessary route to cover the structural implications of ACT already in the design stage. ... The process of the application of multidisciplinary analysis and design methods, which are still underdeveloped, leads already to reassessment of some traditional views in aircraft design and the acceptance of ACT in the initial design.

1985–1999: Combined airframe and controller synthesis

Multidisciplinary Design Optimization (MDO) is a technology that naturally accounts for the multidisciplinary nature of the design problem. The first attempt to integrate control considerations into the aeroelastic MDO of an aircraft was published in 1985 [49]. Zeiler [49] considers an MDO problem where the control law is designed at the same time as the airframe. A four-degree-of-freedom aircraft model, consisting of a three-degree-of-freedom airfoil elastically attached to a fuselage with the pitch degree of freedom is considered, and control synthesis is done by selecting the weights for an LQR design. The structural design variable is the position of the elastic axis in the airfoil, and the objective is to maximize the speed range in which the aircraft is stable (see also [50]).

More complex models were considered starting in 1990. Livne [51], Livne et al. [52, 53] considered an aeroservoelastic MDO problem where the control transfer function is parameterized by the coefficients of its rational representation, which are added as design variables together with parameters for the wing structure. Livne [54] later commented on this choice of parameterization:

This parametrization, while simple, and directly related to the practical implementation of the resulting design, leads to disjoint design spaces and to serious difficulties in gradient-based optimization.

Notably, gradient-based optimization was used in that work, with analytical derivatives for all functions of interest were [51]. This allowed the method to tackle a moderately large problem even for today’s standards, containing “820 constraints and 50 design variables”, with an optimization taking “18 min of CPU time on the UCLA IBM 3090 Model 600J” [53]. In particular, the control modeling is done by parameterizing the polynomial fraction representation of the controller’s transfer functions by its coefficients, which are included

as design variables in the simultaneous optimization. The main function of the controller is flutter suppression. There were no constraints on the robustness of the control loop, and the resulting Nyquist plots reveal poor robustness margins for some cases [53]. The method was later expanded to incorporate planform design variables [55].

Also in 1990, Morris and Kroo [56] proposed a similar approach in which the control-related design variables are the feedback gains in a linear state-space formulation. Designing these gains or the transfer function coefficients is equivalent as long as the controller states are integrated into an augmented space-state description of the plant. A quadratic cost function on the difference between the time history of the system and that of a reference model is introduced to incorporate both flutter and handling qualities into the objective function in a way suitable for using the LQR machinery. For one of the two cases presented, two structural modes were considered along with the longitudinal degrees of freedom and quasi-steady aerodynamics (i.e., a total of 6 states). The analytical computation of gradients was left as future work [56].

In 1992, Karpel [57] presented a “systematic method for database construction, modeling, and structural and/or control optimization of aeroservoelastic systems.” The controller was parameterized by the state-space feedback gains, in the same manner as Morris and Kroo [56] and equivalent to what Livne [51] used within the transfer function formalism. The novelty from the perspective of the control discipline was that Karpel [57] introduced constraints to ensure the robustness of the resulting controller: a gain margin of 6 dB was enforced for every feedback loop, and phase margins were noted as “not critical” in the example considered. Three control gains and 22 structural design variables corresponding to the thicknesses of composite plies were chosen as the design variables. The derivatives necessary for the gradient-based optimization are calculated analytically [58].

Other ways of parameterizing the controller were proposed by Gilbert [59] and Layton and Peterson [60]. The former uses the weighting matrices and noise covariance matrices from an LQG controller as the design variables, while the latter uses the covariance parameterization of all stabilizing controllers [61]. Analytical expressions for gradients were provided for both approaches. Layton and Peterson [60] also includes constraints on the:

- Output saturation limits
- Sensor saturation limits
- Stability robustness
- Performance robustness

The evolution of these combined design approaches through the 1990s is summarized in [54]. Except for the approach of Layton and Peterson [60], they have the drawback of fixing the structure of the controller from the start and, therefore, leave open the possibility that there is a better controller that cannot be represented in the chosen format.

Fundamental limitations of control in MDO

As Livne [54] was aware, the idea of not constraining the structure of the controller had already been tackled by Boyd et al. [41] in 1990, who defined the “Fundamental Problem” of control co-design as:

Given a specific system to be controlled, control configuration, and set of design specifications, either find a control law that meets these design specifications or determine that none exists.

Boyd et al. [41], Boyd and Barratt [42] explored this problem extensively in the linear time invariant (LTI) setting and identified several design specifications that could be written in a “closed-loop convex” form, which is especially well suited for numerical optimization. They also showed that “several important control law specifications cannot be expressed as convex constraints on closed-loop transfer matrices” [41]. This mathematically rigorous approach offers a definitive answer as to whether there exists a control system capable of satisfying the specifications. It is, however, limited to LTI systems, cannot handle all important specifications, and offers little insight into what characteristics of the plant are the cause of the found limit in performance. Even with such limitations, it offers a very nice framework for the problem of control co-design that should be further explored in the field of aeroservoelastic MDO.

Related and more recent approaches can be found in Ruth et al. [62], where the Bode sensitivity integral (see Section 2.1.1) is used to “re-tune the control designs for several NASA science-mission satellites”; Cho and Lee [43], where that integral relation is used to derive the bounds on robustness the robustness of the control loop without control design, using an approach inspired by Stein [39] that has similarities to the ideas presented in this work and could be adapted to become a controller-agnostic constraint for MDO (a parallel to $\mathcal{H}_\infty$ design is also drawn); Micheli [63], which quantifies the achievable performance of an Active Flutter Suppression system accounting for saturation of actuators; and Abbas et al. [64], where a controller/plant co-design methodology is proposed that ensures bounds on the peak of the sensitivity function, which is a measure of robustness. These ideas are further discussed in Sections 1.2.2 and 2.1.

Heuristics based on the open-loop system in MDO

In 1996, Anderson and Mason [36] identified some of the shortcomings of the simultaneous controller and plant design method and suggested an alternative approach for including control in the MDO problem. They argue that “very few, if any, flight control design techniques are automated to the point that they can be embedded in an automated design procedure that does not allow human intervention.” Therefore, “a practical MDO method requires that the control system risk be assigned without actually designing the control system.” They proposed a heuristic function that uses fuzzy logic to link flying quality requirements to a real number between 0 and 100 that represents the required flight control complexity and, therefore, its development risk. This value can then be included in the MDO problem as a constraint or part of the objective. The fuzzy logic rules used in their analysis were developed based on the “MIL-STD-1797 flying qualities requirements for Short Term Pitch Response, Control Anticipation Parameter, and Transient Peak Ratio” [36].

In 2020, Meckstroth and Ordóñez [65] presented a method named they named “control power required” which “gives a finite measure of control moments that would be needed to be produced in order to control the aircraft in a desired manner. Comparing these moments to the values the aircraft is able to produce returns a finite mathematical and measurable representation of how much the aircraft needs to change to become controllable.” The calculations to obtain these moments are based on the dynamic inversion of classical simplified rigid-body dynamic models for short-period, roll, and dutch-roll, in addition to static analysis for take-off roll moment, trim, crosswind rejection, roll initiation, and roll coordination [65].

2000–present: Optimal control co-design

Yet a third approach for control co-design, derived from optimal control, was developed in the 2000s for reentry vehicle design [66], and then in the 2010s with applications on circuit design [67], active suspension design [68], and wind turbine design [69] (see [37] for a summary as of 2014). This approach consists of integrating an optimal control problem into the MDO framework. The time histories of the control inputs are included as new design variables and control-related metrics can be extracted from the optimal control problem and integrated into the multidisciplinary objective and/or constraints. Herber and Allison [70] compared the direct transcription solution with shooting methods and fixed structure controllers.

The optimal control problem can be implemented either as more design variables and

constraints to the MDO problem, utilizing the simultaneous analysis and design (SAND) MDO architecture, or as a nested (i.e., bi-level) optimization, in which case the optimal control problem becomes another discipline in the multidisciplinary feasible (MDF) architecture [70, 71], [72, Chapter 13].

In 2023, Cunis et al. [73] used the approach to tackle the problem of gust rejection during the approach to land flight phase. This allows the optimal control problem to be solved for multiple scenarios and possibly parallelized, it avoids adding design variables to the MDO problem, and adds a single additional constraint, resulting in a less intrusive and more modular architecture.

Approaches to incorporate uncertainty in optimal control co-design were reviewed by Azad and Herber [74] in 2023. This class of problems includes the ones where the model is excited by random noise, such as aircraft flight under continuous gust (Section 1.2.3). They remark that “with all the recent advances and applications of (deterministic) control co-design, significant work is still needed to handle uncertainty when developing effective combined plant and control solutions” [74].

Other recent developments in control co-design

In 2019 Garcia-Sanz [20] published an editor viewpoint that describes control co-design as a “game changer for the control engineer, who will be not only the designer of advanced control algorithms but also the natural leader of the design of new products and systems.” The origins of control co-design are traced back to the Wright brothers and Charles Bush, who “respectively designed the world’s first heavier-than-air powered airplane and the world’s first automatically operating wind turbine for electricity generation.” Garcia-Sanz [20] divides control co-design methodologies into three primary areas, namely:

- *control-inspired paradigms* (A1) for which the CCV concept is a prime example;
- *co-optimization techniques* (A2), which includes the combined plant and controller synthesis as a first generation and optimal control co-design as the second generation; and
- *co-simulation methods* (A3), including the idea of digital twins (e.g., [75]), which is not the focus of this dissertation but finds wide application in the industry.

The methods of fundamental limitations of control in MDO and heuristics based on the open-loop system in MDO lie in the intersection between A1 and A2.

In 2020, Hegseth et al. [76] optimized a linearized aero-hydro-servo elastic floating wind turbine model, which included the structure of the platform, tower, mooring system, and the gains for a proportional (P) controller for below-rated wind speed and a proportional-integral (PI) controller for above-rated wind speed. Hegseth et al. [76] acknowledges that

the selection of this control architecture may have changed the optima of the problem because “including more advanced control strategies could improve the fatigue behavior in the soft-stiff range, and potentially result in a different global minimum for the problem”.

The problem considered in [76] included a fatigue constraint in a gradient-based MDO problem. This constraint is based on using the Dirlik [77] method for estimating the accumulated fatigue damage from the power spectral density (PSD) of the bending moment response to combined wind and wave excitation. The simplifications involved in the method, in particular the linearization of the model and the consideration of only four states, resulted in “reasonably accurate predictions for both fatigue and extreme response” when compared to state-of-the-art simulations [76]. A similar fatigue constraint is presented for aircraft applications in Section 2.2.4 and applied to a linearized flexible aircraft model with 317 states in Section 4.1.

In 2022, Szentannai and Fekete [78] published a systematic review of the use of control in MDO for improved structure life considering multiple fields of application including aerospace and energy.

In 2023, Aksland et al. [79] considered the control co-design of a hybrid unmanned air vehicle. Two feedback loops are considered, one that operates the hybrid power split and another one that controls the propulsion system. The problem is considered both in the combined plant and controller synthesis formulation and in the optimal control co-design formulation, which they refer to as the “closed-loop” and “open-loop” co-design problems, respectively. When the controllers are designed, they are assumed to be PI controllers and are parameterized by their proportional (P) and integral (I) gains. The optimization objective is to minimize the time to complete the mission. Five different cases are solved and compared to a previously published baseline design [80]:

1. The optimal control co-design problem is solved. This is described as a “best-case” scenario and produces an improvement of 35.5% over the baseline design.
2. The optimal control co-design problem is solved first to design the plant, and then another optimization finds the best control gains for that plant. This results in an improvement of 34.1% over the baseline.
3. The combined plant and controller synthesis is solved. This results in 34.7% improvement over the baseline.
4. The plant is optimized with fixed controller gains. This results in 22.6% improvement over the baseline.

5. The controller gains are optimized for a fixed plant. This results in 6.45% improvement over the baseline.

Cases 1–3 are the most interesting. They show that, for this case, designing the plant and controller simultaneously recovers about half of the performance lost when optimizing the plant with an optimal-control based approach and then designing a controller for it. It also suggests that a different controller architecture could obtain better performance, up to that of Case 1.

Also, in 2023, Iori [81] evaluated the impact of control co-design in wind turbine blade design by including the blade’s pitch angle and rotational speed as design variables for each wind speed considered. The disciplines considered were aerodynamics and control. The problem objective is to maximize the annual energy production augmented with a thrust term that is meant as a surrogate for the turbine loads. No significant difference was found between the control strategy found by the optimizer and the conventional control scheme. This may be due to the structural discipline being omitted, which prevents a stronger coupling between the control strategy and blade design, as well as the simple control strategy being employed, as opposed to the state of the art of one blade at a time control. Finally, [81] proposed a post-optimality method based on the shadow cost idea from Lagrange multipliers to evaluate if control co-design would meaningfully change the optimum of a design done with a fixed controller, considering different levels of control integration (i.e., feedforward on wind speed and LQR feedback).

In the next sections, the applications of control co-design for MLA and GLA functions, as well as conventional control design targeting those functions, is considered in more detail.

Applications to maneuver load alleviation (MLA)

The benefits of using maneuver load alleviation (MLA) for improving aircraft performance have been known since the late 1940s [82]. In a paper from 1971, White [9] reported that the use of MLA allowed for a 10% increase in the wingspan or a 13% increase in the payload for a large four-engine transonic transport aircraft. Several MLA controller architectures have been proposed since then, from the scheduling of open-loop deflections based on load factor used in those early studies to model predictive controllers [31, 83]. Nevertheless, most studies involving MLA attempt to improve an existing design instead of considering the use of MLA from the very beginning of the design process.

Recent advancements in multidisciplinary design optimization (MDO) have made it possible to size the airframe while accounting for the effects of MLA. Xu and Kroo [10] used

an extension of the design space with the deflection of control surfaces during a maneuver to allow for MLA and gust load alleviation (GLA) in their MDO problem formulation. The control surface deflections during the maneuver are included as design variables to account for MLA, while the gains of a proportional-derivative (PD) controller are included to account for GLA. Stanford [11] expanded this approach by optimizing the layout of the control surfaces for MLA and GLA of a transport aircraft and replaced the PD controller with a linear quadratic regulator (LQR), where the weighting matrices of the LQR are used as design variables. Binder et al. [12] considered a similar approach with static control deflections during the maneuver accounting for the MLA system and a feedforward control based on angle of attack measurements as the GLA system. Haghighat et al. [13] introduced an approach where a single LQR controller was responsible for performing MLA and GLA.

Applications to gust load alleviation (GLA)

The first time gust load alleviation (GLA) control was considered in the multidisciplinary design optimization (MDO) of aircraft was in 1993 [14], when the structural design of a cantilevered wing was considered simultaneously with the design of a proportional-integral (PI) controller that uses the measurement of the wing tip acceleration to command a control surface also located at the tip of the wing. The chosen gust model was stochastic but simplified since the turbulence input was generated based on a first-order filter driven by Gaussian white noise. A summary of aircraft designed with GLA systems as of 2012 can be found in [15].

An increased number of studies in this area was conducted starting in the late 2000s due to renewed interest in aircraft MDO. Hunten et al. [24] showed results of the optimization of a blended wing-body (BWB) configuration using flap deflection for drag reduction, as well as for maneuver and gust load alleviation, considering only quasi-static gusts and maneuvers. Haghighat et al. [13] performed MDO of an aircraft considering a linear quadratic (LQ) controller for reducing gust loads, which are evaluated in the time domain for a discrete gust. Xu and Kroo [10] performed an aerostructural design optimization of an aircraft considering a proportional derivative control system for GLA, which took the angle of attack as an input, and deflections for the two most outboard control surfaces on the wing being designed as outputs. They considered a discrete gust model and the simulations to obtain the gust loads were performed in the time domain.

Stanford [11] considered the design of a wingbox with trailing edge control surfaces used for GLA, including a static output feedback controller and control surface sizing into the design. To evaluate the gust loads, the turbulence model was added to the state-space representation of the airplane. No time-domain simulation was performed, instead a Lya-

punov equation was solved to obtain the statistics of the stresses due to gust, which were then used to inform constraints for the optimization problem. Krengel and Hepperle [16] proposed a wing optimization problem in which the GLA controller was based on feedback of the lift coefficient, in addition to a pitch damper designed using sequential loop closure on pitch rate and angle of attack. They reported a 14% reduction in bending moment, with a corresponding 11.6% reduction in block fuel consumption, which was the cost function of the proposed MDO problem.¹ The European Flight Phase Adaptive Aero-Servo-Elastic Aircraft Design Methods (FLiPASED) project outlined a method to integrate MLA, GLA, and flutter suppression in the MDO process, using an $\mathcal{H}_\infty$ controller for the MLA function and an MPC controller for the GLA function, based on reduced order models [84]. However, it is not targeted for gradient-based optimization since no attempt is made to differentiate the control design process, which is complicated by the presence of a variable order model order reduction process in the optimization loop.

Meanwhile, there has been much research done on the design of GLA controllers for an already designed aircraft. In particular, Vartio et al. [25] designed a linear quadratic Gaussian (LQG) controller for a half BWB scaled model with multiple trailing edge control surfaces, which was then tested in free pitch and plunge at the NASA Langley’s Transonic Dynamics Tunnel. The closed-loop system exhibited over 50% reduction in gust loads compared to a non-GLA controller. Zeng et al. [28] presented an adaptive feedforward controller capable of significantly reducing gust loads in the 2 to 20 Hz range. Dillsaver et al. [26] designed an LQG controller considering a reduced-order flexible aircraft model and obtained an expressive reduction of wing root curvature in simulation. Haghighat et al. [29] employed a model predictive controller (MPC) for the GLA problem and compared its performance with an LQG controller in simulation. Pereira [30] also considered MPC-based architectures for GLA with LIDAR-based gust preview in a flexible aircraft. Ting et al. [85] designed and performed wind tunnel tests of preview $\mathcal{H}_2$ and $\mathcal{H}_\infty$ controllers, reporting a reduction in the wing root strains for both controllers. Fournier et al. [32] presented structured mixed $\mathcal{H}_2/\mathcal{H}_\infty$ controller designs for gust load alleviation with the use of feedforward LIDAR measurements and achieved 50% bending moment reduction for both discrete and continuous turbulence in simulation, results which were echoed by Cavaliere and Fezans [34], who were able to obtain 75% reduction in wing bending moment at various stations but at the cost of increasing the moment at horizontal tail root to 200% of the original value. Düssler et al. [27] designed a feedback LQG controller that exhibited up to 53% reduction in vertical wingtip displacement of a very flexible aircraft model under discrete gusts in

¹Krengel and Hepperle [16] also considers an MLA system but its design is fixed (i.e., not part of the optimization).

linear simulations and 37% in nonlinear simulations. Stalla et al. [35] designed a GLA controller through μ -synthesis for a wind tunnel experiment capable of reducing the wing-root bending moment by up to 30%.

As can be seen, there is a gap between the techniques employed to design the controllers for an already designed aircraft vs. the ones assumed in the setting of aircraft control co-design. Moreover, there is no agreed-upon choice of control architecture that should be used for GLA. This makes it desirable to develop an approach that guarantees the existence of a controller that gives satisfactory GLA performance for a closed-loop aircraft system without directly designing such a controller. We refer to such an approach as controller-agnostic.

1.2.2 Fundamental limitations of control

The fundamental limitations of control are relations that must be respected by any feedback system independent of the design of the controller. According to Chen et al. [86]:

In their essential spirit, such constraints and limits share much in common with natural laws and principles widely found in science and engineering, such as Shannon’s theorems in communications theory, the Cramer-Rao bound in statistics, and Heisenberg’s uncertainty principle in quantum mechanics, which, while differing in contexts and problems, all address constraints and limits that can neither be overcome nor circumvented, defining boundaries separating what can be achieved and what cannot.

The spirit of these limits and constraints fits well with the idea of controller-agnostic design metrics since they uncover limitations inherent to the plant and not to a particular design:

Knowledge of inherent limitations is useful in designing a control law that achieves a reasonable tradeoff between conflicting system goals. More significantly, such knowledge is useful much earlier in the engineering design cycle, in order to configure the plant and specify sensors and actuators so that the resulting feedback control problem is tractable [87].

The most famous of which is probably the Bode sensitivity integral, which states that the area under the log of the sensitivity function is equal to zero for a feedback system composed by any controller and an open-loop stable plant [88]. This means that any sensitivity decrease in some frequency band will result in a sensitivity increase somewhere else [39].

It has wide applications, from the automatic tuning of PID controllers [89, 90] to better understanding biomolecular systems [91], passing through limitations on the performance of noise-canceling headphones [92], among many others. This and other limitations will be discussed in more detail in Section 2.1.1.

There is a vast literature on the fundamental limitations of control, which starts with the seminal work of Bode [88]. The following references are a good starting point for the interested reader. The most recent review on the topic was done by Chen et al. [86]. The most comprehensive book is by Seron et al. [93], which contains results for both linear (Part 2) and nonlinear (Part 4) systems, the latter based on the theory of “nonlinear operators acting on normed signal spaces”. Good introductions on the topic are given by Freudenberg et al. [87], Looze et al. [94], Åström [95] and the entry on the Encyclopedia of Systems and Control [96]. A control design procedure that uses ideas from fundamental limitations is proposed by Boyd and Barratt [42]. Results on the limitations for the disturbance rejection of unmeasured outputs are provided by Freudenberg et al. [97]. Finally, for practical examples of the implication of these limitations, the 1989 IEEE Bode Lecture delivered by Stein [39] and titled “Respect the Unstable” is highly recommended.

1.2.3 Risk metrics for flying in continuous gust

Accounting for gusts is an essential part of aircraft design. It is important for structural sizing [98], as well as for passenger and crew safety and comfort. Turbulence-related accidents were the most frequent accident type in both 2021 and 2022. They accounted for more than one-third of all accidents involving U.S. scheduled airline flights in the 2009 to 2018 period, resulting in serious injury to passengers and/or crew members [99]. The lack of fatal accidents in recent years attests to the progress made in the technologies and requirements for considering gusts in the aircraft design process.

Gusts are usually modeled as either discrete gusts (usually of the one-minus-cosine type) or continuous gusts (i.e., turbulence), in which case they are idealized as (locally) stationary Gaussian stochastic processes [100]. In the latter case, the gusts are fully described by their power spectral density (PSD), and the most commonly used models are the Dryden [101], von Kármán [102], Kaimal [103], and Mann [104] models. Since the introduction of the power spectral method applied to gusts in 1952–1953 [101], the stochastic modeling has gained wide acceptance [105, Chapter 5] even though this approximation was found to yield non-conservative results in some cases [106]. For this reason, it did not supersede the older discrete gust certification requirements [98, part (a)]. In any case, the stochastic modeling of gusts has been well validated in practice and is enshrined in the certification

procedure for aircraft [98, part (b)] and wind turbines [107]. This analysis has also made its way to MDO (e.g., [108]). Furthermore, Gaussian processes have been extensively studied in the mathematical literature and offer a very tractable framework, both in the time and in the frequency domains, which is not the case for the more recent and much more computationally expensive models based on large-eddy simulations [109–111].

The modeling of gusts as Gaussian processes, however, poses a dilemma because it makes it mathematically impossible to guarantee that the aircraft can reject all gusts it encounters with probability one. This is because Gaussian processes have unbounded support, meaning they can take any real value with a probability greater than zero. In the linear setting that is traditionally used for analyzing the aircraft responses under Gaussian turbulence, these responses (e.g., pitch angle, load factor, etc.) will also be Gaussian processes. Therefore, the probability of them exceeding any limit imposed by the design is greater than zero. A statistical approach must then be employed and was developed to allow the inclusion of this method in the certification requirements.

For the certification of transport aircraft, 14 CFR 25.341(b) Continuous Turbulence Design Criteria [98] requires that the root-mean-square (RMS) value of the gust loads resulting from normalized von Kármán turbulence multiplied by a turbulence intensity specified by the regulation stays below the limit load for the design. As documented in Hoblit [105, Section 5.3.2] and in Advisory Circular: Dynamic Gust Loads [112], this value can be interpreted as the RMS gust velocity multiplied by the gust load peak-to-RMS ratio, that is, the number of standard deviations that the peak gust load is away from the RMS gust load. This requirement is equivalent to ensuring that the load does not exceed the limit load with some preestablished instantaneous probability, which is also known in the literature as a *chance constraint* [113].

The instantaneous probability of exceedance is related to the measure of the trajectories outside of the allowable limits, which, if the process is ergodic, is equal to the proportion of time spent outside the limits [114, Chapter 7]. Due to the nonlinearity present in real systems, it might not be possible to recover from an exceedance event (e.g., low altitude stall, loss of control, etc.), so, in addition to requiring the probability of such events to be small, it is also desirable that they happen with rarely. It is worth noting that these nonlinearities are very complicated in their own right. For example, stall is a dynamic effect [115, Chapter 9]. Thus, it does not happen instantaneously after the critical angle of attack is crossed.

The expected frequency of exceedance (or upcrossings) of design limits by a random process can be evaluated using Kac-Rice’s formula [116, 117], and it provides more information about the likelihood of an exceedance event. In fact, since the point process of upcrossings of increasingly high levels by a stochastic process with finite covariance con-

verges to a Poisson process with a rate given by the expected frequency of upcrossings [118, Chapter 9], this frequency can be used to estimate the joint probability of exceedance, that is, the probability of exceeding a specified level in a period of time, such as the lifetime of the aircraft.

The instantaneous probability and expected frequency of exceedance metrics have been previously considered as safety metrics for airplane flight in turbulence by Richardson et al. [119, 120]. They proposed the concept of a “stationary flight envelope,” that is, a subset of the usual flight envelope in which an aircraft flying under stochastic gust conditions is guaranteed to be able to maintain steady flight within a certain risk level defined by one of these metrics. These same ideas can be applied to the landing flight phase, but for this, the closed-loop effects of having a pilot/controller must be considered. This is done in a controller-agnostic manner in Section 4.2.

Statistical Discrete Gust and Matching Filter Theory

The *statistical discrete gust* method is a technique to evaluate the statistics of an aircraft’s response to continuous gust by simulating its response to a specially constructed discrete gust that depends on the dynamics of the aircraft and on the gust model [121]. It was initially developed in the United Kingdom by Jones, who has a large body of work on this method and the non-Gaussian aspects of atmospheric turbulence, spanning the 40 years between 1969 and 2009 [121–127]. This method aims to overcome the limitations of the usual continuous gust loads calculations based on random process theory, namely the assumption of linear dynamics and Gaussian gust models by drawing on the idea of self-similarity (see [121] and references therein).

In the late 1980s, the stochastic discrete method made its way to the United States and Pototzky et al. [128] evaluated the method as a “candidate gust-load analysis method for complying with FAA certification requirements”. While performing this evaluation, they noticed that the *matching filter theory*, originally used for the detection of returning radar signals could also be used to compute time-correlated gust loads that “are theoretically identical to those of random process theory” [128] (see also [129]). Matched filter theory was identified as a linear application of the stochastic discrete gust method with much greater computational efficiency since it dispensed the use of a search procedure to compute the worst-case gust. The procedure to construct the worst-case gust for this method consists of subjecting the aircraft system, augmented with a filter capable of generating the continuous gust from white noise, to an impulse response, getting the load output of interest and normalizing it. The gust thus constructed can be used to drive the system and obtain a time signal that corresponds to the autocorrelation function of the system driven by a continu-

ous gust. This can be used to derive the PSD of the response (via the Fourier transform). Furthermore, the maximum of the autocorrelation function corresponds to the maximum load that can be created by a gust with the same energy as the input signal, therefore justifying the designation of “worst-case” [130]. This maximum, which happens at time $t = 0$, also corresponds one point on the load ellipse. This procedure must be repeated for each load being analyzed to construct the complete load ellipse.

Extensions of the matched filter theory for nonlinear systems were investigated. Scott et al. [131, 132, 133] showed that, for some nonlinearities, the matched filter theory method was capable of still producing the worse case loads, provided that a search for the critical impulse strength was performed.

Silva and Perry [134] presented a method based on the Volterra theory of nonlinear systems to construct a linearized system that provides bounds on the response of the nonlinear system of interest, which is valid as long as the nonlinearities can be modeled as zero memory non-linear components.

1.2.4 Summary

Based on the exposition on control co-design (Section 1.2.1), there are four approaches for including the control system in the MDO problem of a closed-loop system:

Open-loop metrics augmented with heuristics, in which the open-loop behavior of the plant is used to heuristically estimate the risk in finding a controller that can achieve some level of closed-loop performance (e.g., [36, 65])

Combined plant and controller synthesis, in which the controller is designed simultaneously with the plant (using, e.g., gain selection [10, 12, 14, 16, 49–53, 56, 57, 76, 79], covariance parameterization [60], LQR/LQG [11, 13, 59, 135–137], $\mathcal{H}_\infty$ [138–140], MPC [141, 142])

Optimal control input co-design, in which the control inputs are designed at the same time as the plant, through the solution of an optimal control problem as a part of the optimization or as a suboptimization (e.g., [37, 66–70, 73, 74, 79, 143], for static MLA [10–12, 24])

Closed-loop metrics based on fundamental limitations, in which results of control theory are used to find upper bounds for the performance of the closed-loop system (e.g., [41–43, 62–64])

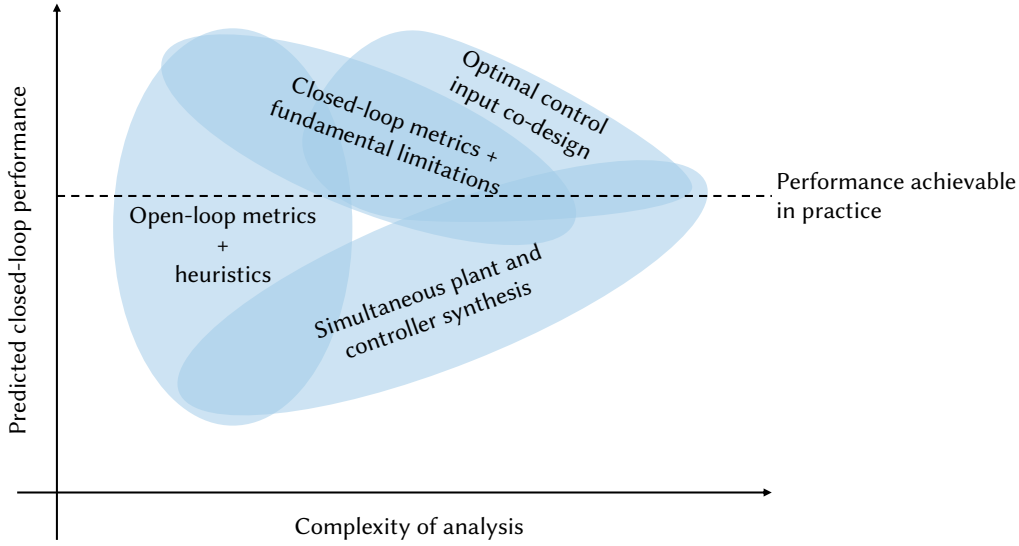


Figure 1.3 Different approaches for including control considerations into MDO

These approaches are schematically represented in Figure 1.3 in terms of the complexity of the analysis process and the predicted closed-loop performance in relation to the best performance that is achievable in practice.

The strategy of using open-loop metrics augmented by domain-specific heuristics provides for the simplest analysis but has high uncertainty, especially when there is not much historical design data to use in developing the heuristics. It presents a high risk that the predicted closed-loop performance will be either not achievable or too conservative.

The method of simultaneous plant and controller synthesis tries to approach the limit of achievable performance from below since it generates a controller design that could, in theory, be realized. Fixing the control architecture from the start risks missing opportunities to achieve improved performance. On the other hand, it risks overestimating the achievable performance if the controller robustness and limitations, such as saturation and bandwidth, are not taken into account. Performing a state-of-the-art controller design at the same time that the plant is designed adds significant complexity to the MDO problem.

The two other approaches, namely optimal control and the fundamental limitations of control, attempt to bound the limit of achievable performance from above. When incorporated into MDO, optimal control will calculate a time history of the inputs that accomplish a pre-determined maneuver or indicate that there is no such control input that does so. Such an approach will generally overestimate the achievable performance because the existence of a controller that implements the calculated behavior is not considered. On the other hand, the basis chosen for the control input might over-constrain the problem and find a conservative solution (e.g., in the case that the control input is constant).

The approach based on the fundamental limitations of control also tries to find an upper bound for the achievable closed-loop performance starting from the open-loop design. It does that by imposing theoretical constraints on the control system, such as the Bode sensitivity integral relation, the Poisson integral relation, bounds on robustness margins, and those limitations derived from the requirement of internal stability. While such an approach will also generally overpredict the achievable performance, since only a subset of all the limitations can be imposed before controller synthesis is effectively performed, it could lead to a tighter upper bound versus the optimal-control-based approaches at a similar level of complexity. As in the previous case, underparameterization—of, e.g., the closed-loop functions being designed—may result in conservative designs.

One particular case in which the maneuver-centric approach derived from optimal control adds too much complexity to an MDO problem is that of stochastic disturbance rejection, as is the case for flight under continuous gust. In this case, the most direct procedure for dealing with the stochastic nature of the disturbance, namely Monte Carlo sampling, would lead to intractably large problems. In this situation, an approach based on the fundamental limitations of control can lead to much more tractable MDO problems, as this thesis will attempt to demonstrate.

1.3 Objectives

This dissertation aims to develop ways to assess the characteristics (e.g., performance, robustness, stability, ...) of a closed-loop system without designing the controller. In this sense, the overarching objective is to develop control-related but controller-agnostic design methods that are suitable for gradient-based MDO. To accomplish this goal, a cross-disciplinary approach is pursued which draws on the fields of optimization, fundamental limitations of control, and stochastic processes, and the following specific objectives are addressed:

1. *Estimate an upper bound for the closed-loop performance from the open-loop plant without controller design while taking into account physical limitations of the control system such as saturation, rate saturation, and bandwidth, as well as limitations on the achievable robustness of the feedback loop itself.* This is accomplished by introducing a parameterization of the sensitivity function (Section 2.1), which can be used to calculate the disturbance rejection performance of the closed-loop system as well as the robustness of the closed-loop itself, subject to constraints on bandwidth imposed via the Bode sensitivity integral (Equation (2.10)). This approach can also be used to

calculate risk metrics for signals of the system, which if applied to the control input signal, can inform constraints on saturation and rate saturation of actuators.

2. *Offer an alternative method to optimal control transcriptions that can more efficiently handle large systems under stochastic disturbances.* As shown in Section 1.2.1, the inclusion of optimal control transcriptions into an MDO problem, in the sense that the control input is designed at the same time as the plant, is a powerful controller-agnostic approach. However, it can substantially increase the size and complexity of the MDO problem. This effect is enhanced when there are stochastic elements in the optimization that are handled via sampling (i.e., Monte Carlo approaches). The sensitivity function parameterization approach (Section 2.1) is an alternative that does not require sampling and scales very well with the size of the dynamical system, as shown by the 317 states example of Section 4.1.
3. *Develop derivative calculation procedures for the proposed metrics so that they are suitable for gradient-based optimization.* The derivative calculation formulae are developed for the PSD moments in Section 2.3, and make use of matrix-free and adjoint techniques to manage the time and space complexity of the calculation. The derivatives for the suboptimization problem are available in the literature and are reviewed in Section 2.4. The accuracy of the derived formulae for analytical derivatives is confirmed by comparison with finite differences approximations for the test cases of Chapter 4.
4. *Evaluate the impact of the proposed metrics when applied to MDO problems.* The proposed approaches are analyzed in three example problems: the MDO of the wingbox of a blended-wing-body aircraft with a GLA system flying under continuous gust (Section 4.1); the analysis of the glideslope tracking capability of a supersonic configuration aircraft flying the approach to land under continuous gust (Section 4.2); and the MDO of a cantilevered wing designed with an MLA system (Section 4.3).

1.4 Outline

The remainder of this dissertation is organized as follows:

Chapter 2 delineates and develops the methods used in this dissertation: **Section 2.1** introduces the approach to parameterize the closed-loop transfer functions, reviews the fundamental limitations of control that can be imposed on these transfer functions, and dis-

cusses the connections between this approach and control design methods. **Section 2.3** develops the framework to calculate the closed-loop PSD moments based on the parameterization of the sensitivity function, including efficient analytical derivatives. **Section 2.2** explains how these moments can be used to calculate relevant metrics for control design, including the probability of exceedance, frequency of exceedance, peak stress, and fatigue life, which were motivated in Section 1.2.3. **Section 2.4** presents an approach to include MLA as a suboptimization problem in a mixed fidelity MDO problem. The procedure to calculate the derivatives of the constrained optimum with respect to parameters of the problem is reviewed.

Chapter 3 discusses the numerical developments that were made to enable the application of the methods introduced in Chapter 2 into aeroservoelastic MDO problems using the geometrically non-linear aeroelastic toolbox UM/NAST [144]. **Section 3.1** details the common implementation of the algorithmic differentiation (AD) for all UM/NAST-based metrics using *policy-based design* [145, Chapter 1] in C++, which allows the creation of new metrics abstracting the AD tool. **Section 3.1.4** describes the integration of UM/NAST into NASA’s MDO framework, OpenMDAO [146]. **Section 3.2** discusses the integration of UM/NAST with the high-fidelity static aero-structural MDO software MACH [147] for MLA applications, building on previous work for a flutter constraint [148].

Chapter 4 applies the methods introduced in Chapter 2 for relevant aircraft design problems. **Section 4.1** presents a new approach for incorporating gust load alleviation (GLA) into the multidisciplinary design optimization (MDO) process. With this approach, closed-loop control considerations are incorporated into the MDO problem formulation through the inclusion of the parameterization of the sensitivity to stress transfer function of the aircraft. The parameters of this function are included as design variables and a constraint based on the Bode integral relation is imposed on it. Two constraints related to the gust loads are added to the MDO problem: one for peak gust stress, based on the design envelope criterion, and the other for fatigue life under continuous turbulence. While the GLA feedback controller does not need to be designed, the optimizer designs the closed-loop sensitivity function directly. The parameterization of the sensitivity function is chosen to guarantee the robustness of the closed-loop system. The use of this new approach is demonstrated by performing multidisciplinary design optimization of a flexible free-flying aircraft model subjected to gust loads. The matrix-free adjoint derivatives are verified against finite differences, and the optimum designs with and without control considerations are compared. Through the use of the proposed approach, it is shown that it is possible to

reduce the optimal wing mass for the closed-loop design in relation to the open-loop design while still satisfying the other constraints. **Section 4.2** considers a surrogate model of the longitudinal dynamics of a supersonic aircraft configuration in the approach-to-land flight phase. A simple optimization problem for designing the sensitivity function is considered and the probabilities and expected frequencies of exceedance of design limits of angle of attack, glideslope deviation, elevator deflection, and elevator deflection rate of the closed-loop system are analyzed. The analytical derivatives of these quantities with respect to design parameters are verified against a finite difference approximation with excellent agreement. **Section 4.3** incorporates a lower-fidelity controller-agnostic MLA analysis into a high-fidelity MDO. The approach is based on expanding the design space with the addition of control surface deflections during a maneuver as design variables, and constraining the resulting stresses. A bi-level optimization architecture is employed to avoid adding design variables to the overall aircraft optimization problem. The inner problem consists of minimizing the aggregated stresses during a maneuver, which are computed in a beam-based structural model coupled with aerodynamics based on the method of sections. The outer optimization problem is an aircraft range maximization MDO, evaluated on a shell-based structure model coupled with Reynolds-averaged Navier-Stokes aerodynamics. This problem is solved for a clamped wing model, with and without a geometrically non-linear flutter constraint. The inclusion of the flutter constraint is shown to impact the trends in the optimal design.

Chapter 5 presents a summary of the developments included in this dissertation. It lists the key contributions resulting from the methodological developments presented herein and their application to different MDO problems. Recommendations for future work are provided.

CHAPTER 2

Methodological Developments

This Chapter provides an overview of methodological developments and advances that have been developed and used for the subsequent case studies (Chapter 4). In this dissertation, two types of problems are investigated:

1. The problem of stochastic disturbance rejection
2. The problem of static input optimization

The first problem is challenging for time-domain techniques due to the need for sampling and large simulation times to converge the statistics of the response. Thus the “optimal control input co-design” approach (Section 1.2.4) may not be the most adequate.

An alternative approach based on parameterizing one of the closed-loop transfer functions is presented in Section 2.1, which also develops methods to incorporate the fundamental limitations of control into the parameterization and in the resulting MDO problem. A class of risk metrics based on the moments of the power spectral density (PSD) of outputs is considered (Section 2.2), including the probability and frequency of exceedance of a level, the expected fatigue life and the peak stress as defined in the certification requirements [98]. The procedure to calculate these moments along with their analytical derivatives is developed in Section 2.3.

The strategy chosen to address the second problem is to be minimally intrusive to a pre-existing MDO problem that does not include optimization of the control input. This is accomplished by proposing a nested (bi-level) optimization approach in Section 2.4. The procedure to compute the derivatives of the function calculated by the inner optimization with respect to the outer design variables is reviewed based on the literature. A model order reduction process is used to exploit the nested design to improve performance.

A common signal of interest in aircraft structures is that of stress. A simplified stress calculation procedure for a beam-based structural model is presented in Section 2.5. Where

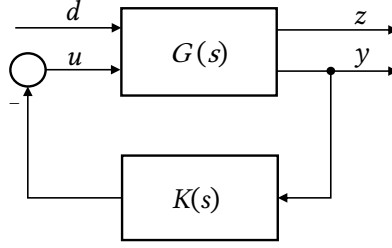


Figure 2.1 Plant G with a feedback controller K

a model order reduction process is used, it is possible that a better stress calculation procedure can be constructed using information from the high-order model. This idea is left as future work. The calculation of the moments of the PSD of von Mises stresses is handled in Section 2.3.2, since it is a non-linear combination of stress components.

2.1 Closed-loop transfer functions

A generic (possibly nonlinear) feedback control system is shown in Figure 2.1. In this representation, the plant, G , has one control input vector, u , and one disturbance input vector, d . Its outputs are separated into two vectors, one for the measured inputs that are used for feedback, i.e., sensor measurements, y , and one for (possibly unmeasured) outputs that are not used for feedback, z . If the plant and controller are represented by linear models, e.g., around a nominal condition, the decomposition shown in Figure 2.2a is possible, where $K(s)$ denotes the transfer function of the controller, $G_{yu}(s)$ is the transfer function from the control inputs to the sensor measurements, $G_{yd}(s)$ is the transfer function from disturbance to sensor measurements, $G_{zu}(s)$ is the transfer function from the control input to the outputs not used for feedback, and $G_{zd}(s)$ is the transfer function from the disturbance to the outputs not used for feedback. To simplify the notation, we define the loop transfer function at the output:

$$L(s) = G_{yu}(s)K(s) \quad (2.1)$$

From the block diagram, the following *closed-loop* transfer functions can be derived:

The **sensitivity function at the output**,

$$S(s) = (I + L(s))^{-1} \quad (2.2)$$

shown in Figure 2.2b.

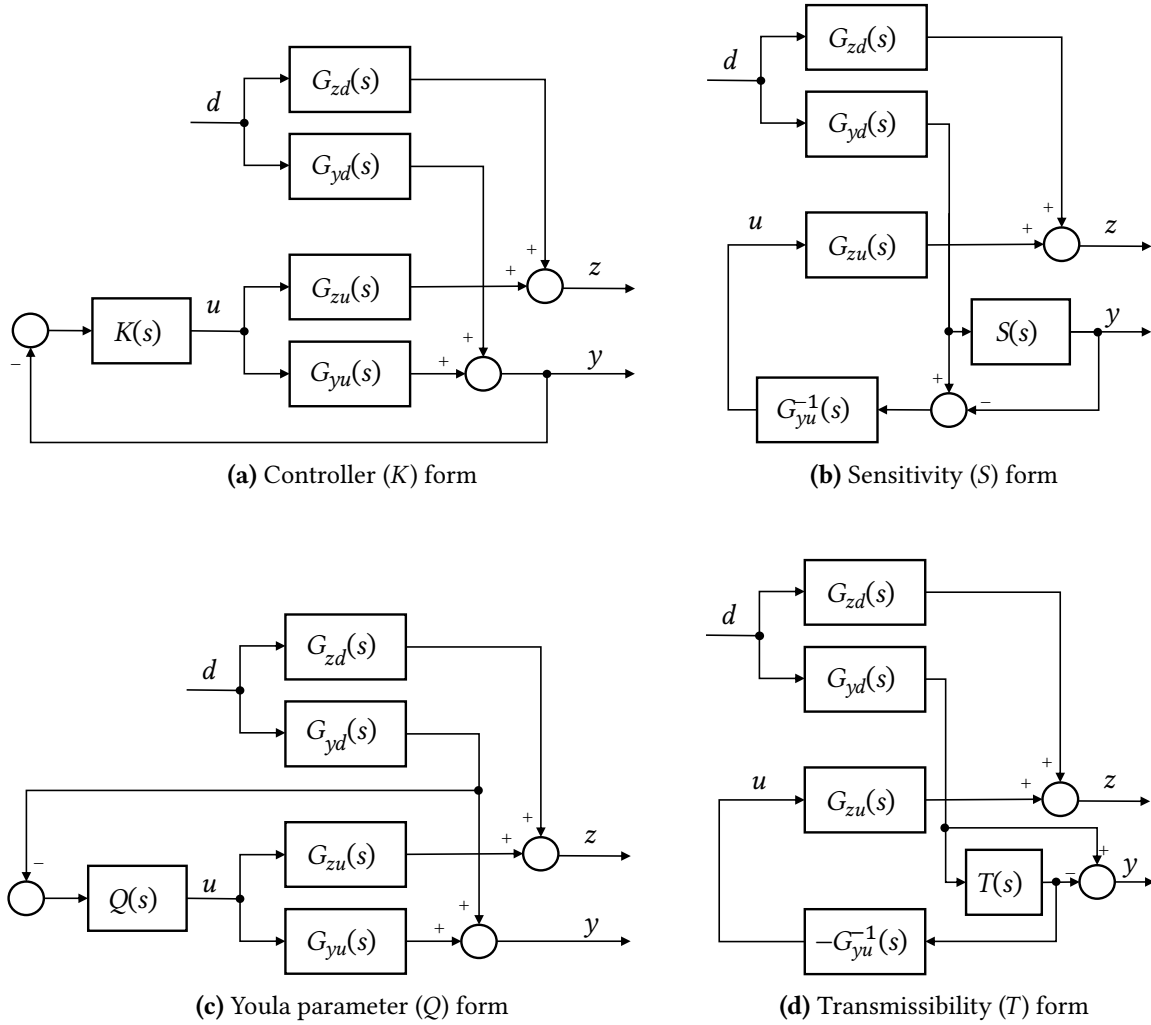


Figure 2.2 Equivalent representations of a linear feedback system

The **transmissibility function at the output**, or complementary sensitivity,

$$T(s) = L(s)(I + L(s))^{-1} \quad (2.3)$$

shown in Figure 2.2d.

The **Youla parameter**, named after the work of Youla et al. [149, 150] and Kučera [151] on the parameterization of all stabilizing controllers for a given plant, defined as

$$Q(s) = K(s)(I + L(s))^{-1} \quad (2.4)$$

and shown in Figure 2.2c.

2.1.1 Constraints on the achievable controllers

Figure 2.2 highlights that the controller has one degree of freedom since once one of the transfer functions denoted by $K(s)$, $Q(s)$, $S(s)$, and $T(s)$ is chosen, it determines the others. In particular, it follows from the definitions that $T(s) + S(s) = 1$ everywhere on the complex plane. Boyd and Barratt [42, Chapter 7] refers to the relations between these transfer functions, i.e., Equations (2.2) to (2.4), as *realizability constraints*.

This is a complete parameterization of the space of stabilizing controllers, in the sense that all stabilizing controllers can be described by any of these transfer functions. The definition of internal stability from Desoer and Chan [152] is adopted here, as was done in [41] and [42, Section 7.2]:

Definition (Internal stability). A negative feedback loop between $K(s)$ and $G_{yu}(s)$ is internally stable if the following closed-loop transfer functions are stable:

$$H_{yy_\Delta}(s) = S(s) \quad (2.5)$$

$$H_{yu_\Delta}(s) = S(s)G_{yu}(s) \quad (2.6)$$

$$H_{uy_\Delta}(s) = -K(s)S(s) \quad (2.7)$$

$$H_{uu_\Delta}(s) = -K(s)S(s)G_{yu}(s) \quad (2.8)$$

where u_Δ and y_Δ are disturbances introduced at the input and output of $G_{yu}(s)$, respectively, and $S(s)$ is defined as in Equation (2.2).

This definition is motivated by the Bounded Input Bounded Output (BIBO) notion of stability since it guarantees that all (internal) signals in the feedback loop remain bounded in the presence of finite additive disturbances. For the SISO case, these conditions are reduced to interpolation conditions that can be written in terms of any closed-loop transfer function, for example in terms of S :

Fact (Interpolation constraints). [41]. Let $S(s)$ be the sensitivity function of a SISO system with strictly proper plant $G_{yu}(s)$. Assume that the plant and the controller are minimal realizations, proper, and all the closed right half-plane (CRHP) poles and zeros of the plant are distinct. Then the closed-loop system is internally stable if and only if the following conditions hold:

1. $S(s)$ is stable
2. $S(s)$ has a zero at all the closed right half plane (CRHP) poles of $G_{yu}(s)$
3. $1 - S(s)$ has a zero at all the closed right half plane (CRHP) zeros of $G_{yu}(s)$
4. $1 - S(s)$ has relative degree greater or equal to $G_{yu}(s)$

Proof. (See also [41] and references therein) Item 1 is equivalent to requiring that 2.5 is stable. Item 2 follows from Equation (2.6) since $S(s)$ has to cancel the unstable poles in $G_{yu}(s)$. Conversely, if $S(s)$ cancels the unstable poles in $G_{yu}(s)$, and they are minimal realizations, then $H_{yu\Delta}$ is stable since requiring $G_{yu}(s)$ and $K(s)$ to be proper implies that $S(s)$ is also proper.

Recall the definition: $S(s) = (1 + G_{yu}(s)K(s))^{-1}$, which can be rearranged to express $K(s)$ as a function of $S(s)$:

$$K(s) = \frac{1 - S(s)}{S(s)G_{yu}(s)} \quad (2.9)$$

Then Equation (2.7) imposes that $(1 - S(s))/G_{yu}$ is stable, thus $1 - S(s)$ must cancel the non-stable zeros of $G_{yu}(s)$ and the resulting transfer function must remain proper, thus requiring item 3 and item 4. Conversely, provided that $S(s)$ is stable, $1 - S(s)$ cancels the CRHP zeros of $G_{yu}(s)$, then $H_{u\gamma\Delta} = -\frac{1-S(s)}{S(s)G_{yu}(s)}S(s)$ has no poles in CRHP and item 4 guarantee that it is proper, therefore it is stable. Finally, Equation (2.8) is satisfied without imposing additional constraints. $\square$

In the case of repeated poles or zeros, $\frac{d^p S(s)}{ds}|_p = 0$ at all CHRP poles p of G_{yu} with multiplicity n_p and $\frac{d^z T(s)}{ds}|_z = 0$ at all CRHP zeros z of G_{yu} with multiplicity n_z [153, Section 3.5]. As remarked by Morari and Zafiriou [154, Chapter 3], the relative degree condition can be ignored in the initial phases of the design since it can be satisfied by adding poles to the controller later.

The goals of feedback control can be expressed in terms of the frequency response of the closed-loop transfer functions [155, Chapters 3 and 4] [153, Section 3.2]. For good disturbance rejection and performance robustness (i.e., insensitivity to changes in the plant), it is desirable that $|S(j\omega)| \ll 1$, for good tracking $|T(j\omega)| \approx 1$, and for low actuator effort $|Q(j\omega)| \ll 1$, possibly frequency weighted to account for the rate of actuation as part of the cost. However, some limitations and tradeoffs prevent all these goals from being achieved simultaneously. The first restriction we turn our attention to is that of bandwidth: the actuators of a physical plant cannot be actuated arbitrarily fast, therefore for sufficiently large frequencies $\omega > \omega_b$, it can be approximated that $S(j\omega) = 1$ and $T(j\omega) = 0$, i.e., the plant behaves as if in open-loop. The constant ω_b denotes the *available bandwidth* [39].

The restrictions imposed by the limited bandwidth can be quantified by the *Bode sensitivity integral*:

Fact (Bode sensitivity integral). [156, 157]. Assume that the open-loop transfer function $L(s)$ has finitely many open right half plane (ORHP) poles ($p_k, k = 1, \dots, n_{\text{ORHP}}$) and its

relative degree is strictly greater than one¹. Then the sensitivity function defined in Equation (2.2) satisfies:

$$\int_{\omega=0}^{\infty} \log |S(j\omega)| d\omega = \pi \sum_{k=1}^{n_{\text{ORHP}}} \Re(p_k) \quad (2.10)$$

for a single-input single-output (SISO) system, or

$$\int_{\omega=0}^{\infty} \log |\det S(j\omega)| d\omega = \pi \sum_{k=1}^{n_{\text{ORHP}}} \Re(p_k) \quad (2.11)$$

for a multiple-input multiple-output (MIMO) system.

The original result, applicable only to stable SISO systems (i.e., the integrated quantity is equal to zero) is due to Bode [88]. The extension to unstable and to MIMO systems is due to Freudenberg and Looze [156, 157].

The Bode integral relation imposes additional restrictions related to the bandwidth of the system because it states that a reduction in sensitivity in one frequency range must be accompanied by an increase in sensitivity in another frequency range. This is made worse for unstable plants since the integral of the log of the magnitude of the sensitivity transfer function must be equal to the sum of the real parts of the unstable poles. The assumptions for the relation to be valid apply to a large class of practical systems since it only requires that both the plant and the controller have finite bandwidth [39]. This is the case, for example, when there are actuators and sensors in the loop. Due to the finite bandwidth of the elements in the control loop, the integral in Equations (2.10) and (2.11) can be approximated by using the available bandwidth, ω_b , as the upper limit of integration, that is, the largest frequency where the modeling assumptions for the plant, sensors, actuators, and the computers that implement the control laws are valid [39].

In addition, stability robustness margins can be stated in terms of S and T [158], [159, Section 2.4.3]. These so-called disk margins are robust versions of the classical gain and phase margins since they account for combined uncertainty in gain and phase. The following definitions and method to calculate the disk margin are reproduced from Seiler et al. [158]:

Definition (Disk in the complex plane). Let $\alpha, \beta \in \mathbb{R}$. The disk with size α and skew β is defined as

$$\mathcal{D}(\alpha, \beta) = \left\{ \frac{1 + \frac{1-\beta}{2}\delta}{1 - \frac{1+\beta}{2}\delta} : \delta \in \mathbb{C} \text{ with } |\delta| < \alpha \right\} \quad (2.12)$$

¹more precisely, $\lim_{R \rightarrow \infty} \sup_{\substack{|s| \geq R \\ \Re(s) \geq 0}} R|L(s)| = 0$ [156].

Definition (Disk margin). Let $\beta \in \mathbb{R}$ and L be the loop transfer function of a stable SISO closed-loop system. Then the disk margin $\alpha_{\max}$ is the maximum value of α such that the closed-loop system with loop transfer function fL is stable for all complex perturbations $f \in \mathcal{D}(\alpha, \beta)$.

Fact (Disk margin calculation). Let $\beta \in \mathbb{R}$ and S be the sensitivity function of a stable SISO closed-loop system. Then the disk margin can be calculated

$$\alpha_{\max}(\beta) = \frac{1}{\max_{\omega \in \mathbb{R}} \left| S(j\omega) + \frac{\beta-1}{2} \right|} \quad (2.13)$$

See [158] for a proof.

In particular, setting $\beta = \pm 1$ relates the disk margin to the peak of the sensitivity and transmissibility functions

Fact (Maximum peak criteria). It follows immediately from Equation (2.13) that:

$$\alpha_{\max}(\beta = 1) = \frac{1}{\max_{\omega \in \mathbb{R}} |S(j\omega)|} \quad (2.14)$$

$$\alpha_{\max}(\beta = 0) = \frac{2}{\max_{\omega \in \mathbb{R}} |S(j\omega) - T(j\omega)|} \quad (2.15)$$

$$\alpha_{\max}(\beta = -1) = \frac{1}{\max_{\omega \in \mathbb{R}} |T(j\omega)|} \quad (2.16)$$

I.e., the size of the disk of complex perturbations that do not destabilize the system for each skew value is inversely proportional to the peak of the related closed-loop transfer function. Furthermore, these peak values can be used to inform lower bounds on the gain and phase margins of the system, as will be discussed next.

Fact (Lower bounds on gain and phase margins). Let $M_S = \max_{\omega} |S(j\omega)|$, $M_T = \max_{\omega} |T(j\omega)|$, and $M_{S-T} = \max_{\omega} |S(j\omega) - T(j\omega)|$. Denote the gain margin by GM and the phase margin by PM. Then the following bounds on the gain and phase margins are guaranteed:

$$\text{GM} \geq \frac{M_S}{M_S - 1} \quad \text{PM} \geq 2 \sin^{-1} \left(\frac{1}{2M_S} \right) \quad (2.17)$$

$$\text{GM} \geq \frac{M_{S-T} + 1}{M_{S-T} - 1} \quad \text{PM} \geq 2 \tan^{-1} \left(\frac{1}{M_{S-T}} \right) \quad (2.18)$$

$$\text{GM} \geq 1 + \frac{1}{M_T} \quad \text{PM} \geq 2 \sin^{-1} \left(\frac{1}{2M_T} \right) \quad (2.19)$$

Proof. For Equation (2.18), note that

$$S(s) - T(s) = \frac{1 - L(s)}{1 + L(s)} \quad (2.20)$$

At the frequency ω_{180} , where the gain margin is calculated, the phase of the loop transfer function is 180 deg and therefore $L(j\omega_{180})$ is a negative real number [159, Section 2.4.3]:

$$L(j\omega_{180}) = -\frac{1}{\text{GM}} \quad (2.21)$$

Therefore,

$$M_{S-T} \geq \frac{1 + 1/\text{GM}}{1 - 1/\text{GM}} \quad (2.22)$$

and the gain margin bound follows.

At the crossover frequency ω_c , where the phase margin is calculated, the loop transfer function has magnitude one and therefore can be written as [153, Equation 4.17]:

$$L(j\omega_c) = -\cos \text{PM} - j \sin \text{PM} \quad (2.23)$$

Then

$$M_{S-T} \geq \left| \frac{1 + \cos \text{PM} + j \sin \text{PM}}{1 - \cos \text{PM} - j \sin \text{PM}} \right| = \sqrt{\frac{2 + 2 \cos \text{PM}}{2 - 2 \cos \text{PM}}} = \frac{1}{\tan(\text{PM}/2)} \quad (2.24)$$

where the last step follows from the trigonometric identities

$$2 \sin(\text{PM}/2) = \sqrt{2 - 2 \cos \text{PM}} \quad 2 \cos(\text{PM}/2) = \sqrt{2 + 2 \cos \text{PM}} \quad (2.25)$$

Equations (2.17) and (2.19) are proved in [159, Section 2.4.3] and follow from similar ideas as the proof of Equation (2.18). $\square$

Skogestad and Postlethwaite [159, Section 2.4.3] remark that $M_S \leq 2$ guarantees a gain margin of at least 6 dB and phase margin of at least 29 deg, and is a usual requirement for control design.

Another important limitation is that the phase of these closed-loop transfer functions is determined by its magnitude and the restrictions imposed by the interpolation conditions:

Fact (Gain-phase relation). For a stable, *minimum phase* (i.e., no time delays and no zeros on the open right half plane) and LTI transfer function, denoted by $G(s)$, the following relation between magnitude and phase holds at any frequency ω_0 :

$$\arg G(j\omega_0) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{d \log |G(j\nu)|}{d\nu} \log \coth \frac{|\nu|}{2} d\nu \quad (2.26)$$

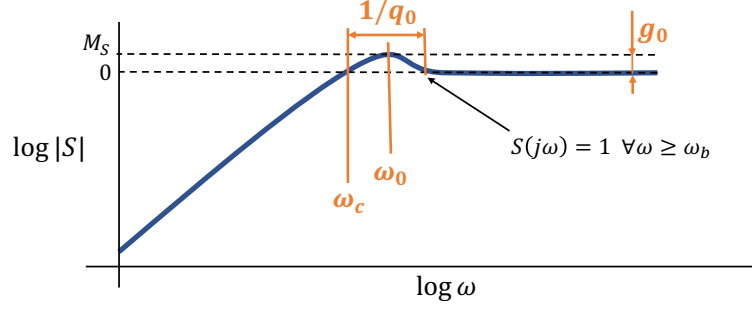


Figure 2.3 Proposed parameterization for the sensitivity function

where $v = \log(\omega/\omega_0)$.

This result is due to Bode [88]. Similar (but more complicated) results can be derived for non-minimum phase as well as for MIMO systems [160].

2.1.2 Parameterization of the sensitivity function

In Section 2.1.1, it was shown that the closed-loop transfer functions encode important information about the closed-loop behaviour of the system, there are limitations on what shapes can be achieved. It is natural, therefore, to consider designing these transfer functions instead of designing the controller. This is the approach taken, for example, in internal model control [154] and neoclassical control [153, Chapter 4]. In both cases, the Youla parameter $Q(s)$ is designed. This approach is effective because the ideal shape of these transfer functions is clearer from first principles than that of the controller transfer function itself.

In particular, the sensitivity function lends itself very well to parameterization because the shape of its magnitude is determined by requirements that are present in virtually every design: at zero frequency, $S(0) = 0 \implies T(0) = 1$ ensures zero steady-state tracking error. Its value must remain low at low frequencies for good disturbance rejection at the plant output and high performance robustness, which are two of the main objectives of feedback [153, Section 4.4.3]. It must then rise above 0 dB to satisfy the Bode integral relation and fall back to 0 dB before the system runs out of available bandwidth. Finally, its maximum value is limited by the requirement of robust stability.

A simple parameterization that satisfies those requirements is a series arrangement of a high-pass and a peaking filter:

$$S(s; \omega_c, \omega_0, g_0, q_0) = \underbrace{\frac{s^2}{s^2 + s\sqrt{2}\omega_c + \omega_c^2}}_{\text{high-pass filter}} \underbrace{\frac{s^2 + sg_0\omega_0/q_0 + \omega_0^2}{s^2 + s\omega_0/q_0 + \omega_0^2}}_{\text{peaking filter}} \quad (2.27)$$

where the parameters are: the cutoff frequency of the high-pass filter, ω_c ; the center frequency of the peaking filter, ω_0 ; the gain of the peaking filter g_0 ; and the *quality factor* of the peaking filter, q_0 , which is related to the sharpness of the peak (analogously, it is inversely related with the width of the peak). All parameters are defined to be positive real numbers. The resulting shape is shown in Figure 2.3, along with a schematic representation of the parameters.

From Equation (2.27), an upper bound on the peak of the magnitude of the sensitivity function, M_S , can be derived. Since the magnitude of the high-pass filter is less than or equal to unity, it follows that:

$$M_S = \max_{\omega \in [0, \infty)} |S(j\omega; \omega_c, \omega_0, g_0, q_0)| \leq \sup_{\omega \in [0, \infty)} \underbrace{\left| \frac{-\omega^2 + j\omega g_0 \omega_0 / q_0 + \omega_0^2}{-\omega^2 + j\omega \omega_0 / q_0 + \omega_0^2} \right|}_{\text{peaking filter}} = \max(1, g_0) \quad (2.28)$$

since the derivative of the gain of the peaking filter with respect to the frequency is zero at $\omega = \omega_0$, in which case its gain is g_0 , and at the extrema of the interval the gain of the peaking filter is one. This motivates imposing a bound on g_0 to guarantee robustness via the maximum peak criterium on the sensitivity function, as discussed in Section 2.1.1.

In addition, the presence of the interpolation constraints due to CHRP poles in the plant could be naturally included in the parameterization of the sensitivity function by adding an all-pass filter known as the Blaschke product:

$$B_p(s; z) = \frac{z - s}{z + s} \quad (2.29)$$

where z is the location of the CRHP zero to be included in the parameterized transfer function. The same idea can be applied to the transmissibility function in the presence of CRHP open-loop zeros. The presence of both open-loop unstable poles and zeros require more involved parameterizations, using, e.g., coprime factorization [161].

2.2 Metrics based on PSD moments

This section summarizes some of the classical results in the theory of probability and stochastic processes and provide the framework for the calculation of the proposed metrics, namely the probability of exceedance and the frequency of exceedance of a given level by a random process. For a more comprehensive treatment and for proofs, the reader is referred to [118, 162–166], among many other references in the topic. Specific references are given for the facts and definitions and care was taken to make the difference in definitions (e.g.,

of the Fourier transform, use of angular vs. natural frequency, etc.) compatible.

We begin by defining a stationary Gaussian random process and stating some results that allow the calculation of its probability of exceedance of a given level. The precise definition of a random process can be found in, e.g., [164, Chapter 1].

Definition (Strict stationarity). [118, Chapter 7]. A random process² $\{\xi(t) : t \geq 0\}$ in continuous time t is said to be *strictly stationary* if the tuple $(\xi(t_1), \dots, \xi(t_n))$ has the same joint probability distribution as $(\xi(t_1 + \tau), \dots, \xi(t_n + \tau))$ for all τ and every choice of $n = 1, 2, \dots$ and $t_1, \dots, t_n$.

Definition (Mean and autocovariance). The mean of a random process $\{\xi(t) : t \geq 0\}$ is

$$\mu_\xi(t) = \mathbb{E}[\xi(t)] \quad (2.30)$$

and its autocovariance function is

$$R_\xi(t, \tau) = \mathbb{E}[(\xi(t) - \mu_\xi(t))(\xi(t + \tau) - \mu_\xi(t + \tau))^T] \quad (2.31)$$

where $\mathbb{E}[\cdot]$ denote the expected value.

Fact. [118, Chapter 7]. A strictly stationary random process has a constant mean and its auto-covariance function is independent of time, i.e., $\mu_\xi(t) = \mu_\xi$ and $R_\xi(t, \tau) = R_\xi(\tau)$.

Definition. [105, Chapter 4]. The *power spectral density* (PSD), $\Phi_\xi(\omega)$, of a stationary random process $\xi(t)$ and its autocovariance function $R_\xi(\tau)$ form a Fourier transform pair, i.e.,

$$\Phi_\xi(\omega) = \frac{2}{\pi} \int_0^\infty R_\xi(\tau) \cos \omega \tau \, d\tau \quad (2.32a)$$

$$R_\xi(\tau) = \int_0^\infty \Phi_\xi(\omega) \cos \omega \tau \, d\omega \quad (2.32b)$$

Note that for a unidimensional random process, the autocovariance is an even real function [118, Chapter 7], which implies that its Fourier transform is also even and real. This allows the use of this “one-sided” version of the Fourier transform, i.e., the integration is only carried out for positive numbers, and the integration of the term which is normally multiplied by $\pm j \sin \omega \tau$ vanishes. Using this particular definition is important since it is compatible with the usual formula of the turbulence PSDs, and the use of different conventions would require rescaling them [105, Section 4.4.3].

²see [164, Section 1.3] for a precise definition of a random process

Definition (Spectral moment). [118, Chapter 7]. The n -th spectral moment of a random process $\xi(t)$ with power spectral density $\Phi_\xi(\omega)$ is defined as

$$\lambda_\xi^n = \int_0^\infty \omega^n \Phi_\xi(\omega) d\omega \quad (2.33)$$

By the properties of the Fourier transform, this is equivalent to stating that

$$\lambda_\xi^{2n} = (-1)^{2n} R_\xi^{(2n)}(0) \quad (2.34)$$

for even $2n$, where the $(2n)$ superscript indicates the $2n$ -th derivative. Note that the variance of the values taken by $\{\xi(t); t \geq 0\}$ is given by $R_\xi(0) = \lambda_\xi^0$, as a direct consequence of Equation (2.31).

Gaussian processes are of special interest in this work since they are frequently used to model continuous gusts (Section 1.2.3) and have the nice property that the output of a linear system driven by a Gaussian process is also a Gaussian process.

Definition. [118, Chapter 7]. A random process $\{\xi(t); t \geq 0\}$ is said to be *Gaussian* or *normal* if the joint distribution of $\xi(t_1), \dots, \xi(t_n)$ is Gaussian for each choice of $n = 1, 2, \dots$ and $t_1, \dots, t_n$.

Fact (Linear filtering). [165, Chapter 8] Let $Y(t)$ be the random process obtained as output of the linear system with transfer function $H(s)$, driven by the strictly stationary Gaussian process $\xi(t)$, with mean and autocovariance as in Equations (2.30) and (2.31). Then $Y(t)$ is Gaussian and strictly stationary, with mean and autocovariance given by

$$\mu_Y = H(0)\mu_\xi \quad (2.35)$$

$$\Phi_Y(\omega) = H(j\omega)\Phi_\xi(\omega)H^H(j\omega) \quad (2.36)$$

In particular, if $H(s)$ is SISO,

$$\Phi_Y(\omega) = |H(j\omega)|^2 \Phi_\xi(\omega) \quad (2.37)$$

2.2.1 Probability of exceedance

The (instantaneous) probability of exceedance corresponds to the probability that, at any instant t , the random process $\xi(t)$ will be above the threshold ℓ . For the case of stationary Gaussian processes that are also *ergodic*³, it corresponds evaluates the fraction of time in which the process is expected to exceed ℓ . It can be calculated as follows.

³a sufficient condition is to have a continuous PSD [167]

Fact. The probability that the stationary Gaussian process $\{\xi(t); t \geq 0\}$, with mean μ_ξ and zeroth spectral moment λ_ξ^0 exceeds the level ℓ is

$$\mathbb{P}\{\xi > \ell\} = \frac{1}{2} - \operatorname{erf}\left(\frac{\ell - \mu_\xi}{\lambda_\xi^0 \sqrt{2}}\right)$$

where erf denotes the error function, i.e., $\operatorname{erf} z = \frac{2}{\sqrt{\pi}} \int_0^z e^{-\tau^2} d\tau$.

Proof. Consider the zero mean and unit covariance random variable $X = (\xi - \mu_\xi)/\lambda_\xi^0$. Its cumulative distribution is given by $\mathbb{P}\{X < x\} = \frac{1}{2} + \operatorname{erf}(x/\sqrt{2})$ [168]. Then this fact can be computed by noticing that $\mathbb{P}\{\xi > \ell\} = 1 - \mathbb{P}\{X < (\ell - \mu_\xi)/\lambda_\xi^0\}$. $\square$

If $\xi(t)$ must be constrained to the interval $[\ell^-, \ell^+]$, $\ell^- \leq \ell^+$, then the probability of it exiting that interval can be obtained by simple sum, i.e., $\mathbb{P}(\{\xi < \ell^-\} \cup \{\xi > \ell^+\}) = \mathbb{P}\{\xi < \ell^-\} + \mathbb{P}\{\xi > \ell^+\}$, since the involved intervals are disjoint.

2.2.2 Expected frequency of exceedance

The expected frequency of exceedance of a level ℓ by a random process $\xi(t)$ is defined as the expected rate of upcrossings of ℓ by $\xi(t)$ in unit time. These terms are defined as follows.

Definition (Upcrossing). [118, Chapter 7]. Let $\mathcal{G}_\ell \subset C^0(\mathbb{R}_+ \rightarrow \mathbb{R})$ denote the class of continuous functions that are not identically equal to $\ell \in \mathbb{R}$ in any subinterval of the real line, i.e.,

$$\mathcal{G}_\ell = \{f \in C^0(\mathbb{R}_+ \rightarrow \mathbb{R}) : f(t) \neq \ell \text{ for some } t \in (a, b) \text{ and for all } (a, b) \subseteq \mathbb{R}_+\}.$$

A function $f \in \mathcal{G}_\ell$ is said to have an *upcrossing* of the level ℓ at the point t_0 if for some $\varepsilon > 0$, $f(t) \leq \ell$ for all $t \in [t_0 - \varepsilon, t_0]$ and $f(t) \geq \ell$ for all $t \in [t_0, t_0 + \varepsilon]$

In particular, processes with finite second spectral moments (see Section 2.3 for the definition) are differentiable in a mean-square sense, and, in this case, the definition of upcrossings becomes equivalent to the requirement that $f(t_0) = \ell$ and $\dot{f}(t_0) > 0$ for f to have an upcrossing of ℓ at time t_0 [118].

Fact. [118, Chapter 7]. Consider a strictly stationary random process $\{\xi(t) : t \geq 0\}$ with a continuous cumulative distribution function. Then all sample paths of ξ are members of $\mathcal{G}_\ell$ with probability one.

Definition (Expected frequency of exceedance). [118, Chapter 7]. Let $N_{\ell, (t_0, t_1)}^+(\xi)$ denote the number of upcrossings of the level ℓ by the strictly stationary random process with continuous cumulative distribution function $\{\xi(t) : t \geq 0\}$ in the interval (t_0, t_1) .

The *expected frequency of exceedance* of a level ℓ by ξ is defined as the expected number of upcrossings of ℓ in unit time and is denoted as $\mathbb{E}[N_{\ell}^+(\xi)]$ (the interval subscript is dropped), i.e.,

$$\mathbb{E}[N_{\ell}^+(\xi)] = \mathbb{E}[N_{\ell, (\tau, \tau+1)}^+(\xi)].$$

Note that since $\xi(t)$ is stationary, this value is independent of the choice of τ .

Fact. (Kac-Rice's formula) [116–118] Let $\{\xi(t), t \geq 0\}$ be a strictly stationary normal random process with mean μ_{ξ} and finite second moment. Then the expected frequency of upcrossings of the mean can be calculated as

$$\mathbb{E}[N_{\mu}^+(\xi)] = \frac{1}{2\pi} \sqrt{\frac{\lambda_{\xi}^2}{\lambda_{\xi}^0}}$$

and the expected frequency of exceedance of an arbitrary constant level ℓ is given by

$$\mathbb{E}[N_{\ell}^+(\xi)] = \mathbb{E}[N_{\mu}^+(\xi)] \exp \left[-\frac{1}{2} \frac{(\ell - \mu_{\xi})^2}{\lambda_{\xi}^0} \right]$$

Due to the linearity of the expectation operator, the expected frequency of exiting an interval $[\ell^-, \ell^+]$, $\ell^- \leq \ell^+$ can be obtained by summing the frequency of upcrossing of the higher limit and the frequency of downcrossings of the lower limit, i.e.,

$$\mathbb{E}[N_{\ell^+}^+(\xi) + N_{\ell^-}^-(\xi)] = \mathbb{E}[N_{\ell^+}^+(\xi)] + \mathbb{E}[N_{\ell^-}^-(\xi)] = \mathbb{E}[N_{\ell^+}^+(\xi)] + \mathbb{E}[N_{-\ell^-}^+(-\xi)] \quad (2.38)$$

In practice, the *probability of exceedance* and *frequency of exceedance* metrics work better when combined, because in the case where there is a high probability of exceedance, the frequency of exceedance might be low simply because the process rarely goes below the prescribed level. It can also be shown that as the level ℓ takes higher values, the distribution of upcrossings approaches a Poisson distribution with a frequency equal to the expected frequency of exceedance [118, Chapter 9], which justifies calling this quantity as a frequency.

2.2.3 Peak stress metric

To constrain the peak stress to which the wingbox is subjected under continuous turbulence, the continuous turbulence design criterium from the certification requirements [98, part (b)] is employed, considering the transfer function from gust to stress as suggested in the Advisory Circular: Dynamic Gust Loads [112, §6.3.2.6.4]. See also [105, Chapter 5] for a more detailed description of the method. This calculation yields the peak gust stress, σ_{gust} , as a function of the 1-g stress, $\sigma_{1\text{-g}}$, and the turbulence intensity, σ_{gust} , which is a design parameter specified by the FAR. All stresses are considered to be von Mises aggregated stresses. The relation⁴ is given by:

$$\sigma_{\text{gust}} = \sqrt{\lambda_0} U_{\text{gust}} \quad (2.39)$$

where λ_0 is calculated by Equation (2.65) considering the normalized von Kármán power spectral density given by Equation (2.63) with $U_g = 0$. The total stress is calculated by summing the 1-g straight and level flight stress to the peak gust stress, in order to calculate the margin of safety (MS), given by

$$\text{MS} = \frac{\sigma_{\text{allowable}}}{|\sigma_{\text{gust}} + \sigma_{1\text{-g}}|} - 1.5 \quad (2.40)$$

where $\sigma_{\text{allowable}}$ is the allowable stress for the application. The margin of safety of the peak gust stress is constrained to be non-negative, i.e.,

$$\text{MS} \geq 0 \quad (2.41)$$

2.2.4 Fatigue metric

When formulating an MDO problem that involves gust loads, it is important to incorporate a fatigue constraint to correctly evaluate designs with large stress power at high frequencies. Even if the peak stress to which these designs are exposed is safe, they may be subjected to a high number of stress cycles that will lead to fatigue damage.

The most widely used method to calculate fatigue damage is based on the Palmgreen-Miner rule [169, 170] combined with the rainflow count of cycles in the stress time history to calculate the damage experienced by the material over time (see e.g., [171, 172]). Since the sensitivity parameterization approach is done in the frequency domain, doing the fatigue analysis also in frequency domain is simpler and more computationally efficient than to generate a representative time signal of the stress (e.g., by designing a filter driven by

⁴The FAR notation for $\sqrt{\lambda_0}$ is $\bar{A}$

white noise). Furthermore, the rainflow counting process is not differentiable, which complicates its inclusion in gradient-based optimization. Instead, the preferred approach is to use Dirlik's empirical method for approximating the rate of damage due to gust [77], and then constraining this damage rate by a limit based on the expected flight hours in turbulence for the lifetime of the airframe. Dirlik's method has been shown to provide good agreement with the time-domain rainflow counting analysis as shown in [173, 174], which provide comparisons with other similar methods, and in [175], for application in floating wind turbine design.

Dirlik's method uses the moments of order 0, 1, 2, and 4 of the stress PSD to estimate the mean damage rate of the rainflow count method denoted by $\dot{D}_{RFC}^{DK}$. The von Mises aggregated stresses are considered as proposed by [176]. The equations that define Dirlik's method are reproduced below for a single point in the structure (i.e., $\lambda_0, \lambda_1, \lambda_2, \lambda_4 \in \mathbb{R}$)

$$\dot{D}_{RFC}^{DK} = \frac{\nu_p}{C} \sqrt{\lambda_0} \left[D_1 Q^k \Gamma(1+k) + 2^{k/2} \Gamma(1+k/2) (D_2 |R|^k + D_3) \right] \quad (2.42)$$

where:

$$\begin{aligned} \nu_0 &= \frac{1}{2\pi} \sqrt{\frac{\lambda_2}{\lambda_0}} & \nu_p &= \frac{1}{2\pi} \sqrt{\frac{\lambda_4}{\lambda_2}} & \alpha_2 &= \frac{\lambda_2}{\sqrt{\lambda_0 \lambda_4}} \\ x_m &= \frac{\lambda_1}{\lambda_0} \sqrt{\frac{\lambda_2}{\lambda_4}} & D_1 &= \frac{2(x_m - \alpha_2^2)}{1 + \alpha_2^2} & D_2 &= \frac{1 - \alpha_2 - D_1 + D_1^2}{1 - R} \\ D_3 &= 1 - D_1 - D_2 & Q &= \frac{1.25(\alpha_2 - D_3 - D_2 R)}{D_1} & R &= \frac{\alpha_2 - x_m - D_1^2}{1 - \alpha_2 - x_m - D_1 + D_1^2} \end{aligned}$$

The parameters ν_0 and ν_p correspond to the frequency of zero crossing and frequency of peaks of the random signal, respectively, $\Gamma(\cdot)$ is the Gamma function, and C and k are the parameters of the material's S-N curve expressed as $N = C\sigma^{-k}$, which can be obtained from, e.g., MIL-HDBK-5J with unity stress ratio.

The expected life in fatigue is then given by

$$E[\text{fatigue life}] = \frac{1}{\dot{D}_{RFC}^{DK}} \quad (2.43)$$

A safety factor of 3 is usually applied to this value, and an additional factor of 1.5 is applied to allow for the variability of loading between different aircraft of the same type [177, Chapter 15]. The appropriate gust intensity for using with the von Kármán model when doing the fatigue life calculation described can be obtained from, e.g., MIL-HDBK-1797.

It was found that the optimization converges better when the logarithm of the expected

fatigue life is considered, i.e. a constraint is imposed on

$$\log_{10} \mathbb{E}[\text{fatigue life}] = \log_{10}[1/\dot{D}_{RFC}^{DK}] \quad (2.44)$$

This model can be efficiently differentiated by using the algorithmic differentiation or the complex step method.

The only other work found in the literature that uses a fatigue metric in gradient-based MDO is by Hegseth et al. [76]. It uses a similar approach to that described herein, applied to floating wind turbine MDO. In that work, Dirlik's method is used to calculate the fatigue damage at selected locations on the tower and platform of the wind turbine using as inputs the PSDs of the wind and waves, which are assumed to be uncorrelated. In other work, fatigue metrics have been included in MDO problems, but either in the frequency domain without calculation of analytical gradients (as will be derived for the PSD moments in Section 2.3) or in the time domain, where, as pointed out, the fatigue calculation may not even be differentiable. For example, Durá et al. [178] apply Dirlik's method to the calculation of damage equivalent loads for use in the MDO of wind turbine blades but does not calculate their derivatives with respect to the model; Miao et al. [179] considered a time-domain fatigue calculation for an automotive MDO problem, which was solved using a gradient-free algorithm. In structural topology optimization, similar fatigue metrics have been included with the use of adjoint derivatives, see, e.g., [180].

2.3 Computation of PSD moments and their derivatives

This section develops formulae to compute the PSD moments (Equation (2.33)) of signals from a linear system driven by a random process, along with their derivatives with respect to the state space matrices of the plant and the parameters of the sensitivity function.

Notation is mostly standard, and the vectorization operation and the Kronecker product are used to avoid dealing with tensors of order larger than two. In what follows, (i) I_n is the identity matrix in $\mathbb{R}^{n \times n}$; (ii) $\mathbf{1}$ denotes a vector in which every element is one; (iii) $\mathbf{0}$ denotes a vector of zeros; (iv) j is the complex unit ($j^2 = -1$) (v) $\bar{Z}$ and $|Z|$ are the element-wise complex conjugate and the element-wise absolute value of $Z \in \mathbb{C}^{m \times n}$, respectively; (vi) $\cdot^\top$ is the transpose operator; (vii) $\cdot^H$ is the conjugate transpose (Hermitian) operator; (viii) $\cdot^{-\top}$ and $\cdot^{-H}$ denote the inverse transpose and inverse conjugate transpose, respectively; (ix) $\text{vec} \cdot$ is the vectorization operation, which stacks the columns of a matrix to produce a column vector; (x) when applied to a vector, $\text{diag} \cdot$ yields a diagonal matrix with the vector elements in the diagonal, and when applied to a matrix it extracts the diagonal elements into a vector;

(xi) $\cdot \circ \cdot$ denotes the element-wise (or Hadamard) product; and (xii) $\cdot \otimes \cdot$ denotes the Kronecker product; (xiii) $\cdot^\circ$ denotes element-wise power (or Hadamard power).

When deriving expressions for directional derivatives, we make use of the simplified notation described in [181, Section 5.16], that is, if $g(x) : U \rightarrow V$ is a vector function; U, V are vector spaces; and $\mathcal{L}(U, V)$ is the space of linear operators that map U to V , then we write $dg = \frac{\partial g}{\partial x}(x) dx$, where $dx \in U$ is a direction in the input space and $dg \in V$ is a direction in the output space. $\frac{\partial g}{\partial x} \in \mathcal{L}(U, V)$ is the derivative of g defined as the linear mapping that locally approximates g , i.e., $\lim_{dx \rightarrow 0} \frac{g(x+dx) - g(x) - \frac{\partial g}{\partial x}(x) dx}{\|dx\|} = \mathbf{0}$. We also define the adjoint derivative $(\frac{\partial g}{\partial x}(x))^* \in \mathcal{L}(V, U)$ such that $\langle dg, \frac{\partial g}{\partial x}(x) dx \rangle = \langle (\frac{\partial g}{\partial x}(x))^* dg, dx \rangle$. For example, if $U = \mathbb{R}^n$ and $V = \mathbb{R}^m$, then $\frac{\partial g}{\partial x}(x) \in \mathbb{R}^{m \times n}$ and $(\frac{\partial g}{\partial x}(x))^* = (\frac{\partial g}{\partial x}(x))^\top$.

Some useful relations are recalled for the convenience of the reader: (xiv) the absolute value squared function, $|\cdot|^2 : \mathbb{C} \rightarrow \mathbb{R}_+$ is not differentiable in the complex plane because it does not satisfy the Cauchy-Riemann equations, but is differentiable when viewed as a $\mathbb{R}^2 \rightarrow \mathbb{R}_+$ mapping, i.e., $|a + jb|^2 = a^2 + b^2$, where $a, b \in \mathbb{R}$, and its differential is given by $d|a + jb|^2 = 2a da + 2b db = 2\Re[(\overline{a + jb}) d(a + jb)]$; (xv) the differential of the matrix inverse is given by $dA^{-1} = -A^{-1} dA A^{-1}$; (xvi) the vectorization operation is distributive over the element-wise product, i.e., $\text{vec}(X \circ Y) = \text{vec}(X) \circ \text{vec}(Y)$; (xvii) if u and v are vectors of the same dimension, then $u \circ v = \text{diag}(u)v$; and (xviii) if X, Y, Z are matrices such that the product XYZ is defined, then $\text{vec}(XYZ) = (Z^\top \otimes X) \text{vec}(Y)$ —this relation is known as the “vec trick”.

2.3.1 Moments of input and output signals PSDs

Calculation

The plant shown in Figure 2.1 is assumed to be linearized around an operating point and to have two inputs, namely one disturbance input (d) and one control input (u), and two outputs, from which one (y) is used for feedback, and the other (z) is purely a performance metric that is not used for feedback. It is assumed that the disturbance input is Gaussian and stationary.

The plant is assumed to have a state space representation given by

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}_u u + \mathbf{B}_d d \\ \begin{bmatrix} y \\ z \end{bmatrix} &= \begin{bmatrix} \mathbf{C}_y \\ \mathbf{C}_z \end{bmatrix} \mathbf{x} + \begin{bmatrix} D_{yu} & D_{yd} \\ D_{zu} & D_{zd} \end{bmatrix} \begin{bmatrix} u \\ d \end{bmatrix} \end{aligned} \quad (2.45)$$

from which the transfer functions shown in Figure 2.1 can be extracted:

$$G_{yd}(s) = C_y(sI - A)^{-1}B_d + D_{yd} \quad (2.46a)$$

$$G_{zd}(s) = C_z(sI - A)^{-1}B_d + D_{zd} \quad (2.46b)$$

$$G_{yu}(s) = C_y(sI - A)^{-1}B_u + D_{yu} \quad (2.46c)$$

$$G_{zu}(s) = C_z(sI - A)^{-1}B_u + D_{zu} \quad (2.46d)$$

Following the proposed controller agnostic approach, the controller $K(s)$ is not designed but, instead, the sensitivity function of the output used for feedback, repeated here for convenience,

$$S(s) = (1 + G_{yu}(s)K(s))^{-1} \quad (2.2)$$

is considered as a design variable, and a suitable parameterization has to be chosen, such as the one proposed in Section 2.1.2. The closed-loop transfer functions can then be expressed in terms of the open-loop transfer functions (2.46) and the sensitivity function:

$$H_{yd}(s) = S(s)G_{yd}(s) \quad (2.47a)$$

$$H_{ud}(s) = G_{yu}^{-1}(s)[S(s) - 1]G_{yd}(s) \quad (2.47b)$$

$$H_{zd}(s) = G_{zd} + G_{zu}(s)G_{yu}^{-1}(s)[S(s) - 1]G_{yd}(s) \quad (2.47c)$$

Due to the involved transfer functions being linear-time-invariant, the outputs of the system will also be Gaussian and stationary, and their PSDs can be readily computed, including the PSD of the rate of the control action, $\dot{u}$:

$$\Phi_y(\omega) = |H_{yd}(j\omega)|^2 \Phi_d(\omega) \quad (2.48a)$$

$$\Phi_z(\omega) = |H_{zd}(j\omega)|^2 \Phi_d(\omega) \quad (2.48b)$$

$$\Phi_u(\omega) = |H_{ud}(j\omega)|^2 \Phi_d(\omega) \quad (2.48c)$$

$$\Phi_{\dot{u}}(\omega) = \omega^2 |H_{ud}(j\omega)|^2 \Phi_d(\omega) \quad (2.48d)$$

where $\Phi_d(\omega)$ denotes the PSD of the disturbance, e.g., the von Kármán turbulence spectrum.

The risk metrics defined in Section 2.2 are functions of the moments of their PSDs, defined as follows.

Definition. The m -th spectral moment of a stationary stochastic process $\{\xi(t); t \geq 0\}$ with PSD $\Phi_\xi(\omega)$ is defined as

$$\lambda_\xi^m = \int_0^\infty \omega^m \Phi_\xi(\omega) d\omega \quad m = 0, 1, \dots$$

For a general dynamical system, these moments can be calculated by three approaches: (i) direct integration of the PSD in the frequency domain; (ii) solution of a Lyapunov equation on the state space matrices (e.g., [182]); (iii) time domain simulation and integration of the autocovariance function, since it and the PSD are Fourier transform pairs Equation (2.32), therefore Parseval's theorem applies.

For the sensitivity function approach, the direct integration of the PSD is more suitable because, in addition to it being a frequency domain approach, it requires the inversion of some transfer functions, namely the transfer function from the control action to the output used for feedback. This inverse transfer function is in general non-causal (relative degree less than zero), and therefore it does not admit a space-state representation.

The convergence of the integral depends on both the relative degree of the closed loop transfer function and the decay of the disturbance PSD at high frequencies. In particular, since the von Kármán spectrum has a power law decay at high frequencies with exponent $-\frac{5}{3}$, and the zeroth and second moments of the PSDs are to be computed, it is required that all closed-loop transfer functions have a relative degree of at least two. This is discussed by Hoblit [105, Appendix E].

Derivatives with respect to the model

To enable gradient-based optimization, it is important to have an efficient way to compute the derivatives of the functions of interest. In particular, the probability of exceedance and the expected frequency of exceedance of y , z , u , and $\dot{u}$ are simple functions of the zeroth and second moments of the PSDs of y , z , u and of the second and fourth moments of u , respectively. The derivatives of these moments with respect to the state-space matrices are calculated as shown next, and they can be integrated into an MDO problem using the chain rule.

The sparsity pattern of these derivatives can be observed in the following block representation, derived from the functional dependency of the closed-loop transfer functions in relation to the open-loop ones, i.e., Equation (2.47):

$$\mathbf{D} \begin{bmatrix} \lambda_y^m \\ \lambda_u^m \\ \lambda_z^m \end{bmatrix} = \begin{bmatrix} \frac{\partial \lambda_y^m}{\partial \text{vec } A} & \frac{\partial \lambda_y^m}{\partial \text{vec } B_d} & \mathbf{0} & \frac{\partial \lambda_y^m}{\partial \text{vec } C_y} & \mathbf{0} & \frac{\partial \lambda_y^m}{\partial D_{yd}} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \frac{\partial \lambda_y^m}{\partial \mathbf{p}} \\ \frac{\partial \lambda_u^m}{\partial \text{vec } A} & \frac{\partial \lambda_u^m}{\partial \text{vec } B_d} & \frac{\partial \lambda_u^m}{\partial \text{vec } B_u} & \frac{\partial \lambda_u^m}{\partial \text{vec } C_y} & \mathbf{0} & \frac{\partial \lambda_u^m}{\partial D_{yd}} & \frac{\partial \lambda_u^m}{\partial D_{yu}} & \mathbf{0} & \mathbf{0} & \frac{\partial \lambda_u^m}{\partial \mathbf{p}} \\ \frac{\partial \lambda_z^m}{\partial \text{vec } A} & \frac{\partial \lambda_z^m}{\partial \text{vec } B_d} & \frac{\partial \lambda_z^m}{\partial \text{vec } B_u} & \frac{\partial \lambda_z^m}{\partial \text{vec } C_y} & \frac{\partial \lambda_z^m}{\partial \text{vec } C_z} & \frac{\partial \lambda_z^m}{\partial D_{yd}} & \frac{\partial \lambda_z^m}{\partial D_{yu}} & \frac{\partial \lambda_z^m}{\partial D_{zd}} & \frac{\partial \lambda_z^m}{\partial D_{zu}} & \frac{\partial \lambda_z^m}{\partial \mathbf{p}} \end{bmatrix} \quad (2.49)$$

The size of this matrix is $3 \times (n_x^2 + 4n_x + 4 + n_p)$.

Expressions for each block can be obtained by evaluating the differential of the mo-

ments. Starting from the output used for feedback, y :

$$d\lambda_y^m = \int_0^\infty \omega^m 2\Re\{\bar{H}_{yd}(j\omega)[dS(j\omega)G_{yd}(j\omega) + S(j\omega)dG_{yd}(j\omega)]\}\Phi_d(\omega)d\omega \quad (2.50)$$

Introducing the shorthand notation $\Psi_{(\cdot)}(s) = C_{(\cdot)}(sI - A)^{-1}$ and $\chi_{(\cdot)} = (sI - A)^{-1}B_{(\cdot)}$ and expressing the differential of the open-loop transfer functions in terms of the differential of the matrices of the state space representation:

$$d\lambda_y^m = \int_0^\infty \omega^m 2\Re\{\bar{H}_{yd}(j\omega)[\frac{\partial S(j\omega)}{\partial \mathbf{p}} d\mathbf{p} G_{yd}(j\omega) + S(j\omega)(dC_y \chi_d(j\omega) + \Psi_y(j\omega) dA \chi_d(j\omega) + \Psi_y(j\omega) dB_d + dD_{yd})]\}\Phi_d(\omega)d\omega \quad (2.51)$$

From inspection of Equation (2.51) and applying the vec-trick, the following partial derivatives may be extracted:

$$\frac{\partial \lambda_y^m}{\partial \text{vec } \mathbf{A}} = \int_0^\infty \omega^m 2\Re\{\bar{H}_{yd}(j\omega)S(j\omega)\chi_d(j\omega)^\top \otimes \Psi_y(j\omega)\}\Phi_d(\omega)d\omega \quad (2.52a)$$

$$\frac{\partial \lambda_y^m}{\partial \text{vec } \mathbf{B}_d} = \int_0^\infty \omega^m 2\Re\{\bar{H}_{yd}(j\omega)S(j\omega)\Psi_y(j\omega)\}\Phi_d(\omega)d\omega \quad (2.52b)$$

$$\frac{\partial \lambda_y^m}{\partial \text{vec } \mathbf{C}_y} = \int_0^\infty \omega^m 2\Re\{\bar{H}_{yd}(j\omega)S(j\omega)\chi_d(j\omega)^\top\}\Phi_d(\omega)d\omega \quad (2.52c)$$

$$\frac{\partial \lambda_y^m}{\partial D_{yd}} = \int_0^\infty \omega^m 2\Re\{\bar{H}_{yd}(j\omega)S(j\omega)\}\Phi_d(\omega)d\omega \quad (2.52d)$$

$$\frac{\partial \lambda_y^m}{\partial \mathbf{p}} = \int_0^\infty \omega^m 2\Re\{\bar{H}_{yd}(j\omega)\frac{\partial S(j\omega)}{\partial \mathbf{p}}G_{yd}(j\omega)\}\Phi_d(\omega)d\omega \quad (2.52e)$$

This procedure is repeated for the moments of the control input u :

$$d\lambda_u^m = \int_0^\infty \omega^m 2\Re\{\bar{H}_{ud}(j\omega)G_{yu}(j\omega)^{-1}[-dG_{yu}(j\omega)H_{ud}(j\omega) + dS(j\omega)G_{yd}(j\omega) + (S(j\omega) - 1)dG_{yd}(j\omega)]\}\Phi_d(\omega)d\omega \quad (2.53)$$

$$\begin{aligned}
d\lambda_u^m = & \int_0^\infty \omega^m 2\Re\{\bar{H}_{yd}(j\omega)G_{yu}(j\omega)^{-1}[\\
& - (dC_y \mathbf{X}_u(j\omega) + \boldsymbol{\Psi}_y d\mathbf{A} \mathbf{X}_u(j\omega) + \boldsymbol{\Psi}_y(j\omega) d\mathbf{B}_u + dD_{yu})H_{ud}(j\omega) \\
& + \frac{\partial S(j\omega)}{\partial \mathbf{p}} d\mathbf{p} G_{yd}(j\omega) \\
& + (S(j\omega) - 1)(dC_y \mathbf{X}_d(j\omega) + \boldsymbol{\Psi}_y(j\omega) d\mathbf{A} \mathbf{X}_d(j\omega) + \boldsymbol{\Psi}_y(j\omega) d\mathbf{B}_d + dD_{yd})]\} \Phi_d(\omega) d\omega \quad (2.54)
\end{aligned}$$

From inspection of Equation (2.54) and applying the vec-trick, the following partial derivatives may be extracted:

$$\begin{aligned}
\frac{\partial \lambda_u^m}{\partial \text{vec } \mathbf{A}} = & \int_0^\infty \omega^m 2\Re\{\bar{H}_{ud}(j\omega)G_{yu}(j\omega)^{-1}[-\mathbf{X}_u(j\omega)^\top \otimes \boldsymbol{\Psi}_y(j\omega)H_{ud}(j\omega) \\
& + (S(j\omega) - 1)\mathbf{X}_d(j\omega)^\top \otimes \boldsymbol{\Psi}_y(j\omega)]\} \Phi_d(\omega) d\omega \quad (2.55a)
\end{aligned}$$

$$\frac{\partial \lambda_u^m}{\partial \text{vec } \mathbf{B}_d} = \int_0^\infty \omega^m 2\Re\{\bar{H}_{ud}(j\omega)G_{yu}(j\omega)^{-1}(S(j\omega) - 1)\boldsymbol{\Psi}_y(j\omega)\} \Phi_d(\omega) d\omega \quad (2.55b)$$

$$\frac{\partial \lambda_u^m}{\partial \text{vec } \mathbf{B}_u} = - \int_0^\infty \omega^m 2\Re\{\bar{H}_{ud}(j\omega)G_{yu}(j\omega)^{-1}\boldsymbol{\Psi}_y(j\omega)H_{ud}(j\omega)\} \Phi_d(\omega) d\omega \quad (2.55c)$$

$$\begin{aligned}
\frac{\partial \lambda_u^m}{\partial \text{vec } \mathbf{C}_y} = & \int_0^\infty \omega^m 2\Re\{\bar{H}_{ud}(j\omega)G_{yu}(j\omega)^{-1}[-\mathbf{X}_u(j\omega)^\top H_{ud}(j\omega) \\
& + (S(j\omega) - 1)\mathbf{X}_d(j\omega)^\top]\} \Phi_d(\omega) d\omega \quad (2.55d)
\end{aligned}$$

$$\frac{\partial \lambda_u^m}{\partial D_{yd}} = \int_0^\infty \omega^m 2\Re\{\bar{H}_{ud}(j\omega)G_{yu}(j\omega)^{-1}(S(j\omega) - 1)\} \Phi_d(\omega) d\omega \quad (2.55e)$$

$$\frac{\partial \lambda_u^m}{\partial D_{yu}} = - \int_0^\infty \omega^m 2\Re\{\bar{H}_{ud}(j\omega)G_{yu}(j\omega)^{-1}H_{ud}(j\omega)\} \Phi_d(\omega) d\omega \quad (2.55f)$$

$$\frac{\partial \lambda_u^m}{\partial \mathbf{p}} = \int_0^\infty \omega^m 2\Re\{\bar{H}_{ud}(j\omega)G_{yu}(j\omega)^{-1}\frac{\partial S(j\omega)}{\partial \mathbf{p}}G_{yd}(j\omega)\} \Phi_d(\omega) d\omega \quad (2.55g)$$

And finally, the procedure is repeated for the moments of the output not used for feed-

back z :

$$\begin{aligned}
d\lambda_z^m = & \int_0^\infty \omega^m 2\Re \left\{ \bar{H}_{zd}(j\omega) \left[dC_z \mathbf{X}_d(j\omega) + \boldsymbol{\psi}_z(j\omega) d\mathbf{A} \mathbf{X}_d(j\omega) + \boldsymbol{\psi}_z(j\omega) d\mathbf{B}_d + dD_{zd} \right. \right. \\
& (dC_z \mathbf{X}_u(j\omega) + \boldsymbol{\psi}_z(j\omega) d\mathbf{A} \mathbf{X}_u(j\omega) + \boldsymbol{\psi}_z(j\omega) d\mathbf{B}_u + dD_{zu}) H_{ud}(j\omega) \\
& + G_{yu}(j\omega)^{-1} \left(-(dC_y \mathbf{X}_u(j\omega) + \boldsymbol{\psi}_y(j\omega) d\mathbf{A} \mathbf{X}_u(j\omega) + \boldsymbol{\psi}_y(j\omega) d\mathbf{B}_u + dD_{yu}) H_{ud}(j\omega) \right. \\
& \left. \left. + \frac{\partial S(j\omega)}{\partial \mathbf{p}} d\mathbf{p} G_{yd}(j\omega) \right) \right] \left. \right\} \Phi_d(\omega) d\omega \quad (2.56) \\
& + (S(j\omega) - 1)(dC_y \mathbf{X}_d(j\omega) + \boldsymbol{\psi}_y(j\omega) d\mathbf{A} \mathbf{X}_d(j\omega) + \boldsymbol{\psi}_y(j\omega) d\mathbf{B}_d + dD_{yd})) \left. \right\} \Phi_d(\omega) d\omega
\end{aligned}$$

From inspection of Equation (2.56) and applying the vec-trick, the following partial derivatives may be extracted:

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial \text{vec } \mathbf{A}} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) [\mathbf{X}_d(j\omega)^\top \otimes \boldsymbol{\psi}_z(j\omega) + \mathbf{X}_u(j\omega)^\top \otimes \boldsymbol{\psi}_z(j\omega) H_{ud}(j\omega) \\
& + G_{zu}(j\omega) (-G_{yu}(j\omega)^{-1} \mathbf{X}_u(j\omega)^\top \otimes \boldsymbol{\psi}_y(j\omega) H_{ud}(j\omega) \\
& + G_{yu}(j\omega)^{-1} (S(j\omega) - 1) \mathbf{X}_d(j\omega)^\top \otimes \boldsymbol{\psi}_y(j\omega))] \} \Phi_d(\omega) d\omega \quad (2.57a)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial \text{vec } \mathbf{B}_d} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) [\boldsymbol{\psi}_z(j\omega) + G_{zu}(j\omega) G_{yu}(j\omega)^{-1} (S(j\omega) - 1) \boldsymbol{\psi}_y(j\omega)] \} \Phi_d(\omega) d\omega \\
& (2.57b)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial \text{vec } \mathbf{B}_u} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) [\boldsymbol{\psi}_z(j\omega) - G_{zu}(j\omega) G_{yu}(j\omega)^{-1} \boldsymbol{\psi}_y(j\omega)] H_{ud}(j\omega)] \} \Phi_d(\omega) d\omega \\
& (2.57c)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial \text{vec } \mathbf{C}_y} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) G_{zu}(j\omega) G_{yu}(j\omega)^{-1} [-\mathbf{X}_u(j\omega)^\top H_{ud}(j\omega) \\
& + (S(j\omega) - 1) \mathbf{X}_d(j\omega)^\top] \} \Phi_d(\omega) d\omega \quad (2.57d)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial \text{vec } \mathbf{C}_z} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) [\mathbf{X}_d(j\omega)^\top + \mathbf{X}_u(j\omega)^\top H_{ud}(j\omega)] \} \Phi_d(\omega) d\omega \quad (2.57e)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial D_{yd}} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) G_{zu}(j\omega) G_{yu}(j\omega)^{-1} (S(j\omega) - 1) \} \Phi_d(\omega) d\omega \quad (2.57f)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial D_{yu}} = & - \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) G_{zu}(j\omega) G_{yu}(j\omega)^{-1} H_{ud}(j\omega) \} \Phi_d(\omega) d\omega \quad (2.57g)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial D_{zd}} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) \} \Phi_d(\omega) d\omega \quad (2.57h)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial D_{zu}} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{zd}(j\omega) H_{ud}(j\omega) \} \Phi_d(\omega) d\omega \quad (2.57i)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \lambda_z^m}{\partial \mathbf{p}} = & \int_0^\infty \omega^m 2\Re \{ \bar{H}_{ud}(j\omega) G_{zu}(j\omega) G_{yu}(j\omega)^{-1} \frac{\partial S(j\omega)}{\partial \mathbf{p}} G_{yd}(j\omega) \} \Phi_d(\omega) d\omega \quad (2.57j)
\end{aligned}$$

2.3.2 Moments of the von Mises stress PSD

It is usual to adopt a failure criterium based on von Mises stresses. However, the random process of von Mises stresses requires special attention because it is not a linear combination of the components of the stress tensor. Aircraft structures are usually much longer than they are wide, tall, or thick, and because of that the stress state is well approximated by a normal, σ , and a shear, τ , stress component. The von Mises stress, σ_{VM} can be then defined:

$$\sigma_{VM} = \sqrt{\sigma^2 + 3\tau^2} \quad (2.58)$$

Calculation

The von Mises stress PSD can be calculated by employing the procedure proposed by Preumont and Piéfort [176]:

$$\Phi_{VM} = \Phi_{\sigma} + 3\Phi_{\tau} \quad (2.59)$$

where the terms correspond to the PSD of the von Mises, normal, and shear stresses respectively.

Consider the transfer functions from one gust input to normal and shear stresses at many points in the structure, given by

$$G_{\sigma}(s) = C_{\sigma}(sI - A)^{-1}B + D_{\sigma} \quad (2.60)$$

$$G_{\tau}(s) = C_{\tau}(sI - A)^{-1}B + D_{\tau} \quad (2.61)$$

where $A \in \mathbb{R}^{n_x \times n_x}$; $B \in \mathbb{R}^{n_x \times 1}$; $C_{\sigma}, C_{\tau} \in \mathbb{R}^{n_y \times n_x}$; $D_{\sigma}, D_{\tau} \in \mathbb{R}^{n_y \times 1}$. The tuples $(A, B, C_{\sigma}, D_{\sigma})$ and $(A, B, C_{\tau}, D_{\tau})$, are the state space representations of the responses from gust to normal stresses and from gust to shear stresses, respectively.

Then, the PSD of the von Mises stress can be calculated by combining Equations (2.37) and (2.59)⁵:

$$\Phi_{VM}(\omega) = (|G_{\sigma}(s)|^2 + 3|G_{\tau}(s)|^2)\Phi_{w_g}(\omega) \quad (2.62)$$

where $\Phi_{w_g}(\omega) \in \mathbb{R}^{n_u}$ is the PSD of the gust that is the input into the system, e.g., given by the von Kármán power spectral density for vertical continuous turbulence Gust and Turbulence Loads [98]:

$$\Phi_{w_g}(\omega) = U_g^2 \frac{L_w}{\pi U_{\infty}} \frac{1 + \frac{8}{3}(1.339L_w\omega/U_{\infty})^2}{[1 + (1.339L_w\omega/U_{\infty})^2]^{\frac{11}{6}}} \quad (2.63)$$

where U_g is the turbulence intensity, L_w is the turbulence scale, and U_{∞} is the free stream velocity. The first two parameters can be found in the FAR.

⁵In the case with multiple inputs for turbulence, Equation (2.36) should be used.

The moments of the PSD can then be calculated by integration:

$$\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Phi_{VM}(\omega) d\omega, \quad m = 0, 1, 2, 4. \quad (2.64)$$

Combining Equations (2.60) to (2.62) and (2.64), it follows that

$$\lambda_m = \int_{\omega=0}^{\infty} \omega^m \left(|C_\sigma(j\omega I - A)^{-1}B + D_\sigma|^2 + 3|C_\tau(j\omega I - A)^{-1}B + D_\tau|^2 \right) \Phi_{w_g}(\omega) d\omega \quad (2.65)$$

For this integral to converge, it suffices that the integrand is bounded by $C^* \omega^{-x}$, for some $C^*, x \in \mathbb{R}$, $x > 1$, and for large values of ω . If the term $\Phi_{w_g}(\omega)$ represents the von Kármán power spectral distribution for turbulence in the vertical direction, it decays proportionally to $\omega^{-5/3}$ at high frequencies, which means that the moment of order m will be finite if the relative degree of each gust-to-stress transfer function is greater or equal to $\sqrt{m}$.

Incorporation of the sensitivity function

Since the von Mises stress relation requires a system with two outputs, a sensitivity function matrix has to be used:

$$S_o(s) = \begin{bmatrix} S_\sigma(s) & S_{\sigma\tau}(s) \\ S_{\tau\sigma}(s) & S_\tau(s) \end{bmatrix} \quad (2.66)$$

For simplicity, consider that this matrix has diagonal structure and that the normal and shear stress sensitivities are identical, i.e.,

$$S_{\sigma\tau}(s) = S_{\tau\sigma}(s) = 0 \quad (2.67)$$

$$S_\sigma(s) = S_\tau(s) = S(s; \mathbf{p}) \quad (2.68)$$

Then Equation (2.65) can be written in closed-loop:

$$\lambda_m = \int_{\omega=0}^{\infty} \omega^m |S(j\omega; \mathbf{p})|^2 \{2 (|C_\sigma(j\omega I - A)^{-1}B + D_\sigma|^2 + 3|C_\tau(j\omega I - A)^{-1}B + D_\tau|^2) \Phi_{w_g}(\omega) d\omega \quad (2.69)$$

The MIMO Bode sensitivity integral relation becomes:

$$\int_0^{\omega_b} \log |\det S_o(j\omega)| d\omega = \int_0^{\omega_b} 2 \log |S(j\omega; \mathbf{p})| d\omega = \pi \sum_{k=1}^{n_u} \Re(p_k) \quad (2.70)$$

Derivatives with respect to the model

The following matrices of frequency response functions are introduced in order to keep the expressions for the derivatives of eq. (2.65) compact:

$$\chi(\omega) = (j\omega I - A)^{-1}B \quad \in \mathbb{R}^{n_x \times 1} \quad (2.71a)$$

$$\psi_\sigma(\omega) = C_\sigma(j\omega I - A)^{-1} \quad \in \mathbb{R}^{n_y \times n_x} \quad (2.71b)$$

$$\psi_\tau(\omega) = C_\tau(j\omega I - A)^{-1} \quad \in \mathbb{R}^{n_y \times n_x} \quad (2.71c)$$

$$G_\sigma(\omega) = C_\sigma(j\omega I - A)^{-1}B + D_\sigma \quad \in \mathbb{R}^{n_y \times 1} \quad (2.71d)$$

$$G_\tau(\omega) = C_\tau(j\omega I - A)^{-1}B + D_\tau \quad \in \mathbb{R}^{n_y \times 1} \quad (2.71e)$$

Differentiating Equation (2.69) with the use of (xiv) and (xv), and swapping the order of the integration with differentiation (the precise assumptions under which this is possible are presented in e.g., [183]):

$$\begin{aligned} d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Re \Big[& 2\overline{G_\sigma}(\omega) \circ (dC_\sigma \chi(\omega) + \psi_\sigma(\omega) dA \chi(\omega) + \psi_\sigma(\omega) dB + dD_\sigma) \\ & + 6\overline{G_\tau}(\omega) \circ (dC_\tau \chi(\omega) + \psi_\tau(\omega) dA \chi(\omega) + \psi_\tau(\omega) dB + dD_\tau) \Big] \Phi_{w_g}(\omega) d\omega \end{aligned} \quad (2.72)$$

This equation allows the calculation of the directional derivatives at a point defined by the values of $(A, B, C_\sigma, C_\tau, D_\sigma, D_\tau)$, given directions for the inputs, i.e., values for dA , dB , dC_σ , dC_τ , dD_σ , and dD_τ . Note that $d\lambda_m$ in eq. (2.72) is linear in the differentials, which are independent of the integration variable ω , allowing it to be written as

$$d\lambda_m = \frac{\partial \lambda_m}{\partial C_\sigma} dC_\sigma + \frac{\partial \lambda_m}{\partial C_\tau} dC_\tau + \frac{\partial \lambda_m}{\partial A} dA + \frac{\partial \lambda_m}{\partial B} dB + \frac{\partial \lambda_m}{\partial D_\sigma} dD_\sigma + \frac{\partial \lambda_m}{\partial D_\tau} dD_\tau \quad (2.73)$$

where the Jacobians are third-order tensors.

This equation can be vectorized by applying the vec operator to both sides of the equal

sign and using the properties presented in Section 2.3, starting with the “vec-trick” (xviii):

$$\begin{aligned} \text{vec } d\lambda_m = & \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \\ & \text{vec} \left\{ \Re \left[2\overline{G_\sigma}(\omega) \circ (dC_\sigma \chi(\omega) + \psi_\sigma(\omega) dA \chi(\omega) + \psi_\sigma(\omega) dB + dD_\sigma) \right. \right. \\ & \left. \left. + 6\overline{G_\tau}(\omega) \circ (dC_\tau \chi(\omega) + \psi_\tau(\omega) dA \chi(\omega) + \psi_\tau(\omega) dB + dD_\tau) \right] \right\} d\omega \end{aligned} \quad (2.74)$$

Using the fact that vec is distributive over the element-wise product (xvi):

$$\begin{aligned} \text{vec } d\lambda_m = & \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \\ & \Re \left\{ 2 \text{vec } \overline{G_\sigma}(\omega) \circ \left[(\chi(\omega)^\top \otimes I_{n_y}) \text{vec } dC_\sigma + (\chi(\omega)^\top \otimes \psi_\sigma(\omega)) \text{vec } dA \right. \right. \\ & \left. \left. + (I_{n_u} \otimes \psi_\sigma(\omega)) \text{vec } dB + \text{vec } dD_\sigma \right] \right. \\ & \left. + 6 \text{vec } \overline{G_\tau}(\omega) \circ \left[(\chi(\omega)^\top \otimes I_{n_y}) \text{vec } dC_\tau + (\chi(\omega)^\top \otimes \psi_\tau(\omega)) \text{vec } dA \right. \right. \\ & \left. \left. + (I_{n_u} \otimes \psi_\tau(\omega)) \text{vec } dB + \text{vec } dD_\tau \right] \right\} d\omega \end{aligned} \quad (2.75)$$

Applying the property that the element-wise product of two vectors is equal to multiplying a vector by the matrix with the elements of the other in the diagonal (xvii):

$$\begin{aligned} \text{vec } d\lambda_m = & \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \\ & \Re \left\{ 2 \text{diag}(\text{vec } \overline{G_\sigma}(\omega)) \left[(\chi(\omega)^\top \otimes I_{n_y}) \text{vec } dC_\sigma + (\chi(\omega)^\top \otimes \psi_\sigma(\omega)) \text{vec } dA \right. \right. \\ & \left. \left. + (I_{n_u} \otimes \psi_\sigma(\omega)) \text{vec } dB + \text{vec } dD_\sigma \right] \right. \\ & \left. + 6 \text{diag}(\text{vec } \overline{G_\tau}(\omega)) \left[(\chi(\omega)^\top \otimes I_{n_y}) \text{vec } dC_\tau + (\chi(\omega)^\top \otimes \psi_\tau(\omega)) \text{vec } dA \right. \right. \\ & \left. \left. + (I_{n_u} \otimes \psi_\tau(\omega)) \text{vec } dB + \text{vec } dD_\tau \right] \right\} d\omega \end{aligned} \quad (2.76)$$

The linear character of this relation can be made explicit by writing it as

$$\begin{aligned} \text{vec } d\lambda_m = & \frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(C_\sigma)} \text{vec } dC_\sigma + \frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(C_\tau)} \text{vec } dC_\tau + \frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(A)} \text{vec } dA \\ & + \frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(B)} \text{vec } dB + \frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(D_\sigma)} \text{vec } dD_\sigma + \frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(D_\tau)} \text{vec } dD_\tau \end{aligned} \quad (2.77)$$

where the Jacobians of the vectorized input-output pairs are given by:

$$\frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(C_\sigma)} = \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \Re \left[2 \text{diag}(\text{vec } \overline{G_\sigma}(\omega)) (\chi(\omega)^\top \otimes I_{n_y}) \right] d\omega \quad (2.78a)$$

$$\frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(C_\tau)} = \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \Re \left[6 \text{diag}(\text{vec } \overline{G_\tau}(\omega)) (\chi(\omega)^\top \otimes I_{n_y}) \right] d\omega \quad (2.78b)$$

$$\begin{aligned} \frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(A)} = & \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \Re \left[2 \text{diag}(\text{vec } \overline{G_\sigma}(\omega)) (\chi(\omega)^\top \otimes \psi_\sigma(\omega)) \right. \\ & \left. + 6 \text{diag}(\text{vec } \overline{G_\tau}(\omega)) (\chi(\omega)^\top \otimes \psi_\tau(\omega)) \right] d\omega \end{aligned} \quad (2.78c)$$

$$\begin{aligned} \frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(B)} = & \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \Re \left[2 \text{diag}(\text{vec } \overline{G_\sigma}(\omega)) (I_{n_u} \otimes \psi_\sigma(\omega)) \right. \\ & \left. + 6 \text{diag}(\text{vec } \overline{G_\tau}(\omega)) (I_{n_u} \otimes \psi_\tau(\omega)) \right] d\omega \end{aligned} \quad (2.78d)$$

$$\frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(D_\sigma)} = \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \Re \left[2 \text{diag}(\text{vec } \overline{G_\sigma}(\omega)) \right] d\omega \quad (2.78e)$$

$$\frac{\partial \text{vec}(\lambda_m)}{\partial \text{vec}(D_\tau)} = \int_{\omega=0}^{\infty} \omega^m (\Phi_{w_g}(\omega)^\top \otimes I_{n_y}) \Re \left[6 \text{diag}(\text{vec } \overline{G_\tau}(\omega)) \right] d\omega \quad (2.78f)$$

The adjoint operators in the vectorized basis are the transpose of the direct operators, shown in Equation (2.78). They can be evaluated and written in the original (not-vectorized)

Table 2.1 Dimensions of explicit and matrix-free derivatives

	Direct	Jacobian	Adjoint
$\frac{\partial \lambda_m}{\partial C_\sigma}$	$\mathbb{R}^{n_y \times n_x} \rightarrow \mathbb{R}^{n_y}$	$\mathbb{R}^{n_y \times n_y \times n_x}$	$\mathbb{R}^{n_y} \rightarrow \mathbb{R}^{n_y \times n_x}$
$\frac{\partial \lambda_m}{\partial C_\tau}$	$\mathbb{R}^{n_y \times n_x} \rightarrow \mathbb{R}^{n_y}$	$\mathbb{R}^{n_y \times n_y \times n_x}$	$\mathbb{R}^{n_y} \rightarrow \mathbb{R}^{n_y \times n_x}$
$\frac{\partial \lambda_m}{\partial A}$	$\mathbb{R}^{n_x \times n_x} \rightarrow \mathbb{R}^{n_y}$	$\mathbb{R}^{n_y \times n_x \times n_x}$	$\mathbb{R}^{n_y} \rightarrow \mathbb{R}^{n_x \times n_x}$
$\frac{\partial B}{\partial \lambda_m}$	$\mathbb{R}^{n_x \times 1} \rightarrow \mathbb{R}^{n_y}$	$\mathbb{R}^{n_y \times n_x \times n_y}$	$\mathbb{R}^{n_y} \rightarrow \mathbb{R}^{n_x \times 1}$
$\frac{\partial D_\sigma}{\partial \lambda_m}$	$\mathbb{R}^{n_y \times 1} \rightarrow \mathbb{R}^{n_y}$	$\mathbb{R}^{n_y \times n_y \times 1}$	$\mathbb{R}^{n_y} \rightarrow \mathbb{R}^{n_y \times 1}$
$\frac{\partial D_\tau}{\partial \lambda_m}$	$\mathbb{R}^{n_y \times 1} \rightarrow \mathbb{R}^{n_y}$	$\mathbb{R}^{n_y \times n_y \times 1}$	$\mathbb{R}^{n_y} \rightarrow \mathbb{R}^{n_y \times 1}$

basis, yielding the following matrix-free adjoint equations:

$$\left(\frac{\partial \lambda_m}{\partial C_\sigma} \right)^* d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Re \left\{ \left[2\overline{G_\sigma}(\omega) \circ (d\lambda_m \Phi_{w_g}(\omega)^\top) \right] \chi(\omega)^\top \right\} d\omega \quad (2.79a)$$

$$\left(\frac{\partial \lambda_m}{\partial C_\tau} \right)^* d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Re \left\{ \left[6\overline{G_\tau}(\omega) \circ (d\lambda_m \Phi_{w_g}(\omega)^\top) \right] \chi(\omega)^\top \right\} d\omega \quad (2.79b)$$

$$\begin{aligned} \left(\frac{\partial \lambda_m}{\partial A} \right)^* d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Re \left\{ \psi_\sigma(\omega)^\top \left[2\overline{G_\sigma}(\omega) \circ (d\lambda_m \Phi_{w_g}(\omega)^\top) \right] \chi(\omega)^\top \right. \\ \left. + \psi_\tau(\omega)^\top \left[6\overline{G_\tau}(\omega) \circ (d\lambda_m \Phi_{w_g}(\omega)^\top) \right] \chi(\omega)^\top \right\} d\omega \end{aligned} \quad (2.79c)$$

$$\begin{aligned} \left(\frac{\partial \lambda_m}{\partial B} \right)^* d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Re \left\{ \psi_\sigma(\omega)^\top \left[2\overline{G_\sigma}(\omega) \circ (d\lambda_m \Phi_{w_g}(\omega)^\top) \right] \right. \\ \left. + \psi_\tau(\omega)^\top \left[6\overline{G_\tau}(\omega) \circ (d\lambda_m \Phi_{w_g}(\omega)^\top) \right] \right\} d\omega \end{aligned} \quad (2.79d)$$

$$\left(\frac{\partial \lambda_m}{\partial D_\sigma} \right)^* d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Re \left[2\overline{G_\sigma}(\omega) \circ (d\lambda_m \Phi_{w_g}(\omega)^\top) \right] d\omega \quad (2.79e)$$

$$\left(\frac{\partial \lambda_m}{\partial D_\tau} \right)^* d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Re \left[6\overline{G_\tau}(\omega) \circ (d\lambda_m \Phi_{w_g}(\omega)^\top) \right] d\omega \quad (2.79f)$$

The direct derivatives, Equation (2.72), allow the calculation of the directional derivative of the function given a direction in the input space, while the adjoint derivatives, Equation (2.79), allow the calculation of this directional derivative given a direction in the output space. They can be used to calculate the total derivatives for the optimization problem without forming the potentially very big Jacobian matrices of Equation (2.78). Table 2.1 summarizes the sizes of the matrices involved in these different forms of obtaining derivatives.

Derivatives with respect to the sensitivity function

Differentiating Equation (2.69) yields

$$d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \left\{ d[|S(j\omega)|^2] \Phi_{VM}(\omega) + |S(j\omega)|^2 d\Phi_{VM}(\omega) \right\} d\omega \quad (2.80)$$

from which the direct matrix-free directional derivative can be obtained:

$$\frac{\partial \lambda_m}{\partial |S(j\omega)|^2} d[|S(j\omega)|^2] = \int_{\omega=0}^{\infty} \omega^m d[|S(j\omega)|^2] \Phi_{VM}(\omega) d\omega \quad (2.81)$$

where $d[|S(j\omega)|^2] \in \{f|f : \mathbb{R}_+ \rightarrow \mathbb{R}^{n_y \times n_y}\}$ is a matrix of functions which represent the direction in which the derivative is taken, and $\frac{\partial \lambda_m}{\partial |S(j\omega)|^2} : \{f|f : \mathbb{R}_+ \rightarrow \mathbb{R}^{n_y \times n_y}\} \rightarrow \mathbb{R}^{n_y}$ maps a direction in the space of matrices of magnitude squared of sensitivity functions evaluated in frequency to a direction in the space of PSD moments.

Equation (2.81) makes more sense when the sensitivity function is parameterized by parameters $p \in \mathbb{R}^{n_p}$, in which case the chain rule can be applied, yielding:

$$\frac{\partial \lambda_m}{\partial p} = \frac{\partial \lambda_m}{\partial |S(j\omega; p)|^2} \frac{\partial |S(j\omega; p)|^2}{\partial p} \quad (2.82)$$

The adjoint matrix-free directional derivative can be readily calculated by transposition:

$$\left(\frac{\partial \lambda_m}{\partial |S(j\omega)|^2} \right)^* d\lambda_m = \int_{\omega=0}^{\infty} \omega^m \Phi_{VM}(\omega)^\top d\lambda_m d\omega \quad (2.83)$$

2.4 Sub-optimization for static stress computation considering MLA

The methods developed in the previous sections can efficiently handle stochastic systems, but another approach is needed for deterministic systems, such as aircraft performing pre-determined maneuvers. In particular, maneuver load alleviation (MLA) systems work by deflecting control surfaces during a maneuver to change the lift distribution in such a way that the total lift is preserved but the stress on the aircraft's structure is reduced.

The maneuver load alleviation (MLA) metric considered here is based on the addition of control surface deflections as design variables to the calculation of the maximum stress during aircraft maneuvers, which is used to inform a failure constraint in the optimization problem. The origins of this approach can be found in the literature [10, 11], and it belongs in the category of “optimal control input co-design” approaches for including control in

MDO (Section 1.2.4).

One concern with this type of approach is that it adds a substantial number of design variables to the optimization problem, corresponding to the inputs that are designed. To avoid this, a bi-level optimization approach is proposed. This approach has the additional advantage that it can benefit from different levels of computational complexity present in the optimization problem. For instance, the example introduced in Section 4.3 uses a model reduction process which makes the stress calculations much faster than the calculation of the objective function, and the procedure described here can provide enhanced performance.

A sub-problem is defined to calculate both the aircraft trim and the optimal control deflections within allocation limits per control surface for reducing maneuver loads at two points in the V-g diagram, namely maneuver and dive, with positive load factors. The output of the MLA analysis is the minimal failure metric (KS_{failure}), whose calculation is discussed in Section 2.5. The derivatives of the optimal solution of the sub-problem with respect to the outer design variables are calculated by post-processing the solution, according to the derivations in Section 2.4.1. The formulation of the inner optimization problems is given by

Maneuver (A)		Dive (D)	
minimize	KS_{failure}	minimize	KS_{failure}
with respect to	V, δ	with respect to	α, δ
subject to	$F_Z = n \text{ MTOW}$	subject to	$F_Z = n \text{ MTOW}$
	$ \delta_i \leq \delta_{\text{max}}, i = 1, \dots, n_\delta$		$ \delta_i \leq \delta_{\text{max}}, i = 1, \dots, n_\delta$

where α is the body angle of attack, V is the flight speed, $\delta = [\delta_1, \dots, \delta_{n_\delta}]^T$ is the vector of control surface deflections, δ_{max} is the maximum deflection allocation for MLA, F_Z is the total vertical force, n is the load factor, and MTOW is the maximum weight of the aircraft. The difference between the maneuver and dive conditions resides in the selection of the trim variable: in the former, the root angle of attack is prescribed, and the flight speed is solved for, while in the latter, the flight speed is prescribed, and the angle of attack solved for. A block diagram of the sub-problem is shown in Figure 2.4.

The proposed approach can be compared with the Asymmetric Subspace Optimization (ASO) distributed MDO architecture described in [72, Section 13.5.4]. In ASO, the cheaper disciplinary analysis is replaced by a copy of the outer optimization problem which is solved only for one discipline. On the other hand, the approach described herein uses a suboptimization to minimize an inequality constraint of the outer problem with respect to the

variables of one discipline, i.e., MLA control surface deflections. Both approaches result in the removal of design variables from the outer problem, but ASO is not directly applicable in this case since the coupling between the MLA discipline and the aircraft design problem happens through the failure constraint and not through the objective function.

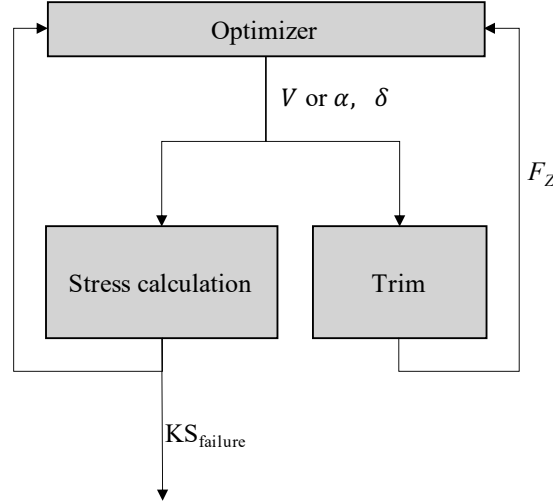


Figure 2.4 Sub-optimization problem for trim and calculation of optimal MLA deflections.

2.4.1 Derivatives of optimal value of the objective function with respect to the design variables

When performing gradient-based optimization in which one of the functions of interest is the output of an optimization problem, it is desirable to calculate the sensitivity of that function evaluated at the solution (i.e., of the value function) with respect to the design variables of the outer problem, which are given for the inner problem. Consider the following parameter-dependent constrained optimization problem

$$\begin{aligned}
 &\text{minimize} && f(x, \theta) \\
 &\text{with respect to} && x \\
 &\text{subject to} && g(x, \theta) = 0 \\
 & && h(x, \theta) \leq 0
 \end{aligned}$$

where f , g , and h are twice continuously differentiable (C^2) functions of $x \in \mathbb{R}^{n_x}$ and $\theta \in \mathcal{U} \subseteq \mathbb{R}^{n_\theta}$, where $\mathcal{U}$ is an open set. The vector θ represents the problem parameters, which correspond to the outer optimization design variables, $g(x, \theta) = 0$ are equality constraints and $h(x, \theta) \leq 0$ are inequality constraints.

Suppose, for a given value of the parameters $\theta = \theta_0$, there is a solution to this problem, denoted by $x^*(\theta_0)$ that is locally continuously differentiable in a neighborhood of θ_0 (this property holds, for instance, under strong second-order sufficient conditions [184]). Let $\lambda^*(\theta_0)$ denote the vector of Lagrange multipliers corresponding to the equality constraints and let $\mu^*(\theta_0)$ denote the vector of Lagrange multipliers corresponding to the inequality constraints. This section summarizes the procedure to compute the derivative of the optimum value of the objective function with respect to changes in the value of the parameters, i.e., $\frac{df}{d\theta}(x^*(\theta), \theta)$ evaluated at θ_0 . The treatment follows [184] and [185, Chapter 2], which demonstrates that only the first-order derivative information is necessary to compute the sensitivity of the objective function, unlike the sensitivity of the solution itself that requires second-order derivatives. In any case, the existence of these sensitivities depends on both the objective and constraint functions being twice differentiable (see the aforementioned references).

Let $G^a(x, \theta)$ denote the vector function which stacks up all active constraint functions (i.e., all equality constraints and active inequality constraints) of the problem and let λ^a denote the vector of corresponding Lagrange multipliers. For instance, if there are two equality constraints, $g_1(x, \theta) = 0$ and $g_2(x, \theta) = 0$ and two inequality constraints, $h_1(x, \theta) \leq 0$ and $h_2(x, \theta) \leq 0$, but only h_1 is active at θ_0 , then $G^a = [g_1, g_2, h_1]^T$ and λ^a is the three-element vector consisting of the Lagrange multipliers corresponding to g_1 , g_2 , and h_1 , i.e., $\lambda^a = [\lambda_1, \lambda_2, \mu_1]^T$. With this notation, the Lagrangian of the problem can be written as

$$\mathcal{L}(x, \theta, \lambda^a) = f(x, \theta) + (\lambda^a)^T G^a(x, \theta). \quad (2.84)$$

Since the stationarity condition must hold at the minimizer, it follows that

$$\frac{\partial \mathcal{L}}{\partial x}(x^*(\theta), \theta, \lambda^{a*}(\theta)) = \frac{\partial f}{\partial x}(x^*(\theta), \theta) + (\lambda^{a*}(\theta))^T \frac{\partial G^a}{\partial \theta}(x^*(\theta), \theta) = 0. \quad (2.85)$$

Also note that the problem constraints must be satisfied for all parameters $\theta \in U$, i.e.,

$$G^a(x^*(\theta), \theta) = 0 \quad \forall \theta \in \mathcal{U}. \quad (2.86)$$

Then the value of the Lagrangian at the minimizer is a function only of the problem parameters and its total derivative can be calculated as

$$\begin{aligned} \frac{d\mathcal{L}}{d\theta}(x^*(\theta), \theta, \lambda^{a*}(\theta)) &= \left[\frac{\partial f}{\partial x}(x^*(\theta), \theta) + (\lambda^{a*}(\theta))^T \frac{\partial G^a}{\partial x}(x^*(\theta), \theta) \right] \frac{dx^*}{d\theta}(\theta) \\ &+ \frac{\partial f}{\partial \theta}(x^*(\theta), \theta) + (\lambda^{a*}(\theta))^T \frac{\partial G^a}{\partial \theta}(x^*(\theta), \theta). \end{aligned} \quad (2.87)$$

From the stationarity condition (2.85), the first term is zero so it follows that

$$\frac{d\mathcal{L}}{d\theta}(x^*(\theta), \theta, \lambda^{a*}(\theta)) = \frac{\partial \mathcal{L}}{\partial \theta}(x^*(\theta), \theta, \lambda^{a*}(\theta)). \quad (2.88)$$

On the other hand, Equation (2.86) implies that

$$\frac{dG^a}{d\theta}(x^*(\theta), \theta) = \frac{\partial G^a}{\partial x}(x^*(\theta), \theta) \frac{dx^*}{d\theta}(\theta) + \frac{\partial G^a}{\partial \theta}(x^*(\theta), \theta) = 0 \quad \forall \theta \in \mathcal{U}. \quad (2.89)$$

Applying Equation (2.89) to Equation (2.87), it follows that

$$\frac{d\mathcal{L}}{d\theta}(x^*(\theta), \theta, \lambda^{a*}(\theta)) = \frac{\partial f}{\partial x}(x^*(\theta), \theta) \frac{dx^*}{d\theta}(\theta) + \frac{\partial f}{\partial \theta}(x^*(\theta), \theta) = \frac{df}{d\theta}(x^*(\theta), \theta). \quad (2.90)$$

Finally, combining results (2.88) and (2.90) and noticing that the Lagrange multipliers for inactive constraints are zero, it follows that

$$\frac{df}{d\theta}(x^*(\theta), \theta) = \frac{\partial \mathcal{L}}{\partial \theta}(x^*(\theta), \theta, \lambda^*(\theta)), \quad (2.91)$$

which represents a simple and effective way of calculating the derivative of the optimum objective function with respect to the problem parameters using only the first derivative information. The sensitivity depends only on the knowledge of the optimum design point (x^*), Lagrange multipliers at the optimum (λ^*), and the sensitivity of the functions of interest with respect to parameters ($\frac{\partial f}{\partial \theta}$, $\frac{\partial g}{\partial \theta}$, and $\frac{\partial h}{\partial \theta}$).

For the particular case of the inner optimization problem considered here, $x = [\alpha, \delta^T]^T$, where α is the body angle of attack and δ is the vector of control deflections. From Equation (2.91),

$$\frac{df}{d\theta}(\alpha^*(\theta), \delta^*(\theta), \theta) = \frac{\partial f}{\partial \alpha}(\alpha^*(\theta), \delta^*(\theta), \theta) + \lambda_\alpha^* \frac{\partial F_Z}{\partial \theta}(\alpha^*(\theta), \delta^*(\theta), \theta) + \lambda_\delta^* \frac{\partial B_\delta}{\partial \theta}(\delta^*(\theta)), \quad (2.92)$$

where $F_Z(\alpha, \delta, \theta)$ is the trim constraint and $B_\delta(\delta)$ are the bounds on control surface deflection. Since $B_\delta(\delta)$ is not a function of θ , it follows that $\frac{\partial B_\delta}{\partial \theta} = 0, \forall \theta$ and thus Equation (2.92)

reduces to

$$\frac{df}{d\theta}(\alpha^*(\theta), \delta^*(\theta), \theta) = \frac{\partial f}{\partial \theta}(\alpha^*(\theta), \delta^*(\theta), \theta) + \lambda_\alpha^* \frac{\partial F_Z}{\partial \theta}(\alpha^*(\theta), \delta^*(\theta), \theta). \quad (2.93)$$

The application for the case where flight speed is used as the trim variable and the angle of attack is held constant is completely analogous.

2.5 Stress calculation

Usually, the stress constraint used for maneuver loads in high-fidelity static aerostructural optimization is derived from the von Mises stress (or an equivalent for composite constructions) of the wingbox walls at various points along the span [186]. To obtain a similar metric from a beam-based model, it is necessary to reintroduce information about the wingbox cross section, which was lost during the condensation process. Hence, a simplified stress model is described in what follows.

From the Euler-Bernoulli beam formulation for isotropic materials (see e.g., [187]), the normal stress due to bending at a given spanwise station and at a point (x, z) in the cross section is given by

$$\sigma = E[\kappa_x z - \kappa_z x] \quad (2.94)$$

where E is the Young's modulus of the material and κ_x and κ_z are the out-of-plane and in-plane bending curvatures, respectively.

The shear stresses are calculated based on the cross sectional shear flow. First, the shear flow q_ξ is divided into the basic shear flow $q_{\xi,b}$ and closed-section shear flow $q_{\xi,0}$, so that

$$q_\xi = q_{\xi,b} + q_{\xi,0} \quad (2.95)$$

The basic shear flow is obtained by introducing a fictitious cut in the upper right corner of the idealized wingbox:

$$q_{\xi,b}(\xi) = \int_0^\xi \left(-\frac{V_x I_{xx} - V_z I_{xz}}{I_{xx} I_{zz} - I_{xz}^2} x(\xi) + \frac{V_z I_{zz} - V_x I_{xz}}{I_{xx} I_{zz} - I_{xz}^2} z(\xi) \right) t(\xi) d\xi, \quad \xi \in [0, 2(w + h)] \quad (2.96)$$

where ξ is the coordinate that parameterizes the cross section shape by arc length, V_x and V_y are the shear force resolved in the cross sectional local coordinate frame, and I_{xx} , I_{zz} , I_{xz} are the moments of area of the cross section. These moments are obtained by dividing the stiffness matrix of the condensed beam model by the material Young's modulus and not by a direct calculation from the cross section shape.

Finally, the closed-section shear flow is obtained by enforcing the relation between shear flow and torsion, i.e.,

$$\kappa_y = \frac{1}{2whG} \oint \frac{q(\xi)}{t(\xi)} ds \quad (2.97)$$

Applying this relation to the idealized cross section and solving for $q_{\xi,0}$ yields

$$q_{\xi,0} = 2whG\kappa_y \frac{\oint q_{\xi,b}(\xi)/t(\xi) d\xi}{\oint 1/t(\xi) ds} \quad (2.98)$$

Then the shear stresses in each skin are obtained from the shear flow by the relation

$$\tau = \frac{q_\xi}{t} \quad (2.99)$$

This value is then combined with the normal stress from Equation (2.94) using the von Mises criterion:

$$\sigma_{VM} = \sqrt{\sigma^2 + 3\tau^2} \quad (2.100)$$

2.6 Flutter metric

For aircraft, it is important to guarantee that the dynamics are open-loop stable and, in particular, flutter-free. For this purpose, this work adopts the approach of constraining the real part of the eigenvalues of the A matrix (poles of the system) to be negative [148, 188]. These constraints are aggregated using the Kreisselmeier-Steinhauser (KS) function [189–191] to achieve greater numerical efficiency in the calculation of the derivatives in the adjoint mode, as well as removing the need for mode tracking. The KS function is defined as

$$KS(g, \rho) = \frac{1}{\rho} \log \sum_j e^{\rho g_j} \quad (2.101)$$

where g is the vector of constraints to be aggregated and ρ is a positive aggregation parameter. The use of the KS function is motivated by the useful property that it is smooth and approximates the maximum function, in the sense that $\lim_{\rho \rightarrow \infty} KS(g, \rho) = \max(g)$.

$$KS_{\text{flutter}} = KS(\Re(\text{diag } \Lambda) - \mathbf{1}\epsilon, 100) \quad (2.102)$$

where $\Re(\text{diag } \Lambda)$ is a vector containing the real part of the eigenvalues of A , $\mathbf{1}$ is a vector of ones of appropriate size and ϵ is a small number of allowable positive damping to compensate for the conservativeness introduced by the KS function combined with the aforementioned eigenvalues at the origin.

The contribution of this work is to compute the derivatives of this metric with respect to the A matrix analytically, using the observation that the derivative of the eigenvalues with respect to the A matrix can be obtained by the method proposed by Nelson [192]

Let $A \in \mathbb{R}^{n_x \times n_x}$ be diagonalizable, i.e., $A = V\Lambda V^{-1}$, where Λ is the diagonal matrix of eigenvalues and V is the matrix of eigenvectors. Then:

$$d\Lambda = V^{-1} dAV \quad (2.103)$$

The vec-trick (xviii) can be applied to this equation

$$\text{vec } d\Lambda = (V^\top \otimes V^{-1}) \text{vec } dA \quad (2.104)$$

From Equation (2.104), the adjoint vectorized Jacobian can be computed by taking its conjugate transpose,

$$\left(\frac{\partial \text{vec } \Lambda}{\partial \text{vec } A} \right)^* \text{vec } d\Lambda = (V^\top \otimes V^{-1})^H \text{vec } d\Lambda = (\bar{V} \otimes V^{-H}) \text{vec } d\Lambda = \text{vec}(V^{-H} d\Lambda V^H) \quad (2.105)$$

Therefore,

$$\left(\frac{\partial \Lambda}{\partial A} \right)^* d\Lambda = V^{-H} d\Lambda V^H \quad (2.106)$$

The analytical and matrix-free derivative calculation in direct and adjoint modes stated in Equations (2.103) and (2.106), respectively, can be used as part of the chain rule for computing the derivatives of the flutter metric.

CHAPTER 3

Software Developments

A considerable part of the complexity involved in this work comes from integrating the different tools involved. This chapter discusses the implementation of an interface for algorithmic differentiation (AD) in the University of Michigan Nonlinear Aeroelastic Simulation Toolbox (UM/NAST) [144], which facilitates the use of this analysis software in gradient-based optimization. UM/NAST is integrated with NASA’s generic MDO framework, OpenMDAO [146], and with the high-fidelity aircraft MDO framework MACH [186].

3.1 New interface for UM/NAST metrics in MDO

UM/NAST was originally developed to be a simulation code [144]. It was originally written in MATLAB and later ported to C++ with a Python interface [193]. Following the development and open-source availability of powerful AD tools based on operator overloading, the capability of derivative calculation was implemented in UM/NAST using CoDiPack [194], a development which was initially applied to simplified aircraft MDO with a geometrically nonlinear flutter constraint [193, 195].

The AD capabilities of UM/NAST were enhanced by the author to support its integration with MACH, which was demonstrated by incorporating a beam-based geometrically nonlinear flutter constraint into a high fidelity MACH MDO problem [148, 188]. These developments also enabled the implementation of the metrics described in this work.

To maintain a common interface for the different metrics, a modular interface inspired by OpenMDAO was developed (Figure 3.1). This interface is defined by the abstract base class `Component`. This interface consists of two functions, one to calculate the metric itself (`compute`) and one to compute its derivatives (`compute_partials`). They take their inputs as OpenMDAO vectors, which are a set of regular vectors labeled by strings (see the C++ definition in Listing 3.1). The `compute` function returns an OpenMDAO vector

Code Snippet 3.1 Definitions of “OpenMDAO” vectors and matrices

```
template<typename Scalar>
using OMVector = std::map<std::string, Eigen::Array<Scalar, Eigen::Dynamic, 1>>;

template<typename Scalar>
using OMMatrix = std::map<std::pair<std::string, std::string>,
                        Eigen::Array<Scalar, Eigen::Dynamic, Eigen::Dynamic>>;
```

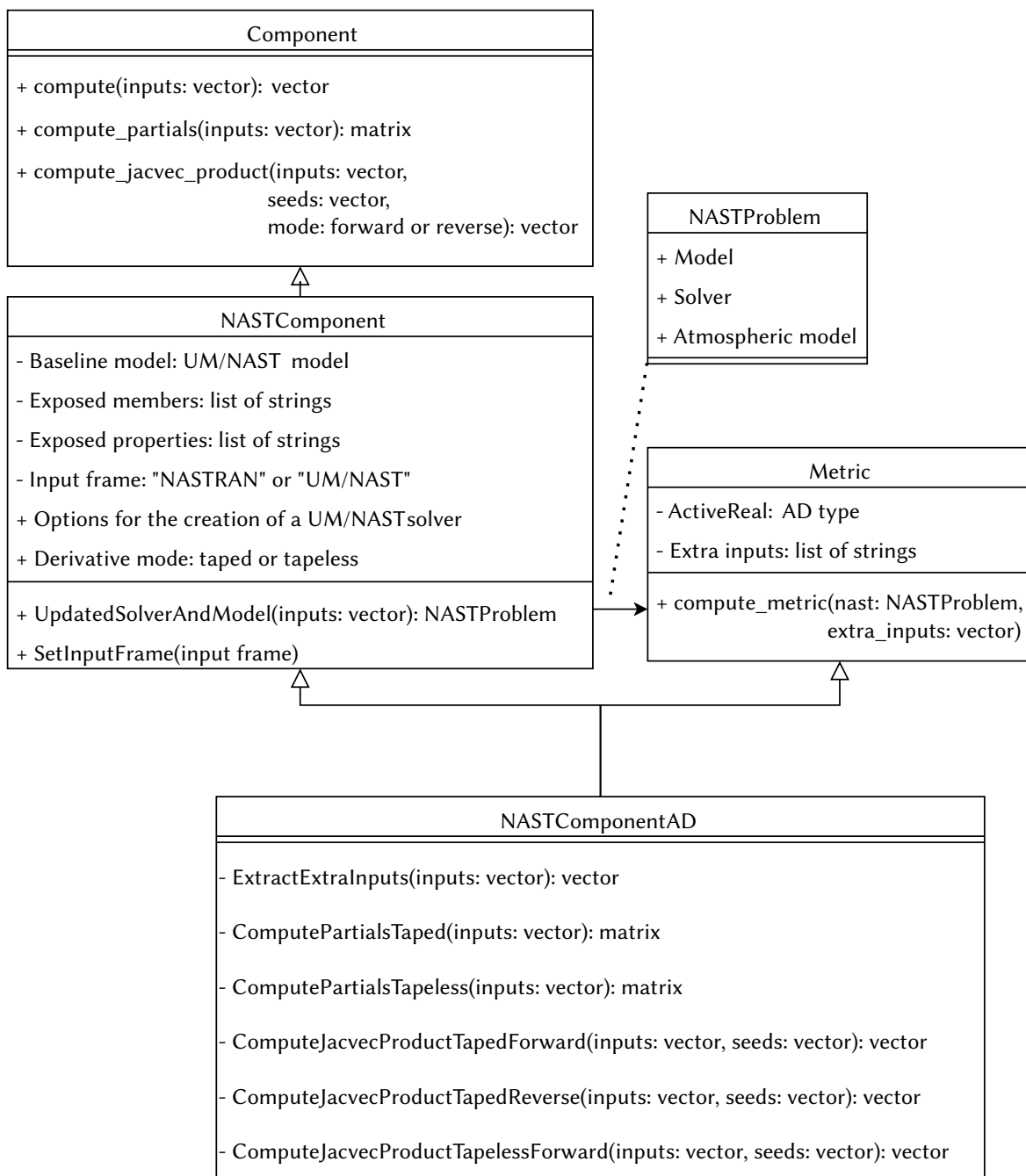


Figure 3.1 Unified Modeling Language (UML) diagram for the new AD interface of UM/NAST

of outputs, while the `compute_partials` function returns an OpenMDAO matrix, which represents the Jacobian of an OpenMDAO vector, and thus is a double-labeled set of matrices. The first label is the name of the output variable of which the derivative is taken, while the second label is the input variable name with respect to which the derivative is taken.

3.1.1 Input handling: the UM/NAST model and solver update

This generic interface is specialized for usage in UM/NAST by the `NASTComponent` class. Its responsibility is to update the UM/NAST solver and a baseline model provided by the user, with the inputs that vary with each iteration in the optimization. For this, it holds the following information:

- a list of the names of the members of the UM/NAST model that are being designed
- a list of the names of the properties of the UM/NAST model that are being designed
- the options for creating the UM/NAST solver
- the frame of reference convention in which the data is provided, which can be the NASTRAN frame (x pointing backward and z up) or the UM/NAST frame (y forward and z up)
- the type of calculation of derivatives desired (taped or tapeless)

Table 3.1 show which of the currently available inputs are related to the solver, members, and properties of UM/NAST. More inputs can be added to `UpdatedModelAndSolver`, and the choice of the ones currently available was driven by examples presented in Chapter 4. The `UpdatedModelAndSolver` function checks which of these options are available in the input vector passed from OpenMDAO, using a naming convention, and uses this data to update the model and solver. This input handling works for any UM/NAST model.

It is worth noting that the way of passing the baseline model differs from how the baseline solver is created. The baseline model is indeed an instantiation of the `UM/NAST Model` of type `double`, which relies on its copy operator (`Model::operator=`) to be transformed to the required AD type. This operator copies all data fields across the different types. On the other hand, the solver creation process relies on a *factory* class that holds the options for the solver and the function to create the solver of the required type given those options. The advantage of the second approach is that it guarantees consistency between solvers of different types, while there is the possibility of bugs in the copy process for the model. The update process is common for the model and solver. There is a check that ensures that all

Table 3.1 Inputs that can currently be updated using the UM/NAST AD interface

<i>Related to:</i>		
Exposed member	Exposed property	Solver
x, y, z position of keypoints	stiffnesses	flight speed
	masses	angle of attack
	concentrated masses	control surface deflections
	airfoil position	
	chord lengths	
	aerodynamic twist	
	structural twist	

inputs have been used during the model and solver update process, which helps to catch bugs in this step.

3.1.2 Output calculation: UM/NAST metrics

Each output of interest (“metric”) is defined in the code as a separate class. The responsibility of this class is to define a function that takes a UM/NAST model and solver as input and returns the output of interest. It does not need to implement the derivative calculation, which is implemented only once for all metrics and is described in Section 3.1.3. Examples of metrics are shown in Listings 3.3 and 3.4. Since the AD procedure is implemented only once and should work for all metrics, each metric must conform to the following requirements:

- The metric must be copy-constructable
- The metric must define a type template called `ActiveReal`, which is instantiable for both `double` and `codi::RealReverse`. This requirement exists to support metrics that require AD themselves. If the metric does not require AD (e.g., Listing 3.3, it suffices to define

```
template <typename Scalar> using ActiveReal = Scalar;
```

If the metric requires AD itself, for example, if it linearizes the UM/NAST model at some flight condition (e.g., Listing 3.4), the real number type required for AD must be defined in the metric:

```
template <typename Scalar>
using ActiveReal = codi::RealForwardGen<Scalar>;
```

Code Snippet 3.2 The MetricPolicy concept contains the requirements for a class that defines a UM/NAST metric

```
template <typename Metric>
concept MetricPolicy =
std::copy_constructible<Metric> &&
requires(Metric m) {
    typename Metric::ActiveReal<double>;
    typename Metric::ActiveReal<codi::RealReverse>;
} &&
requires(const Metric m,
    const NASTProblem<typename Metric::ActiveReal<double>> prob,
    const OMVector<typename Metric::ActiveReal<double>> extra_inputs) {
    { m.compute_metric(prob, extra_inputs) } -> std::same_as<OMVector<double>>;
} &&
requires(const Metric m,
    const NASTProblem<typename Metric::ActiveReal<codi::RealReverse>> prob,
    const OMVector<typename Metric::ActiveReal<codi::RealReverse>> extra_inputs){
    { m.compute_metric(prob, extra_inputs) }
    -> std::same_as<OMVector<codi::RealReverse>>;
} &&
requires(Metric m) {
    { typename decltype(m.extra_inputs)::value_type() }
    -> std::convertible_to<std::string>;
};
```

- The metric define a function, `compute_metric`, which takes a UM/NAST model and solver (`NASTProblem`) and returns a vector (i.e., the output of interest). This function should work for both for both `double` and `codi::RealReverse`, that is, it must be able to perform both primal and AD calculations.
- The metric must define a list of extra inputs that it needs, for example, the dimensions of the wingbox in the case of the stress metric (Listing 3.3). These will be inputs of the resulting Component and will be differentiated, but will bypass the model update procedure (Section 3.1.1).

These requirements are implemented as the C++ *concept*¹ `MetricPolicy` shown in Listing 3.2, which helps to produce better error messages in supporting compilers.

3.1.3 Policy-based implementation of algorithmic differentiation

The derivatives are calculated through algorithmic differentiation (AD) using the C++ library CoDiPack [194]. CoDiPack supports AD in both forward and reverse mode through

¹*Concepts* were introduced in C++20, so they are a fairly new language feature not yet supported by all compilers. This is handled in the code by preprocessor directives, which will use concepts if available, but if not, will maintain compatibility with C++14. The preprocessor directives were omitted here for brevity.

Code Snippet 3.3 Example metric for calculation of stresses

```
class StressMetric
{
public:
    std::string member_name_; // Name of member on where to compute the stresses
    std::string prop_name_; // Name of property on where to compute the stresses
    double E_ = 0.0; // Young's modulus [Pa]
    double G_ = 0.0; // Shear modulus [Pa]
    double rho_ = 100.0; // Parameter for KS aggregation
    double allowable_stress_ = 0.0; // Value to normalize metric
                                   // (before KS aggregation) [Pa]

    static constexpr std::initializer_list<const char*> extra_inputs_ = {
        "_h",
        "_w",
        "_thickness_LE_spar",
        "_thickness_TE_spar",
        "_thickness_upper_skin",
        "_thickness_lower_skin"};

    template <typename Scalar>
    using ActiveReal = Scalar; // 1st order AD

    template <typename Scalar>
    OMVector<Scalar> compute_metric(
        NASTProblem<ActiveReal<Scalar>> nast,
        const OMVector<ActiveReal<Scalar>>& extra_inputs) const;
};
```

primal value tapping, or if desired it can be used without a tape in forward mode only. The usage of CoDiPack in taped mode requires nine steps:

1. Set the tape² as active
2. Register inputs to the tape
3. Perform the desired computation to record it on the tape
4. Register outputs to the tape
5. Set the tape as passive
6. Set direction of gradient evaluation (in direct or reverse mode)
7. Evaluate the tape to calculate gradients (in direct or reverse mode)

²The “tape” is a data structure used in algorithmic differentiation to record the operations performed during a function evaluation, such that it can be used later to calculate the derivatives.

Code Snippet 3.4 Example metric for the linearization of the UM/NAST system

```
class LinearizationMetric
{
    public:
        static constexpr std::initializer_list<const char*> extra_inputs_ = {};

        LinearizationMetric() = default;

        template<typename Scalar>
        using ActiveReal = codi::RealForwardGen<Scalar>;

        template<typename Scalar>
        OMVector<Scalar> compute_metric(
            NASTProblem<ActiveReal<Scalar>> nast,
            const OMVector<ActiveReal<Scalar>>& extra_inputs) const;
};

template<typename Scalar>
OMVector<Scalar> LinearizationMetric::compute_metric(
    NASTProblem<ActiveReal<Scalar>> nast,
    const OMVector<ActiveReal<Scalar>>& extra_inputs) const
{
    using Vector = Eigen::Matrix<Scalar, Eigen::Dynamic, 1>;
    using Matrix = Eigen::Matrix<Scalar, Eigen::Dynamic, Eigen::Dynamic>;

    Matrix A = LinearizeNASTModelusingFAD(nast);
    Vector A_flattened = Eigen::Map<Vector>(A.data(), A.size());

    return {{ "A_flattened", A_flattened }};
}
```

8. Extract gradients from variables

9. Clear the tape

The theory of algorithmic differentiation is reviewed in [196].

To provide a common implementation of the derivative calculation process for all UM/-NAST based metrics, a programming technique known as *policy-based design* is employed [145, Chapter 1]. In this design, the class `NASTComponentAD` implements the AD computation of any UM/NAST metric, i.e., the procedure outlined above is implemented only once, in the `compute_partials_taped` function (Figure 3.1). To do so, `NASTComponentAD` inherits publicly from two classes: `NASTComponent`, which implements the model and solver update process (Section 3.1.1), and the class that implements a given metric (Section 3.1.2). This metric class is also a template of `NASTComponentAD`, therefore its declaration is

```
template<MetricPolicy Metric>
class NASTComponentAD: public NASTComponent, public Metric;
```

Because of this, the `Metric` is the “policy” of the policy-based design. It is responsible for implementing the generic (i.e., templated) function `compute_metric`, which performs the calculation of the output of interest for any real number-type inputs, e.g., `type double` and `type codi::RealReverse`, used to compute the metric value and its AD derivatives, respectively.

Two example metrics are shown in Listings 3.3 and 3.4. First, the metric class is defined with its non-differentiable options as class members, and a list of inputs that are handled by the metric directly, as opposed to being used by the model and solver update process is defined. Then the metric calculation itself is defined in the function `compute_metric`, without concern for the derivatives at this point. Finally, it is just a matter of instantiating the `NASTComponentAD` template with the metric class, and an AD-enabled `NASTComponent` is created. This process is advantageous because the AD algorithm implementation is only done once, and there is no requirement for the programmer to learn the AD tool to implement a new metric.

Another feature of this implementation is that it will automatically choose whether to perform forward or reverse AD based on the number of inputs and outputs of the metric, to guarantee the minimal function of tape evaluations. That is, it will perform forward AD when there are more inputs than outputs and reverse AD otherwise. Furthermore, it provides functions for the matrix-free evaluation of directional derivatives through the `compute_jacvec_product` function.

3.1.4 Integration of UM/NAST with NASA’s OpenMDAO

The C++ interface described in the preceding sections and summarized in Figure 3.1 is available in Python through the use of `pybind11`³. When compared to the previous C++/Python interfaced used, `Cython`⁴, it was found that `pybind11` made coding faster and easier to debug since the Python wrapping is done purely in C++, without the need to introduce a third language in the mix, namely `Cython`.

With the functions available in Python, each instantiation of `NASTComponentAD` can be wrapped in an `OpenMDAO ExplicitComponent`. This is done by defining the intermediate class `NastComponent`, which declares the options available in UM/NAST in the `OpenMDAO` format and adds the inputs for the `OpenMDAO Component` following the input naming scheme. Then each instantiation of `NASTComponentAD` is wrapped in a python class inherited from `NastComponent`. For example, the wrapping for the metric that computes the B matrix from a gust input to the UM/NAST system is shown in Listing 3.5. The

³<https://github.com/pybind/pybind11>, accessed April 14, 2024

⁴<https://cython.org/>, accessed April 14, 2024

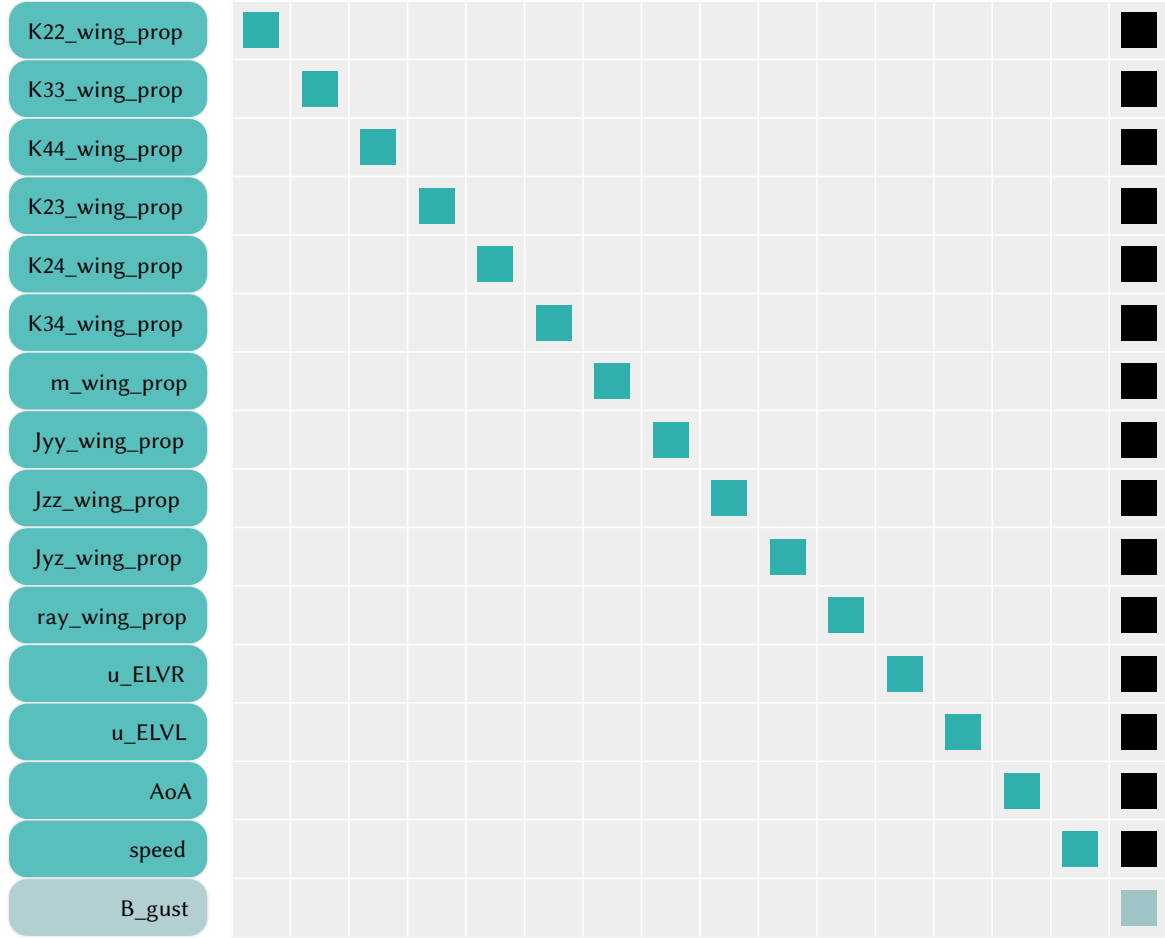


Figure 3.2 OpenMDAO N^2 diagram for the LinearizedGustComp

N^2 diagram generated by OpenMDAO for this component is shown in Figure 3.2, and it gives an idea of the input naming convention. Note that not all available inputs are present in this component, only the ones that were used in this particular problem, which is the one described in Section 4.1.

The metrics available in this interface that were used in this dissertation to implement the problems described in Sections 4.1 and 4.3 are:

TrimComp evaluates the residual forces and moments in the UM/NAST model, that can be used by the nonlinear solvers in OpenMDAO to trim the aircraft model in different flight conditions.

StressComp implements the simplified stress model described in Section 2.5.

LinearizedStressComp calculates a state space C matrix based on the AD linearization of the simplified stress model described in Section 2.5.

Code Snippet 3.5 Sample OpenMDAO wrapping of a metric

```
# Python side of the new interface
from .nast_components import NastComponent
from .nast_components import NastFlightCondition, NastFlightConditionMatching
# pybind11 bindings from the C++ side of the new interface
from .flutter_component import LinearizedGustComponent, state_size, DerivativeMode

class LinearizedGustComp(NastComponent):
    """Calculates linearized B matrix for vertical gust on aircraft"""

    def initialize(self):
        super().initialize()
        self.options.declare("flight_condition", desc="Flight condition",
                             types=(NastFlightCondition, NastFlightConditionMatching))

    def setup(self):
        super().setup() # Inputs are declared in NastComponent
        model = self.options["baseline_model"]
        opt = self.create_dynamic_options(self.options["flight_condition"])
        n_states = state_size(model, opt)
        self.add_output(
            "B_gust",
            shape=(n_states,),
            desc="B matrix for vertical gust input"
        )

    def setup_partials(self):
        self.declare_partials("*", "*")

    def create_impl(self):
        impl = LinearizedGustComponent(
            self.options["baseline_model"],
            self.exposed_members,
            self.exposed_properties,
            self.options["input_frame"],
        )
        impl.options = self.create_dynamic_options(self.options["flight_condition"])
        impl.derivative_mode = DerivativeMode.kTapeless
        return impl

    def compute(self, inputs, outputs):
        impl = self.create_impl()
        output_dict = impl.compute(dict(inputs))
        for k in output_dict.keys():
            outputs[k] = output_dict[k]

    def compute_partials(self, inputs, partials):
        impl = self.create_impl()
        partials_dict = impl.compute_partials(dict(inputs))
        for k in partials_dict.keys():
            partials[k] = partials_dict[k]
        impl = self.create_impl()
```

`LinearizedGustComp` calculates a state space B matrix for a vertical gust input based on an AD linearization of the UM/NAST equations of motion.

`LinearizationComp` calculates the state space A matrix based on an AD linearization of the UM/NAST equations of motion.

`NASTFlutter` evaluates the flutter metric described in Section 2.6.

3.2 Generalization of the MACH-UM/NAST framework

The MACH (framework for MDO of Aircraft Configurations with High fidelity) [147] couples the parallel finite element model (FEM) structural solver TACS (Toolkit for the Analysis of Composite Structures) [197] with the Reynolds averaged Navier-Stokes computer fluid dynamics (CFD) solver ADFlow [198]. It also includes pyGeo [199] to handle the geometry parameterization.

Jonsson et al. [148] integrated UM/NAST with MACH to incorporate a geometrically non-linear flutter constraint calculated by UM/NAST in the MDO problem. In the process of doing so, [148] demonstrated the use of the University of Michigan’s Enhanced FEM to Stick (UM/EF2S) [200, Appendix C] (see also [201–204]) model order reduction process in an MDO problem for the first time. UM/EF2S converts a built-up FEM model, such as those from TACS or NASTRAN, to an equivalent beam model that can be used in UM/NAST analyses, and thus enables mixed-fidelity optimization with these tools.

In this work, the MACH-UM/NAST integration has been generalized so that multiple UM/NAST metrics can be included in a high-fidelity MACH optimization. Figure 3.3 illustrates the change in the framework necessary for this expansion in capability. In particular, the methodology to incorporate MLA into high-fidelity MDO using bi-level optimization from Section 2.4 is demonstrated using MACH-UM/NAST in Section 4.3. Figure 3.4 shows a flowchart of this integration.

Previously, the required input to the framework from the user was a `NASTFlutter` component from UM/NAST’s OpenMDAO interface, which was then assembled into an OpenMDAO model with an aggregation component within MACH-UM/NAST (denoted by the dashed box in Figure 3.3a). In order to support arbitrary UM/NAST analyses and outputs, the updated MACH-UM/NAST framework requires the user to provide a complete OpenMDAO model, which must accept the same inputs describing a UM/NAST model as were previously used within the framework, namely:

- Concentrated nodal masses, $\hat{m}$

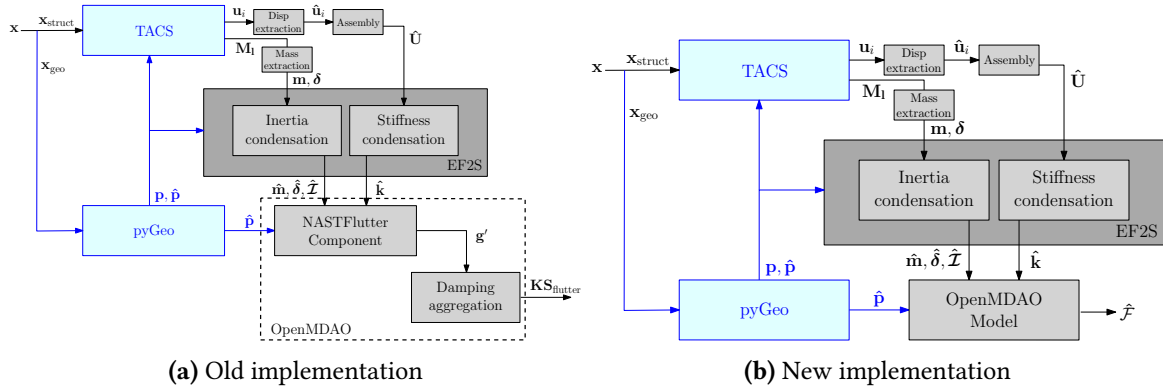


Figure 3.3 The updated version of the MACH-UM/NAST framework can be used to include any function(s) of interest computed by any user-provided OpenMDAO model that includes OpenMDAO_NAST components.

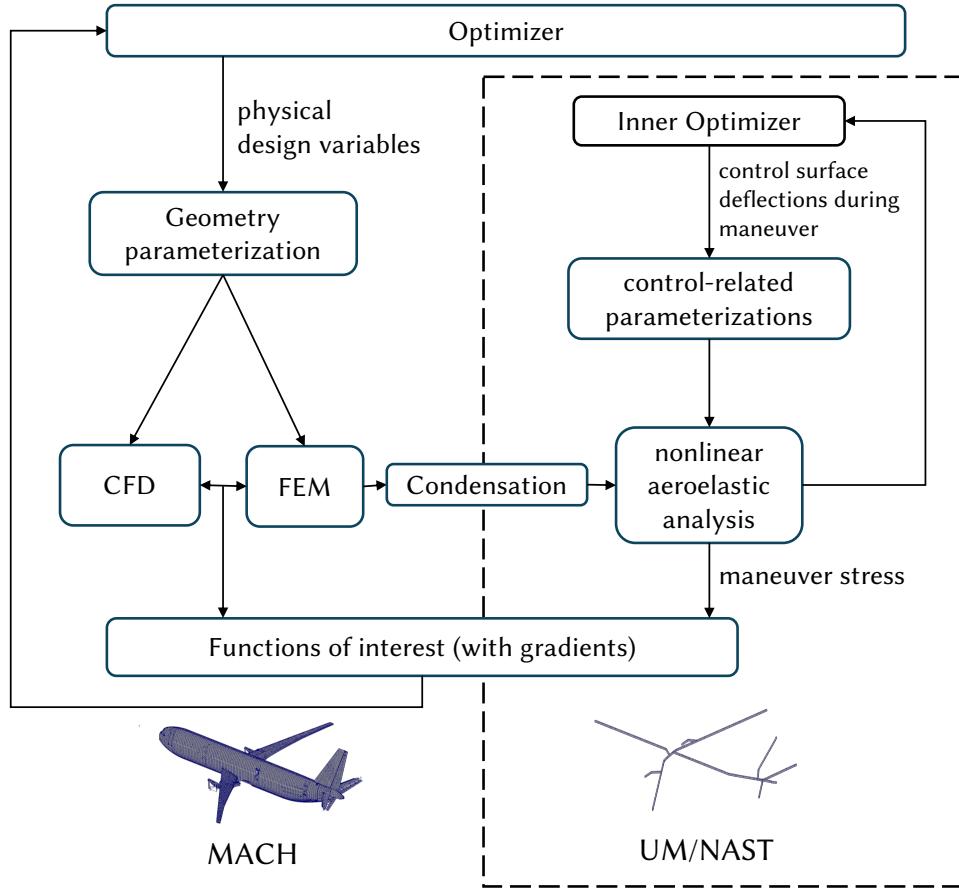


Figure 3.4 MACH-UM/NAST integration for MLA

Code Snippet 3.6 Old interface for MACH-UM/NAST

```
# Create the aeroProblem used by MACH-UM/NAST
ap = AeroProblem(
    name="UMNAST",
    altitude=0,
    mach=0.0
)

# Create a flutter component
FlutterComponent = NastFlutter( ... )

# Create MACH-UM/NAST solver
UMNAST_MACH_SOLVER = UMNAST_MACH(
    EF2SSolver, # EF2S solver instance
    FlutterComponent, # The NastFlutter component created above
    FEASolver, # pyTACS solver instance
)
```

- Concentrated mass offsets, $\hat{\delta}$
- Concentrated mass inertia tensors, $\hat{I}$
- Element stiffness matrices, $\hat{k}$
- Node coordinates, $\hat{p}$

Additionally, the user must provide a list of names of the values within the OpenMDAO model that should be treated as the outputs of interest, $\hat{\mathcal{F}}$. With these changes, a single instance of a MACH-UM/NAST solver can be used to compute the values and derivatives of an arbitrary number of outputs from an arbitrary combination of UM/NAST analyses. For example, in the optimization presented in Section 4.3, a single MACH-UM/NAST solver instance is used to compute a flutter constraint and two separate MLA stress constraints.

Listings 3.6 and 3.7 illustrate how this change affects the user, showing the code the user is responsible for writing to set up a MACH-UM/NAST solver instance with the old and new versions of the framework. The old interface (Listing 3.6) sets up a MACH-UM/NAST solver to compute a flutter constraint, while the new interface (Listing 3.7) example sets up a solver to compute both a flutter and MLA stress constraints.

The derivative computation procedure within MACH-UM/NAST remains largely identical to that described in [148, 188]. In this work, only minor modifications to the code were necessary to support the computation of multiple functions, rather than a single flutter constraint value.

Code Snippet 3.7 New interface for MACH-UM/NAST

```
# Create the aeroProblem used by MACH-UM/NAST, need to define which values are to be
# evaluated as outputs
ap = AeroProblem(
    name="UMNAST",
    altitude=0,
    mach=0.0,
    evalFuncs=["KS_flutter.KS", "Failure_Vd"]
)

# Create a flutter component
FlutterComponent = NastFlutter( ... )

# Create an MLA component
MLAComponent_Vd = MLAComp( ... )

# Assemble components into OpenMDAO group
OpenMDAOModel = om.Group()

OpenMDAOModel.add_subsystem(
    "mla_Vd",
    MLAComponent_Vd,
    promotes_inputs=["*"],
    promotes_outputs=[("y", "Failure_Vd")],
)

# Flutter component needs additional KS component to aggregate damping values
OpenMDAOModel.add_subsystem(
    "flutter",
    FlutterComponent, promotes=["*"],
    proc_group=f"UMNAST_{ptID}"
)

OpenMDAOModel.add_subsystem(
    "KS_flutter",
    om.KSComp(width=len(sampling_points))
)

OpenMDAOModel.connect("damping_aggregates", "KS_flutter.g")

# Create MACH-UM/NAST solver
UMNAST_MACH_SOLVER = UMNAST_MACH(
    EF2SSolver, # EF2S solver instance
    OpenMDAOModel, # The OpenMDAO group created above
    FEASolver, # pyTACS solver instance
    FlutterComponent, # Any one of the OpenMDAO_NAST components used above
)
```

CHAPTER 4

Numerical Results

In this chapter, the methods from Chapter 2 are applied to the following MDO problems:

- Estimating the impact of a gust load alleviation (GLA) system on a blended-wing-body aircraft design (Section 4.1)
- Evaluating the turbulence rejection capability of a supersonic configuration aircraft in the approach to land phase of flight (Section 4.2)
- Estimating the impact of a maneuver load alleviation (MLA) system on a cantilevered wing (Section 4.3)

4.1 Gust load alleviation

In order to illustrate the integration of the proposed combination of sensitivity function parameterization with a constraint on the Bode sensitivity integral (Section 2.1.2) and gust stress constraints (both peak and fatigue, Section 2.2), the optimization of the wingbox of a blended wing body aircraft (BWB), shown in Figure 4.1, is considered.

4.1.1 Optimization problem

The wingbox is assumed to be rectangular and uniform along the wing. It is parameterized in terms of its height, h , width w , side surfaces thickness, t_h , upper and lower surfaces thickness, t_w , and distance between the airfoil leading edge and the reference axis of the wingbox normalized by the chord, r_a , which is defined as the geometric center of the rectangle. In particular, if $r_a = 0$, then the reference axis is located at the leading edge, and if $r_a = 1$, then it is located at the trailing edge.

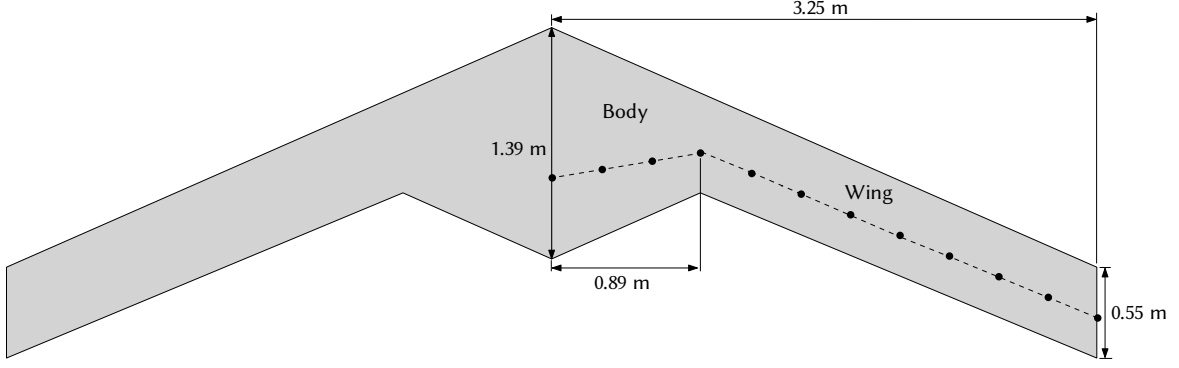


Figure 4.1 BWB model planform, first considered by Su and Cesnik [144]
(dashed line: beam reference axis; markers: beam element ends)

The sensitivity function is parameterized considering a series arrangement of a second-order Butterworth high-pass filter and a peaking filter. It is defined by Equation (2.27), which is repeated here:

$$S(s; \omega_c, \omega_0, g_0, q_0) = \underbrace{\frac{s^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2}}_{\text{high-pass filter}} \underbrace{\frac{s^2 + g_0\omega_0 s/q_0 + \omega_0^2}{s^2 + \omega_0 s/q_0 + \omega_0^2}}_{\text{peaking filter}} \quad (4.1)$$

The design variables of the sensitivity function are the high-pass filter's cutoff frequency ω_c , and the peaking filter's gain g_0 , center frequency ω_0 , and quality factor q_0 .

For the proposed parameterization, the peak of the magnitude of the sensitivity function bounded by the value of g_0 , i.e., $M_s \leq \max(1, g_0)$. Therefore, g_0 is constrained (via a bound in the optimizer) to be less or equal to 2 to ensure the robustness margin suggested in Skogestad and Postlethwaite [159, Section 2.4.3].

The Bode integral is added to the problem as an additional constraint as described in Section 2.1.2. A single sensitivity function is designed for all points where the stress is evaluated. For a stable plant, which is the case considered in this paper due to the flutter constraint, this is equivalent to assuming a diagonal structure in the MIMO matrix of sensitivity functions, with repeated entries in the diagonal.

Additionally, the trim angle-of-attack, α_{1g} , is included as a design variable to find the equilibrium point around which the system is linearized to obtain the state space representation used in the calculations of the stochastic risk metrics.

The objective is to minimize the wingbox mass, which is calculated by multiplying the cross-sectional mass by the length of the wingbox, L .

The aircraft model chosen for this study is a free-flying BWB configuration shown in Figure 4.1, first considered by Su and Cesnik [144], which is simulated using the Univer-

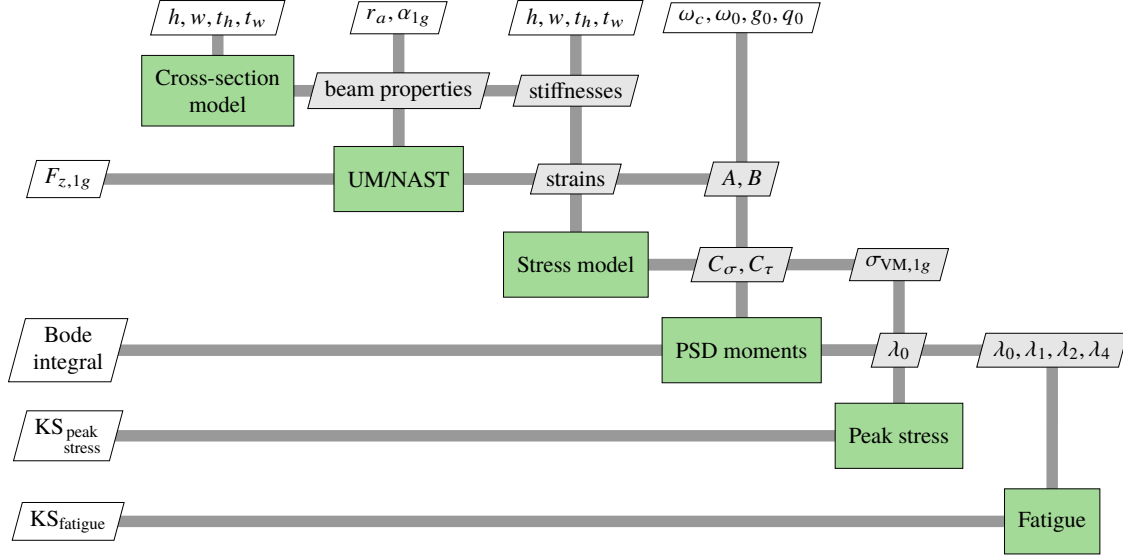


Figure 4.2 XDSM diagram of the proposed gust metrics

sity of Michigan’s Nonlinear Aeroelastic Simulation Toolbox (UM/NAST) [144]. The simple aeroelastic representation is composed of a geometrically exact, strain-based, beam structural model with strip theory aerodynamics. The BWB model consists of three rigid elements with an 80-kg concentrated mass at the nose (the body) and 8 flexible elements with parameterized distributed mass and stiffness (the wing). The aerodynamic coefficients along the span are computed based on the local effective angle of attack using a lookup table for the NACA0012 airfoil. Unsteady aerodynamic effects are considered using the 2D Peters’ theory [205]. It has 176 elastic states, 132 aerodynamic states, and 9 rigid body states, totaling 317 states. This model was previously used for aeroelastic optimization studies in [206]

The calculation process of the constraints is represented graphically in Figure 4.2, making use of the eXtended Design Structure Matrix (XDSM) diagram [207]. First the equivalent beam properties (in the strain basis) of the parameterized cross section are calculated using standard formulae (see, e.g., [187]). These properties are the axial stiffness, the torsional stiffness, the in- and out-of-plane bending stiffnesses, the mass linear density, and the moments of inertia linear densities. These, along with α_{1g} and r_a , are then input into UM/NAST for the calculation of the residual vertical force $F_{z,1g}$, which is used to inform a trim constraint, and for the calculation of the 1-g strains which are input into the stress model described in Section 2.5 to calculate the 1g von Mises aggregated stress $\sigma_{VM,1g}$ for the points of interest (which are also described in Section 2.5). The vertical gust to strain system is linearized using algorithmic differentiation (AD), as described in Chapter 3, to yield

the state space matrices A and B . The stress model is also linearized using AD to obtain the linear relations from strains to normal and shear stresses at the points of interest, C_σ , and C_τ . The state space representation thus obtained is then used to calculate the closed-loop moments of the von Mises stress PSD using Equation (2.69). The left-hand side of the Bode integral relation is also computed by numerical integration.

The integrals involved in these calculations are computed numerically using the trapezoidal rule with 400 equally spaced points and truncated to the interval $\omega \in [0.001, 50]$ Hz. The truncation to this finite interval avoids numerical challenges with infinite integration intervals and with the poles at the origin present in the UM/NAST linearized system due to the states representing the aircraft reference frame position in space, its heading, and the use of quaternions for attitude representation.

The moments of the stress PSDs are used to compute the log expected fatigue life, as described in Section 2.2.4, and, combined with the 1g stresses, the margin of safety peak gust stresses (Section 2.2.3) at the points of interest. The values thus obtained are aggregated using the KS function, Equation (2.101). The following aggregated constraints are considered:

$$\text{KS}_{\text{peak stress}} = \text{KS}(-\text{MS}, 100) \quad (4.2)$$

$$\text{KS}_{\text{fatigue}} = \text{KS}(\log_{10}[\text{design fatigue life}] - \log_{10} \text{E}[\text{fatigue life}], 100) \quad (4.3)$$

The choice of the aggregation parameter (second argument) tries to balance the smoothness of the KS function against introducing too much conservatism in the design.

Although not shown in Figure 4.2, the eigenvalues of the A matrix are calculated and used to inform a flutter metric described in Section 2.6.

Finally, the packaging constraints described in Section 4.1.2 are imposed. They consist of the distance that the wingbox extends vertically over the airfoil outer mold line for both the fore and aft corners, denoted respectively h_{fore} and h_{aft} . Only the upper surface is considered since both the airfoil and the wingbox cross sections are symmetric.

The complete problem is stated as follows, and relevant parameters are given in Table 4.1.

$$\begin{aligned}
& \text{minimize} && 4\rho(t_h h + t_w w)L && \text{(wingbox mass)} && (4.4a) \\
& \text{with respect to} && h, w, t_h, t_w, r_a, \alpha_{1g} && \text{(physical design variables)} && (4.4b) \\
& && \omega_c, \omega_0, g_0, q_0, && \text{(sensitivity function design variables)} && (4.4c) \\
& \text{subject to} && \text{KS}_{\text{peak stress}} \leq 0 && \text{(peak gust stress constraint)} && (4.4d) \\
& && \text{KS}_{\text{fatigue}} \leq 0 && \text{(fatigue constraint)} && (4.4e) \\
& && \text{KS}_{\text{flutter}} \leq 0 && \text{(flutter constraint)} && (4.4f) \\
& && F_{z,1g} = 0 && \text{(trim constraint)} && (4.4g) \\
& && \int_{\omega_{\min}}^{\omega_{\max}} \log |S(j\omega)| d\omega = 0 && \text{(Bode integral relation constraint)} && (4.4h) \\
& && g_0 \leq 2 && \text{(Robustness constraint via eq. 2.28)} && (4.4i) \\
& && \hbar_{\text{fore}} \leq 0 && \text{(fore packaging constraint)} && (4.4j) \\
& && \hbar_{\text{aft}} \leq 0 && \text{(aft packaging constraint)} && (4.4k)
\end{aligned}$$

Table 4.1 Optimization parameters

Parameter	Value	Source
Flight speed	120 m/s	Design requirements
Flight Altitude	20000 ft	
Threshold (flutter)	0.07	
Design fatigue life	100 000 h	
Turbulence scale	2500 ft	Typical choice for altitude
Fatigue gust intensity	10 ft/s	MIL-STD-1797A
Peak gust intensity	70 ft/s	Hoblitt [105, Figure 5.5]
Young's modulus	70 GPa	Aluminum 7075-T6
Density (ρ)	2700 kg/m ³	
Shear modulus	26 GPa	
$\sigma_{\text{allowable}}$	430 MPa	
k	5.80	MIL-HDBK-5J p. 3-409 Fig. 3.7.6.1(d)
$\log_{10} C$	14.86 $\sqrt[k]{\text{ksi}}$	
$\omega_{\min}$	0.001 Hz	Representative range for this aircraft
$\omega_{\max}$	50 Hz	

This MDO problem was implemented in the OpenMDAO optimization framework [146] and solved using SciPy's SLSQP optimizer.¹

¹<https://scipy.org>

4.1.2 Packaging constraint

The shape and position of the rectangular wingbox are also used as design variables, namely the wingbox height h , width w , and the distance of its geometric center to the leading edge of the airfoil, denoted by r_a . Therefore, it is necessary to define constraints that ensure that the wingbox cross section fits inside the airfoil.

A symmetric airfoil is fully defined by a function that maps the chord to the thickness, i.e., $t_c(x/c) : [0, 1] \rightarrow \mathbb{R}$, where t is the airfoil thickness, c is its chord and x is the coordinate that runs along the chord. For a NACA0012 airfoil, this function is

$$t_c(x/c) = 0.12/0.20[0.2969\sqrt{x/c} - 0.1260x/c - 0.3516(x/c)^2 + 0.2843(x/c)^3 - 0.1015(x/c)^4] \quad (4.5)$$

The chordwise limits of the wingbox parameterized as described above are $x = r_a \pm w/2$, both at a perpendicular distance of $h/2$ from the chord. The vertical excesses of these two points, normalized by the chord, are then given by

$$\hbar_{\text{fore}} \triangleq \frac{h}{2c} - t_c(r_a - \frac{w}{2c}) \quad (4.6)$$

$$\hbar_{\text{aft}} \triangleq \frac{h}{2c} - t_c(r_a + \frac{w}{2c}) \quad (4.7)$$

Furthermore, the linear constraints $r_a - \frac{w}{2c} \geq 0.05$ and $r_a + \frac{w}{2c} \leq 0.8$ are included to prevent computing t_c outside of the interior of its domain, where it is differentiable. The derivatives of these constraints are calculated using the complex-step method [208] for precision and ease of implementation.

4.1.3 Verification of derivatives of the proposed gust analysis

In order to numerically verify the procedure for derivative calculation described in Section 2.3, the total derivatives of the proposed constraints with respect to the design variables were calculated using the adjoint method for the arbitrary design point given by: $h = 55$ mm, $w = 165$ mm, $t_h = 1.0$ mm, $t_w = 1.0$ mm, $r_a = 0.30$, $\alpha_{1g} = 1.0$ deg, $\omega_c = 0.002$ Hz, $\omega_0 = 8.0$ Hz, $g_0 = 2.0$, $q_0 = 3.0$. These total derivatives were then compared against ones calculated using finite differences (Table 4.2) and using the direct method (Table 4.3). For the case of the flutter metric, $\text{KS}_{\text{flutter}}$ (Section 2.6), the derivatives calculated using the direct method were also included in Table 4.2 due to the poor matching between direct and adjoint analytical derivatives. A relative step size of 1×10^{-6} was used when computing the forward finite differences. This step size was selected base on a step-size study, and was found to

Table 4.2 Verification of exact derivatives against finite differences (not matching digits are underlined)

(a) Derivatives with respect to physical variables							
of	wrt mode	h (m)	w (m)	t_h (mm)	t_w (mm)	r_a (chord fr.)	α_{1g} (deg)
KS _{peak stress}	Adjoint	-50.13202772	-14.33102859	-0.41305936	-2.08192256	2.49292277	0.29274164
	FD	-50.10366989	-14.32064796	-0.41284357	-2.08149928	2.49432521	0.28969245
KS _{fatigue}	Adjoint	-67.02255179	-19.10264589	-0.47102252	-2.90887711	3.68266921	0.03308077
	FD	-66.97753730	-19.08519747	-0.47072755	-2.90829519	3.68514846	0.02794816
KS _{flutter}	Adjoint	0.02622839	0.00431653	0.00010136	0.00066053	-0.00117842	-0.00026081
	Direct	0.01871342	0.00288407	0.00006185	0.00047869	-0.00107073	-0.00013206
	FD	0.01906667	0.00288560	0.00005308	0.00047050	-0.00105289	-0.00027511
$F_{z,1g}$ (kN)	Adjoint	-0.36771291	-1.31224912	-0.07095762	-0.03193474	0.61907804	1.69712136
	FD	-0.36771793	-1.31224835	-0.07095741	-0.03193466	0.61907809	1.69712117

(b) Derivatives with respect to control-related variables					
of	wrt mode	ω_c (rad/s)	ω_0 (rad/s)	g_0 (-)	q_0 (-)
KS _{peak stress}	Exact	-0.00009598	-0.00494446	0.09541288	-0.04261832
	FD	-0.00009597	-0.00494445	0.09541291	-0.04261827
KS _{fatigue}	Exact	-0.00010181	-0.00967206	0.18211993	-0.07998246
	FD	-0.00010180	-0.00967300	0.18212893	-0.07998605
Bode integral	Exact	-58.96978592	0.46865546	24.50766684	-7.86823638
	FD	-58.96976280	0.46865543	24.50766594	-7.86822896

give the best overall matching between finite differences and analytical derivatives.

The derivative of KS_{flutter} had a poor match between the finite differences approximation and also between the direct and adjoint calculations. The sign of the derivative agrees, but it often has no significant digits in agreement. It is known from the literature [209, Chapter 2] [210] that the spectral abscissa (maximum real part of the eigenvalues) is a differentiable function almost everywhere, with the singularity points being where the eigenvalue with the largest real part (the “active” eigenvalue) changes and where there are repeated eigenvalues. The use of the KS aggregation deals with the first issue but not with the second. The steps in the proposed calculation of KS_{flutter} (Section 2.6) were verified for small random matrices (30×30), and good agreement was obtained.

For the other constraints, there were at least two significant digits in agreement between the derivatives obtained using finite differences and the ones obtained using the adjoint method, which was deemed satisfactory given the loss of precision inherent to finite difference schemes [72, Section 6.4.2]. An agreement of approximately ten significant digits was obtained between the direct and adjoint calculation of the derivatives.

A more qualitative form of derivative verification is presented in Figure 4.3 for the flut-

Table 4.3 Relative errors between direct and adjoint derivatives

(a) With respect to physical variables						
wrt of	h (m)	w (m)	t_h (mm)	t_w (mm)	r_a (chord fr.)	α_{1g} (deg)
$KS_{\text{peak stress}}$	-1.7×10^{-12}	-3.4×10^{-11}	-3.5×10^{-11}	-7.5×10^{-12}	-1.7×10^{-10}	5.7×10^{-14}
KS_{fatigue}	2.1×10^{-12}	1.5×10^{-11}	2.6×10^{-11}	3.3×10^{-12}	1.1×10^{-10}	1.9×10^{-13}
KS_{flutter}	-2.9×10^{-1}	-3.3×10^{-1}	-3.9×10^{-1}	-2.8×10^{-1}	-9.1×10^{-2}	-4.9×10^{-1}

(b) With respect to control-related variables				
wrt of	ω_c (rad/s)	ω_0 (rad/s)	g_0 (-)	q_0 (-)
$KS_{\text{peak stress}}$	0.0	0.0	1.3×10^{-15}	-1.1×10^{-15}
KS_{fatigue}	-4.0×10^{-16}	-1.8×10^{-16}	0.0	-6.9×10^{-16}

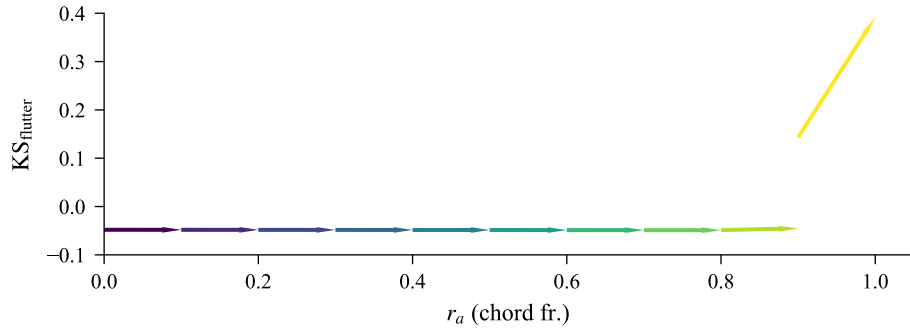
Note: 0.0 indicates identical 64-bit floating point representations

ter metric and Figure 4.4 for the other gust metrics. These figures plot sweeps in r_a of the functions of interest in which the base of the arrow marks the function value at each point, and the tip of each arrow is at the linear approximation for the next sampled point. Therefore, for a linear function, the tip of one arrow should touch the base of the next arrow and, for a function with mild non-linearities, they should be close. The functions in Figure 4.4 are smooth especially in the stable region, $r_a < 0.8$, as is the behavior of the individual eigenvalues shown in Figure 4.3b. However, the aggregated flutter metric, shown in Figure 4.3a shows strong non-linearities due to the eigenvalues close to the origin contributing the most to the aggregated value until very close to flutter, when the “active” eigenvalue changes and this causes a sharp change in the slope of KS_{flutter} .

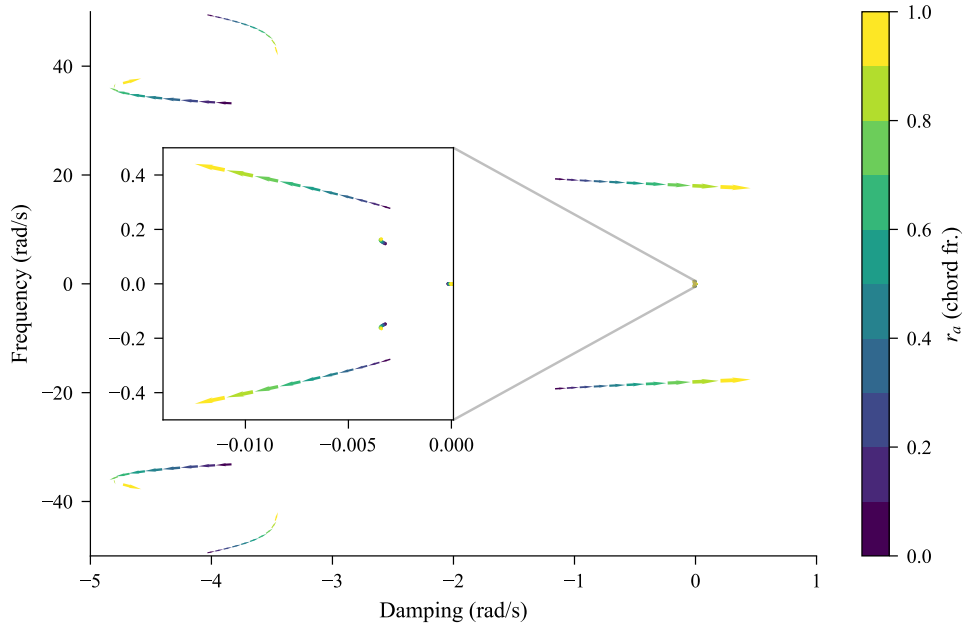
4.1.4 Optimization results

Two simplified versions of the optimization problem proposed in Equation (4.4) converged using the SNOPT optimizer [211], both in open-loop ($S(j\omega) = 1 \forall \omega$) and in closed-loop ($S(j\omega)$ given by Equation (2.27)). The resulting cases are described by Table 4.4, and convergence plots are shown in Figure 4.5. There is substantial increase in computational cost when the sensitivity function is designed, which is partially explained by the increase in the dimensionality of the problem, but also suggests that the specific parameterization of the sensitivity function employed leads to a hard optimization problem. The computational cost for these optimizations is shown in Figure 4.6 A search for other local minimizers was not attempted.

The first simplified version of the optimization problem (cases A and B) includes all



(a) KS-aggregated flutter constraint



(b) Root locus. An inset zooming in on the roots close to the origin was added

Figure 4.3 Flutter metric with derivatives for varying relative position of the airfoil and wingbox

Table 4.4 Optimization problems solved

Case	Design variables		Constraint
	h, w, r_a	$\omega_c, \omega_0, g_0, q_0$	KS_{flutter}
A			✓
B		✓	✓
C	✓		
D	✓	✓	

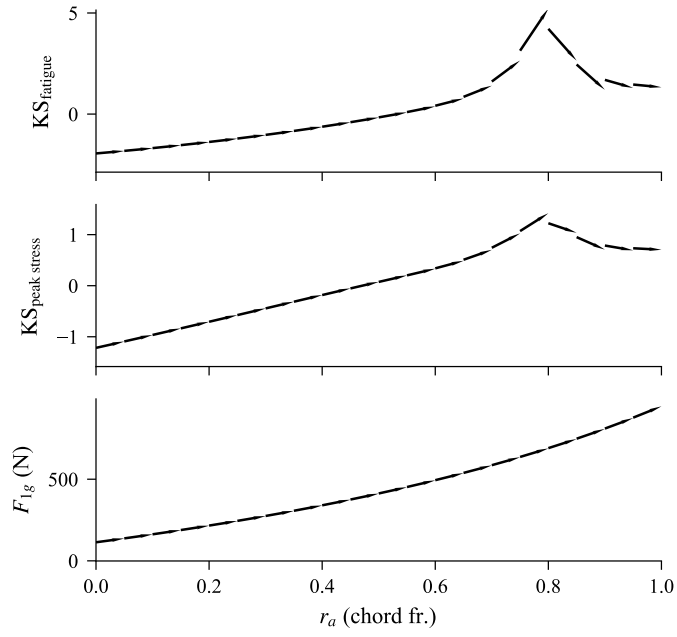


Figure 4.4 Parametric study of the gust stress and trim constraints with varying relative position between the wingbox and the airfoil

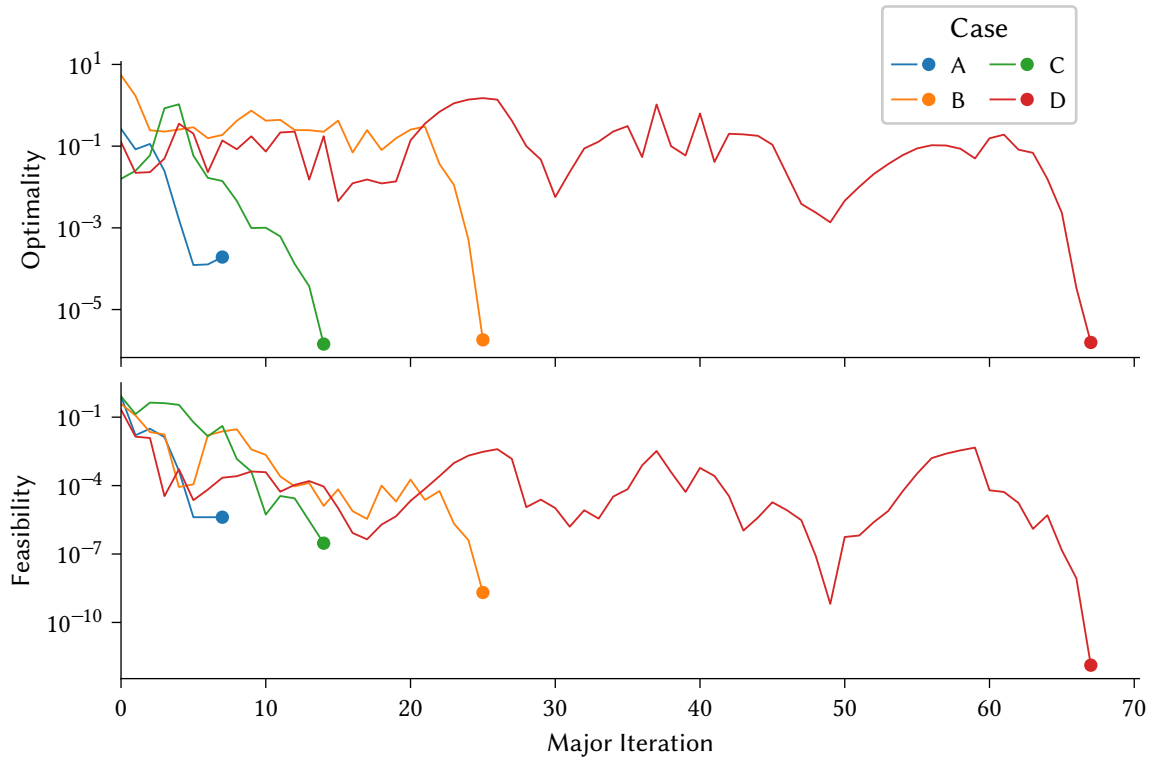


Figure 4.5 Convergence plots for GLA problems

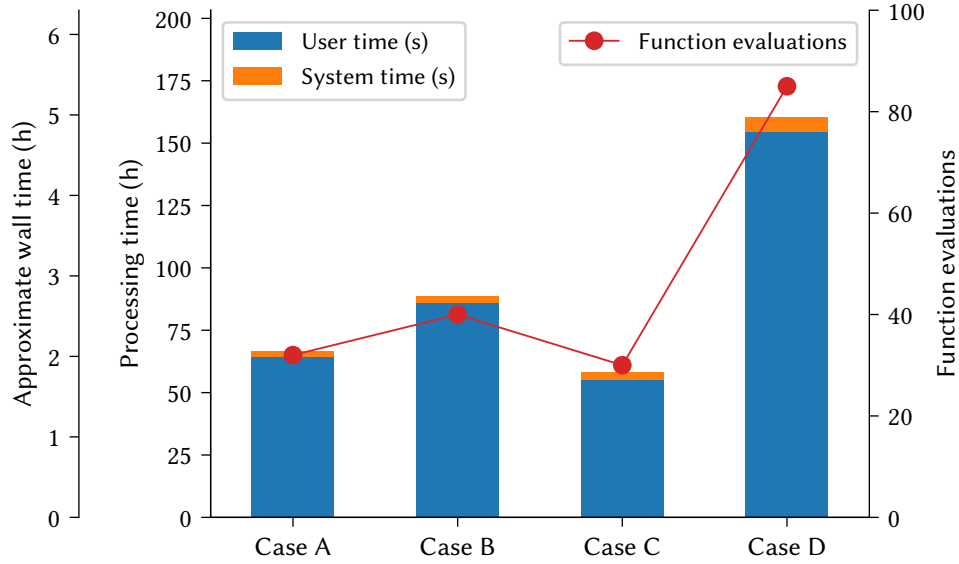


Figure 4.6 Time taken for the GLA MDO, on a system with 4×16 cores Intel® Xeon® CPU E7-4850 v4 @ 2.10GHz. A parallel speedup of from 28× to 37× was observed due to the parallel computation of the AD derivatives in forward mode. The maximum memory usage was of 16GB.

Table 4.5 Optimization results for the problem with fixed wingbox shape (A,B)

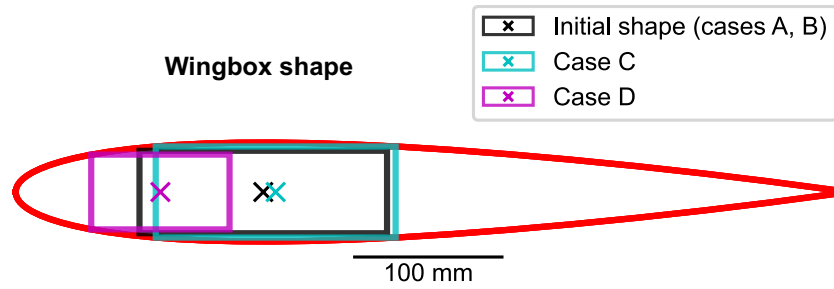
(a) Design variables					
	Open-loop design	Closed-loop design	Δ [%]	Lower bound	Upper bound
t_h (mm)	0.657	0.456	−30.6%	0.100	2.000
t_w (mm)	0.851	0.455	−46.5%	0.100	2.000
α_{1g} (deg)	0.811	0.790	−2.6%	−10.00	10.000
ω_c (Hz)	-	5.000	-	0.000	5.000
ω_0 (Hz)	-	10.00	-	5.000	10.000
g_0 (-)	-	2.000	-	1.000	2.000
q_0 (-)	-	1.139	-	1.000	10.000

(b) Response variables					
	Open-loop design	Closed-loop design	Δ [%]	Lower bound	Upper bound
$KS_{\text{peak stress}}$	1.1×10^{-8}	1.0×10^{-7}	-	-	0
KS_{flutter}	−0.048	−0.048	-	-	0
KS_{fatigue}	−0.346	−0.938	-	-	0
$F_{z,1g}$ (N)	3.5×10^{-7}	-3.2×10^{-6}	-	0	0
Bode integral	-	-8.5×10^{-14}	-	0	0
Mass (kg)	2.591	1.470	−43.3%	-	-

Table 4.6 Optimization results for the problem without the flutter constraint (C, D)

(a) Design variables					
	Open-loop design	Closed-loop design	Δ [%]	Lower bound	Upper bound
h (m)	0.061	0.050	-18.0%	0.027	0.066
w (m)	0.159	0.092	-42.1%	0.005	0.549
t_h (mm)	0.590	0.522	-11.5%	0.100	2.000
t_w (mm)	0.807	0.563	-22.2%	0.100	2.000
r_a (chord fr.)	0.315	0.175	-30.2%	0.100	2.000
α_{1g} (deg)	0.796	0.785	-1.4%	-10.00	10.000
ω_c (Hz)	-	7.451	-	0.000	10.000
ω_0 (Hz)	-	15.00	-	10.000	15.000
g_0 (-)	-	2.000	-	1.000	2.000
q_0 (-)	-	1.000	-	1.000	10.000

(b) Response variables					
	Open-loop design	Closed-loop design	Δ [%]	Lower bound	Upper bound
$KS_{\text{peak stress}}$	4.9×10^{-8}	1.6×10^{-7}	-	-	0
KS_{fatigue}	-0.354	-2.157	-	-	0
$F_{z,1g}$ (N)	6.5×10^{-6}	-4.2×10^{-5}	-	0	0
Bode integral	-	-8.5×10^{-14}	-	0	0
$\dot{h}_{\text{fore}}$ (chord fr.)	1.8×10^{-13}	2.0×10^{-8}	-	-	0
$\dot{h}_{\text{aft}}$ (chord fr.)	1.2×10^{-13}	-0.014	-	-	0
Mass (kg)	2.421	1.144	-52.7%	-	-

**Figure 4.7** Optimal wingbox size and position

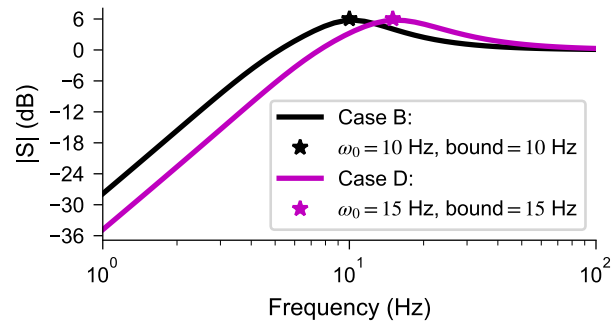


Figure 4.8 Optimal sensitivity function

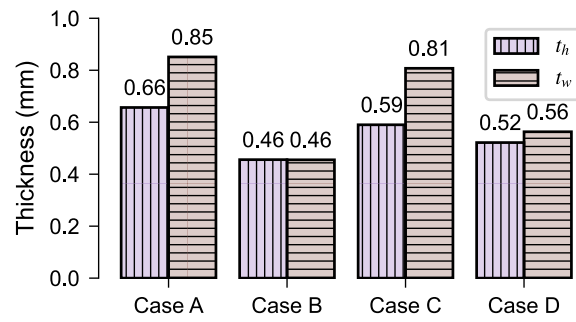


Figure 4.9 Optimal thicknesses

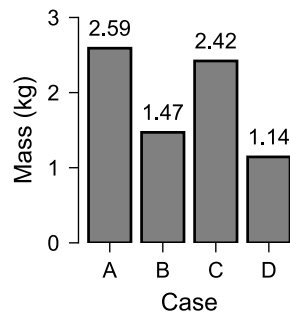


Figure 4.10 Optimal mass (objective)

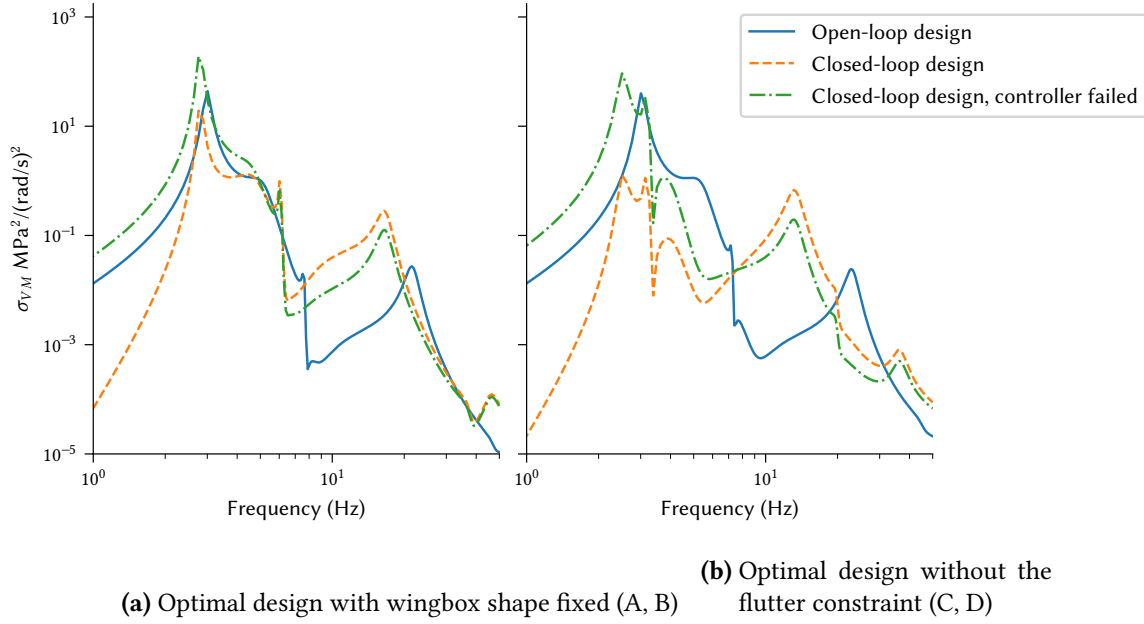


Figure 4.11 Stress PSD for the critical point in peak stress

constraints but drops the design variables that position shape the wingbox, i.e., h , w , and r_a . These variables are fixed at the values described in Section 4.1.3. The optimum design and response for this version of the problem are shown in Table 4.5. The second version of the problem (cases C and D) includes all the design variables but drops the flutter constraint, KS_{flutter} . The optimum design and response for this version of the problem are shown in Table 4.6. Adding the design variables for the wingbox shape caused a reduction in mass for the open-loop designs (i.e., between cases A and C) of 6.6% (Figure 4.10). Cases B and D are not directly comparable because of changes in the bounds for ω_c and ω_0 , which correspond to different assumed bandwidths for the control systems. These results are also shown graphically in Figures 4.7 to 4.10.

Figure 4.11 shows the von Mises stress PSD for the optimal designs found. The closed-loop design exhibits a reduction in the first peak in the PSD (around 2 Hz) thanks to the sensitivity function, which in turn allows the peak between 10–20 Hz to be moved closer to 10 Hz, where the von Kármán PSD has more energy and therefore makes the peak higher, while maintaining the same level of peak stress.

Figure 4.11 also highlights what would happen to the closed-loop design in case of a controller failure by showing the stress PSD of the closed-loop design calculated with the sensitivity function assumed to be one at all frequencies. The stress levels would increase substantially for low frequencies, where the value of the sensitivity function is smaller than one, and decrease slightly at high frequencies, where the sensitivity function is greater than

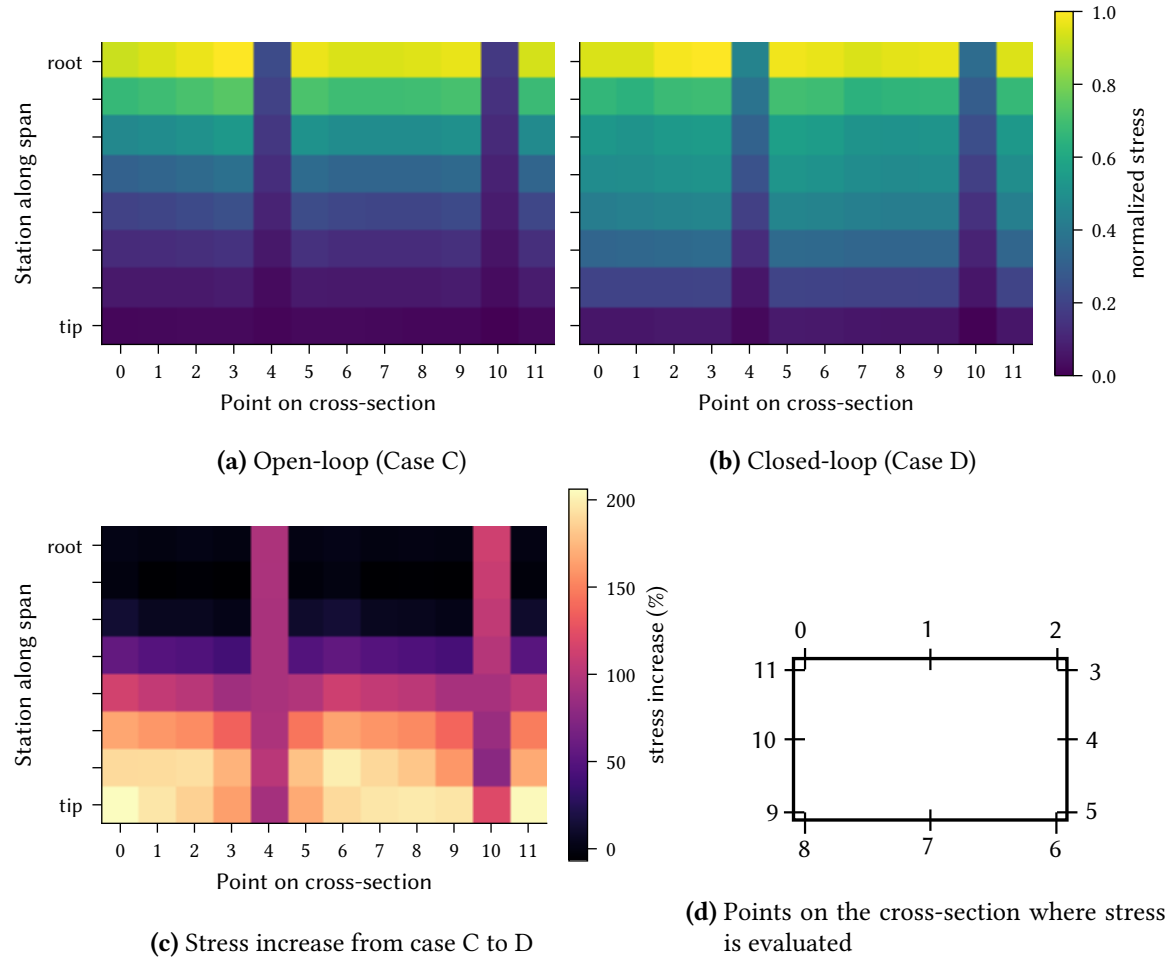
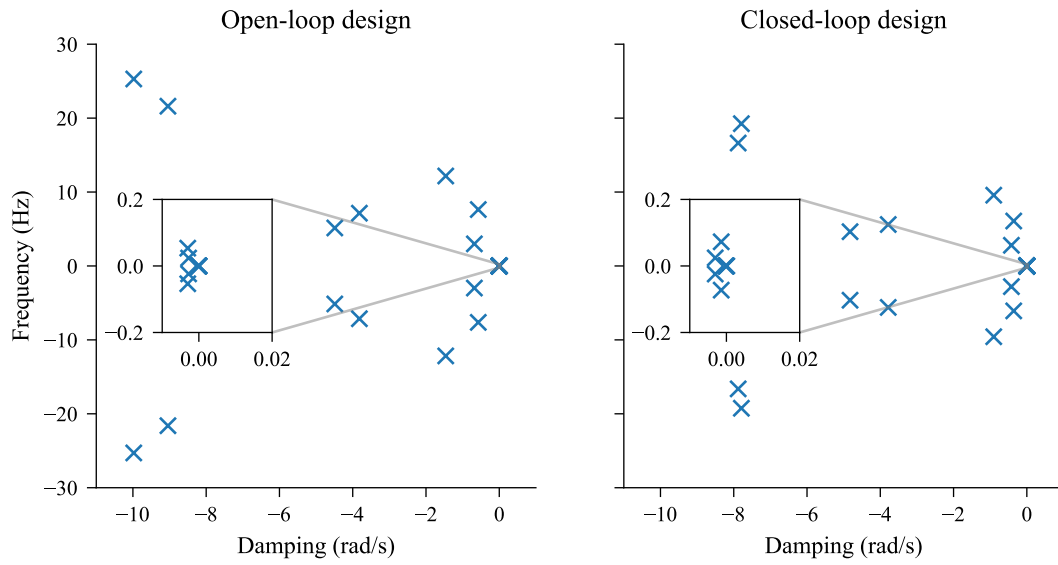


Figure 4.12 Peak stress at different points in the wingbox

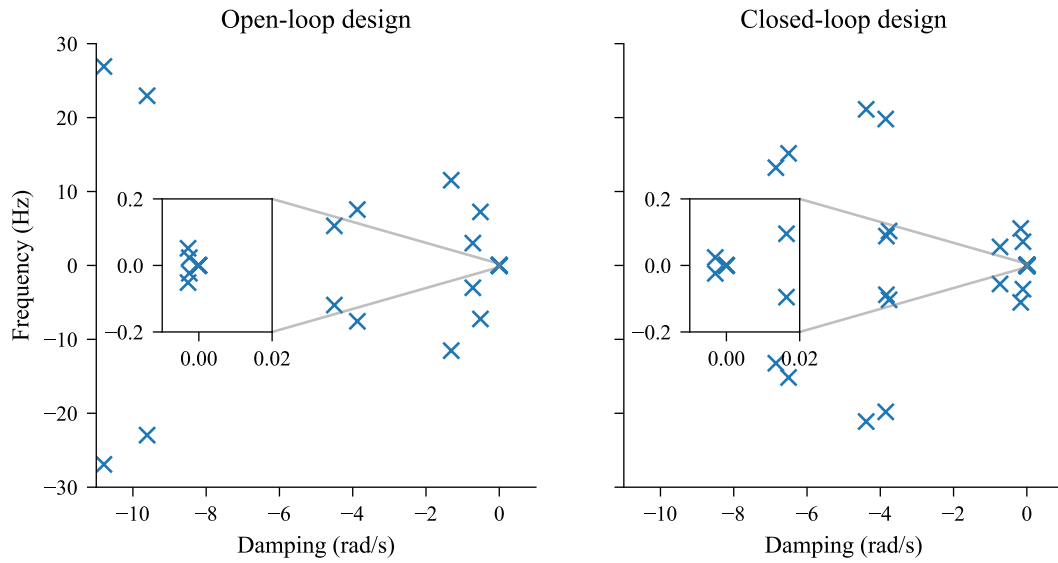
one. Even though this was not done here, a case with a failed controller must be included in the optimization of a realistic aircraft with GLA. This can be done by including an open-loop analysis in the closed-loop MDO problem, but with a reduced safety factor, e.g., from 1.5 to 1.3 as was done in [212].

The stress distribution in the wingbox also changes, as can be seen in Figure 4.12. In the closed-loop case, the stress becomes higher towards the tip of the wing. If the wingbox had variable thickness along the span, the stresses would be more constant throughout the wingbox, and it is likely that the difference in mass between the open- and the closed-loop design would be smaller.

In all cases, the peak stress constraint is active while the fatigue constraint is not, despite the stress having greater magnitudes at higher frequencies for the closed-loop cases (B and D). The open-loop poles for the various designs are shown in Figure 4.13. Designs A, B, and C are open-loop stable but D is not. This highlights the desirability of including the flutter



(a) Optimal design with fixed wingbox shape (A, B)



(b) Optimal design without the flutter constraint (C, D)

Figure 4.13 Pole map for the different designs, with insets zooming in close to the origin

Table 4.7 Derivatives at the optimum point with fixed wingbox shape

(a) Open-loop design (case A)				
w.r.t. of	t_h/t_h^*	t_w/t_w^*	$\alpha_{1g}/\alpha_{1g}^*$	
KS _{dynamic stress}	-0.422	-1.594	0.164	
KS _{fatigue}	-0.450	-2.990	0.031	
KS _{flutter}	0.000	0.000	0.000	
$F_{z,1g}$ (kN)	-0.107	-0.026	1.341	
Mass (kg)	0.530	2.061	0.000	
(b) Closed-loop design (case B)				
w.r.t. of	t_h/t_h^*	t_w/t_w^*	$\alpha_{1g}/\alpha_{1g}^*$	
KS _{dynamic stress}	-0.392	-1.128	0.216	
KS _{fatigue}	-0.547	-1.798	0.031	
KS _{flutter}	0.000	-0.001	0.000	
$F_{z,1g}$ (kN)	-0.078	0.034	1.197	
Mass (kg)	0.368	1.103	0.000	
w.r.t. of	ω_c/ω_c^*	ω_0/ω_0^*	g_0/g_0^*	q_0/q_0^*
KS _{dynamic stress}	-1.229	-0.139	0.321	-0.178
KS _{fatigue}	-3.252	-0.273	1.037	-0.543
Bode integral	-69.755	57.538	135.074	-58.296

constraint in the problem.

4.1.5 Sensitivity analysis

Since the Jacobian of the optimization problems is readily available, a sensitivity analysis was conducted at the optimum, and the results are shown in Tables 4.7 and 4.8, normalized by the optimum value of the design variables (denoted by a $\star$ superscript). These figures are color-coded based on the value of the derivatives, with stronger shades of red denoting larger positive values and shades of blue denoting negative values.

These derivatives show that an increase in the dimensions of the wingbox causes a reduction of peak stress and an increase in mass. However, increasing the height and width of the wingbox decreases the fatigue damage in the open-loop optimum design of case C but mildly increases it in the closed-loop optimum design of case D. This is likely due to

Table 4.8 Derivatives at the optimum point without the flutter constraint

(a) Open-loop design (case C)						
w.r.t. of	h/h^*	w/w^*	t_h/t_h^*	t_w/t_w^*	r_a/r_a^*	$\alpha_{1g}/\alpha_{1g}^*$
KS _{dynamic stress}	-2.505	-2.130	-0.478	-1.665	0.846	0.169
KS _{fatigue}	-4.391	-3.648	-0.522	-3.185	1.596	0.036
$F_{z,1g}$ (kN)	-0.102	-0.330	-0.129	-0.032	0.210	1.344
Mass (kg)	0.526	1.895	0.526	1.895	0.000	0.000
(b) Closed-loop design (case D)						
w.r.t. of	h/h^*	w/w^*	t_h/t_h^*	t_w/t_w^*	r_a/r_a^*	$\alpha_{1g}/\alpha_{1g}^*$
KS _{dynamic stress}	-0.835	-0.988	-0.404	-0.732	0.399	0.208
KS _{fatigue}	0.196	0.227	0.012	-0.768	0.862	0.036
$F_{z,1g}$ (kN)	0.341	-0.230	-0.111	0.142	0.151	1.084
Mass (kg)	0.382	0.762	0.382	0.762	0.000	0.000
w.r.t. of	ω_c/ω_c^*	ω_0/ω_0^*	g_0/g_0^*	q_0/q_0^*		
KS _{dynamic stress}	-0.347	-0.132	0.340	-0.044		
KS _{fatigue}	-1.430	-0.683	1.958	-0.243		
Bode integral	-103.882	65.774	189.270	-70.346		

the competing effects that increasing stiffness shifts the stresses to higher frequencies while adding material decreases the stresses in all frequencies. Moving the wingbox towards the leading edge is favorable for both peak stresses and fatigue metrics. Finally, the flutter metric is largely insensitive to changes in the design because designs A and B are flutter-free, and therefore, the eigenvalues with the largest real part are the ones fixed at the origin, related to the position and heading of the aircraft reference frame, and they dominate the KS aggregation.

Notably, the open-loop designs (cases A and C) are more sensitive to changes in the structural design variables than the closed-loop designs (cases B and D), supporting the well-known idea that adding feedback control to the system makes its performance less sensitive to variations in the plant.

4.2 Turbulence rejection during final approach

One particular situation where gust disturbances are a critical factor in guaranteeing the safety of new aircraft designs is the low-speed flight phases of commercial supersonic aircraft. This type of aircraft has received renewed attention in recent years due to the development of new technology that reduces the loudness of the sonic boom, thereby promising to enable supersonic flight over land (see e.g., [213] and references therein). The culmination of these efforts is the design and fabrication of the X-59 low-boom supersonic aircraft demonstrator, which NASA is planning to fly over U.S. communities, starting in 2024, to evaluate the public's response to the improved sonic-boom signature [214, 215].

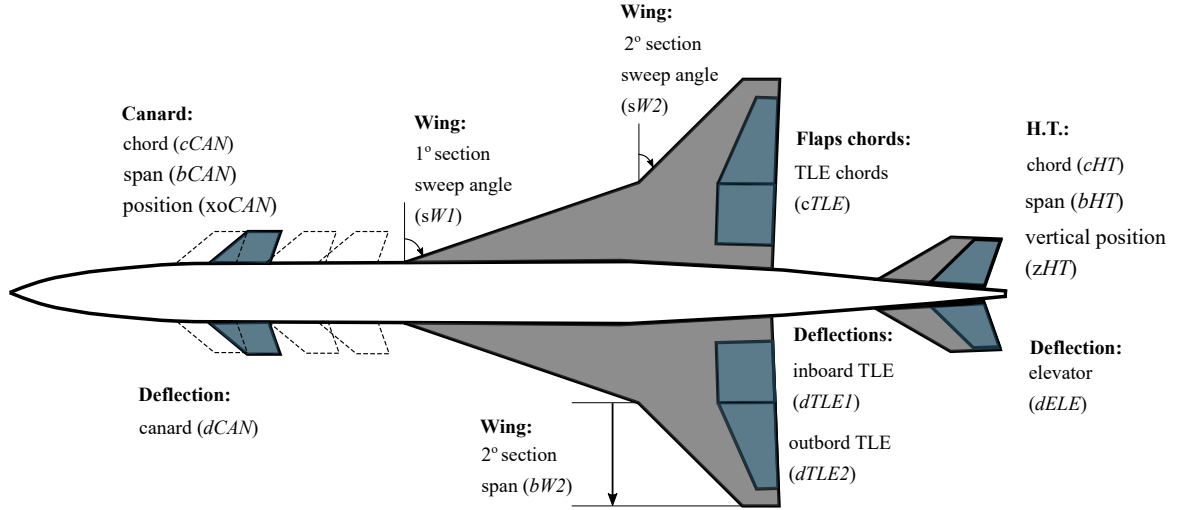
However, to be viable and certifiable, a supersonic aircraft must also operate satisfactorily at low-speed flight conditions, especially during take-off, initial climbing, final approach, and landing. These flight regimes are very different from the supersonic cruise for which these aircraft are usually optimized, and, due to the lower altitude, while flying in these conditions, the aircraft is more susceptible to weather phenomena such as gusts.

In this section, the closed-loop performance of a supersonic configuration aircraft flying the approach to land phase of flight under stochastic atmospheric turbulence is analyzed. The effect of the pilot or autopilot closing the loop is modeled by parameterizing the sensitivity function, as was done in Section 4.1.

4.2.1 Aircraft model

The proposed risk metrics are used to analyze a longitudinal model of a rigid aircraft with steady aerodynamics in the approach-to-landing condition. The choice of a rigid

model is based on previous studies that showed that flexibility effects do not greatly impact the global behavior of the Supersonic Configuration at Low Speeds (SCALOS) aircraft [216]. The aerodynamic coefficients are obtained from the parameterized surrogate model of the SCALOS aircraft described in [217], and the CG position is moved aft of the datum by 1% of the chord. A schematic drawing of the aircraft along with a description of the available design variables is shown in Figure 4.14. This section presents both the nonlinear and lin-



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Figure 4.14 SCALOS UW-S-15W configuration with the parameterization variables highlighted

earized models employed. The nonlinear model is used for trim, while the linearized model around the trimmed condition is used to compute the control-related metrics. The inputs to the models are vertical gust velocity, w , and elevator deflection, η , while the outputs are the displacement normal to the approach ramp, ΔGL (which depends on the approach ramp angle γ_R) and the angle of attack, α . The states are the velocity components in the body axis, U and W , pointing forward and down respectively, the pitch rate Q and the pitch attitude θ . The notation used is based on [218, Chapter 2].

Nonlinear model

The nonlinear model is given by the following system of ordinary differential equations (ODEs):

$$\dot{U} = -QW - g \sin \theta + (X + T)/m \quad (4.8a)$$

$$\dot{W} = QU + g \cos \theta + Z/m \quad (4.8b)$$

$$\dot{Q} = M/J \quad (4.8c)$$

$$\dot{\theta} = Q \quad (4.8d)$$

$$\frac{d}{dt} \Delta GL = V^{GL} = U \sin(\theta - \gamma_R) - W \cos(\theta - \gamma_R) \quad (4.8e)$$

where g is the acceleration of gravity; X and Z are the aerodynamic forces pointing forward and down, respectively; m is the mass of the aircraft; M is the aerodynamic pitch-up moment; J is the aircraft's moment of inertia; and V^{GL} is the velocity component normal to the approach ramp.

This ODE system is linked to the surrogate aerodynamic model via the auxiliary equations:

$$\alpha = \tan^{-1} \left(\frac{W - w \cos(\theta)}{U + w \sin(\theta)} \right) \quad (4.9a)$$

$$\bar{q} = \frac{1}{2} \rho \left((U + w \sin(\theta))^2 + (W - w \cos(\theta))^2 \right) \quad (4.9b)$$

$$X = \sin(\alpha) \bar{q} SC_L(\alpha, Q, \eta) - \cos(\alpha) \bar{q} SC_D(\alpha, \eta) \quad (4.9c)$$

$$Z = -\cos(\alpha) \bar{q} SC_L(\alpha, Q, \eta) - \sin(\alpha) \bar{q} SC_D(\alpha, \eta) \quad (4.9d)$$

$$M = \bar{q} S c C_M(\alpha, Q, \eta) \quad (4.9e)$$

where $\bar{q}$ is the dynamic pressure; ρ is the air density; S is the wing reference area; c is the mean aerodynamic chord; and C_L , C_D and C_M are the aerodynamic coefficients of lift, drag, and moment respectively.

Linearized model

The nonlinear equations of motion derived for the body-axis frame and presented in the previous section can be linearized for any particular equilibrium point (trim point), to yield a state space representation in the form of (Section 2.3.1), where $\mathbf{x} = [U \ W \ Q \ \theta \ \Delta GL]^T$ is the state vector, $d = w$ is the disturbance, $u = \eta$ is the control input, $y = \Delta GL$ is the output used for feedback, and $z = \alpha$ is the output not used for feedback. The state space's matrices

are given by

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c|cc} A & B_d & B_u \\ \hline C_y & D_{yd} & D_{yu} \\ C_z & D_{zd} & D_{zu} \end{array} \right] = \left[\begin{array}{ccccc|cc} X_U & X_W & X_Q - W_{\text{trim}} & -g \cos \theta_{\text{trim}} & 0 & X_w & X_\eta \\ Z_U & Z_W & Z_Q + U_{\text{trim}} & -g \sin \theta_{\text{trim}} & 0 & Z_w & Z_\eta \\ M_U & M_W & M_Q & 0 & 0 & M_w & M_\eta \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ V_U^{\text{GL}} & V_W^{\text{GL}} & 0 & V_\theta^{\text{GL}} & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \alpha_U & \alpha_W & 0 & 0 & 0 & \alpha_w & 0 \end{array} \right] \quad (4.10)$$

The elements of these matrices include the dimensional derivatives defined in Table 4.9, which, in turn, are functions of the model parameters, the trim condition (denoted by variables with the subscript “trim”), the aerodynamic coefficients, and the aerodynamic derivatives. The terms related to $C_{D\alpha}$ and $C_{D\eta}$ are neglected since their inclusion does not substantially change the dynamics of the system.

4.2.2 Integration limits

Due to the presence of the non-zero passthrough term D_{zd} , the output z will not have a finite second moment. However, the existence of this term is due to modeling simplifications such as not considering the unsteady aerodynamic effects, especially the roll-off of the gust-to-outputs transfer functions when the gust spacial frequency gets close to the dimensions of the wing of the aircraft (i.e., the “averaging of gust velocity” over chord and span), the faster decrease of the PSD in the “viscous subrange”, and the effect of flexibility in the aircraft structure at high frequencies. These simplifications are discussed in [105, Appendix E]. Due to these modeling inaccuracies at high frequencies, the integration of the PSDs to obtain the spectral moments of the signals of interest is not carried out to infinity, but rather to the frequency of 20 Hz. Replacing the surrogate aerodynamic model of [217] by linearized unsteady vortex lattice method (UVLM) aerodynamics enhanced with vortex lift, such as the one described in [219] will help to eliminate some of these simplifications and is left as future work.

4.2.3 Sensitivity function design

The sensitivity function is parameterized by a second-order high-pass filter and a peaking filter, as per Equation (2.27). A simplified optimization problem is considered for computing the parameters of the sensitivity function. It consists of minimizing the sum of the

Table 4.9 Dimensional derivatives for linearized longitudinal rigid aircraft model

A matrix	
$X_U = \frac{1}{m} S \rho \left((-C_D U_{\text{trim}} - \frac{1}{2} W_{\text{trim}} C_{L_{\text{trim}}}) \cos(\alpha_{\text{trim}}) + (-\frac{1}{2} C_D W_{\text{trim}} + U_{\text{trim}} C_{L_{\text{trim}}} - \frac{1}{2} W_{\text{trim}} C_{L_{\alpha}}) \sin(\alpha_{\text{trim}}) \right)$	
$X_W = \frac{1}{m} S \rho \left((-C_D W_{\text{trim}} + \frac{1}{2} U_{\text{trim}} C_{L_{\text{trim}}}) \cos(\alpha_{\text{trim}}) + (\frac{1}{2} C_D U_{\text{trim}} + \frac{1}{2} U_{\text{trim}} C_{L_{\alpha}} + W_{\text{trim}} C_{L_{\text{trim}}}) \sin(\alpha_{\text{trim}}) \right)$	
$X_Q = \frac{1}{m} \bar{q}_{\text{trim}} S C_{L_Q} \sin(\alpha_{\text{trim}})$	
$Z_U = \frac{1}{m} S \rho \left((-C_D U_{\text{trim}} - \frac{1}{2} W_{\text{trim}} C_{L_{\text{trim}}}) \sin(\alpha_{\text{trim}}) + (\frac{1}{2} C_D W_{\text{trim}} - U_{\text{trim}} C_{L_{\text{trim}}} + \frac{1}{2} W_{\text{trim}} C_{L_{\alpha}}) \cos(\alpha_{\text{trim}}) \right)$	
$Z_W = \frac{1}{m} S \rho \left((-C_D W_{\text{trim}} + \frac{1}{2} U_{\text{trim}} C_{L_{\text{trim}}}) \sin(\alpha_{\text{trim}}) + (-\frac{1}{2} C_D U_{\text{trim}} - \frac{1}{2} U_{\text{trim}} C_{L_{\alpha}} - W_{\text{trim}} C_{L_{\text{trim}}}) \cos(\alpha_{\text{trim}}) \right)$	
$Z_Q = -\frac{1}{m} \bar{q}_{\text{trim}} S C_{L_Q} \cos(\alpha_{\text{trim}})$	
$M_U = -\frac{1}{2J} S c \rho W_{\text{trim}} C_{M_{\alpha}}$	
$M_W = \frac{1}{2J} S c \rho U_{\text{trim}} C_{M_{\alpha}}$	
$M_Q = \frac{1}{J} \bar{q}_{\text{trim}} S c C_{M_Q}$	
$V_U^{GL} = \sin(\theta_{\text{trim}} - \gamma_R)$	
$V_W^{GL} = -\cos(\theta_{\text{trim}} - \gamma_R)$	
$V_{\theta}^{GL} = U_{\text{trim}} \cos(\theta_{\text{trim}} - \gamma_R) + W_{\text{trim}} \sin(\theta_{\text{trim}} - \gamma_R)$	
B matrix	
$X_w = \frac{1}{m} S \rho \left(-\frac{1}{2} C_{L_{\text{trim}}} (U_{\text{trim}} \cos(\theta_{\text{trim}}) + W_{\text{trim}} \sin(\theta_{\text{trim}})) \cos(\alpha_{\text{trim}}) \right. \\ \left. + (-\frac{1}{2} C_D (U_{\text{trim}} \cos(\theta_{\text{trim}}) + W_{\text{trim}} \sin(\theta_{\text{trim}})) - \frac{1}{2} C_{L_{\alpha}} (U_{\text{trim}} \cos(\theta_{\text{trim}}) + W_{\text{trim}} \sin(\theta_{\text{trim}}))) \sin(\alpha_{\text{trim}}) \right. \\ \left. + (-C_D \cos(\alpha_{\text{trim}}) + C_{L_{\text{trim}}} \sin(\alpha_{\text{trim}})) (U_{\text{trim}} \sin(\theta_{\text{trim}}) - W_{\text{trim}} \cos(\theta_{\text{trim}})) \right)$	
$Z_w = \frac{1}{m} S \rho \left(-\frac{1}{2} C_{L_{\text{trim}}} (U_{\text{trim}} \cos(\theta_{\text{trim}}) + W_{\text{trim}} \sin(\theta_{\text{trim}})) \sin(\alpha_{\text{trim}}) \right. \\ \left. + (\frac{1}{2} C_D (U_{\text{trim}} \cos(\theta_{\text{trim}}) + W_{\text{trim}} \sin(\theta_{\text{trim}})) + \frac{1}{2} C_{L_{\alpha}} (U_{\text{trim}} \cos(\theta_{\text{trim}}) + W_{\text{trim}} \sin(\theta_{\text{trim}}))) \cos(\alpha_{\text{trim}}) \right. \\ \left. - (C_D \sin(\alpha_{\text{trim}}) + C_{L_{\text{trim}}} \cos(\alpha_{\text{trim}})) (U_{\text{trim}} \sin(\theta_{\text{trim}}) - W_{\text{trim}} \cos(\theta_{\text{trim}})) \right)$	
$M_w = \frac{1}{J} S c \rho \left(-\frac{1}{2} C_{M_{\alpha}} (U_{\text{trim}} \cos(\theta_{\text{trim}}) + W_{\text{trim}} \sin(\theta_{\text{trim}})) - W_{\text{trim}} \cos(\theta_{\text{trim}}) \right)$	
$Z_{\eta} = -\frac{1}{m} \bar{q}_{\text{trim}} S C_{L_{\eta}} \cos(\alpha_{\text{trim}})$	
$X_{\eta} = \frac{1}{m} \bar{q}_{\text{trim}} S C_{L_{\eta}} \sin(\alpha_{\text{trim}})$	
$M_{\eta} = \frac{1}{J} \bar{q}_{\text{trim}} S c C_{M_{\eta}}$	
C matrix	D matrix
$\alpha_U = -\frac{W_{\text{trim}}}{U_{\text{trim}}^2 + W_{\text{trim}}^2}$	$\alpha_w = -\frac{U_{\text{trim}} \cos(\theta_{\text{trim}}) + W_{\text{trim}} \sin(\theta_{\text{trim}})}{U_{\text{trim}}^2 + W_{\text{trim}}^2}$
$\alpha_W = \frac{U_{\text{trim}}}{U_{\text{trim}}^2 + W_{\text{trim}}^2}$	

probabilities of the signals of interest exceeding pre-established limits while enforcing a limit on g_0 for the robustness, as discussed in the previous section, and a constraint on the Bode integral relation. It can be stated as:

$$\begin{aligned}
& \text{minimize} \quad \mathbb{P}(\{\Delta\text{GL} < -18 \text{ m}\} \cup \{\Delta\text{GL} > 18 \text{ m}\}) + \mathbb{P}(\{\alpha < -10 \text{ deg}\} \cup \{\alpha > 40 \text{ deg}\}) \\
& \quad + \mathbb{P}(\{\eta < -30 \text{ deg}\} \cup \{\eta > 30 \text{ deg}\}) + \mathbb{P}(\{\dot{\eta} < -60 \text{ deg/s}\} \cup \{\dot{\eta} > 60 \text{ deg/s}\}) \\
& \text{with respect to} \quad \omega_c, \omega_0, g_0, q_0 \\
& \text{such that} \quad g_0 \leq 2 \\
& \quad \int_0^\infty \log |S(j\omega)| d\omega = \pi \sum_{k=0}^{n_u} \Re(p_k^+)
\end{aligned}$$

where the sensitivity function is parameterized as proposed in Equation (2.27). The von Kármán turbulence model parameters were extracted from the MIL-F-8785C low altitude model, considering an altitude of 500 ft, which in this model is equal to the characteristic scale of the turbulence, and a wind level at 20 ft of 45 knots, which is severe. This resulted in a turbulence intensity of 7.6 ft/s. The allowable glideslope deviation was calculated for a gate at 500 ft with an allowed angular deviation of 1 “dot”, which corresponds to 0.35 deg [220–222]. This approximately results in the allowable linear deviation of ± 18 m

An XDMS of the complete analysis and optimization problem proposed is shown in Figure 4.15.

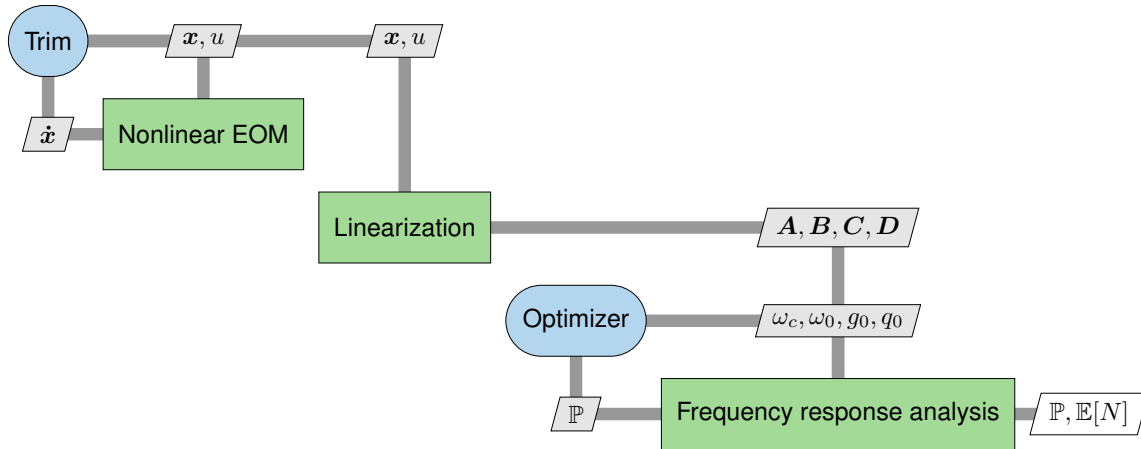


Figure 4.15 XDMS for the turbulence rejection analysis

4.2.4 Trim condition and poles of the linearized system

The nonlinear Equations of Motion 4.8 were solved using NASA OpenMDAO's Newton solver [146] in order to find an equilibrium condition for flying an approach ramp of -3 deg with a 20 deg angle of attack. The equilibrium states along with related quantities are:

$$\begin{array}{lll}
 U_{\text{trim}} = 93 \text{ m/s} & W_{\text{trim}} = 34 \text{ m/s} & T_{\text{trim}} = 25 \text{ kN} \\
 \theta_{\text{trim}} = 17 \text{ deg} & \alpha_{\text{trim}} = 20 \text{ deg} & \eta_{\text{trim}} = -17 \text{ deg} \\
 C_{D\text{trim}} = 0.21 & C_{L\text{trim}} = 0.63 & \\
 C_{L\alpha\text{trim}} = 1.58 & C_{L\eta\text{trim}} = 0.08 & C_{LQ\text{trim}} = -0.03 \\
 C_{M\alpha\text{trim}} = -0.01 & C_{M\eta\text{trim}} = -0.15 & C_{MQ\text{trim}} = -0.02
 \end{array}$$

The dynamical system was linearized around this point using Equation (4.10) and Table 4.9, which resulted in a open-loop system with the following poles:

$$\begin{aligned}
 p_1 &= 0.00 \text{ rad/s} \\
 p_2 &= (-0.19 + j0.22) \text{ rad/s} \\
 p_3 &= (-0.19 - j0.22) \text{ rad/s} \\
 p_4 &= (-0.01 + j0.09) \text{ rad/s} \\
 p_5 &= (-0.01 - j0.09) \text{ rad/s}
 \end{aligned}$$

The first pole is a pure integrator and is related to the integration of the velocity of glideslope deviation into the glideslope deviation itself, the first conjugate pair corresponds to the short-period mode, and the second pair is related to the phugoid mode. Figure 4.16a shows the Bode plots for the open-loop linearized system.

4.2.5 Turbulence rejection metrics

The optimization problem for sensitivity function design shown in Section 4.2.3 was solved using SciPy's SLSQP optimizer [223] to a tolerance of 10^{-8} , which converged in 19 iterations. The optimization took 50 s of processing time and 25 s of wall time on a system with 4×16 cores Intel® Xeon® CPU E7-4850 v4 @ 2.10GHz. The optimization used a maximum of 150 MB of memory. The optimal values for the sensitivity function parameters were $\omega_c = 0.58 \text{ rad/s}$, $\omega_0 = 1.14 \text{ rad/s}$, $g_0 = 1.43$, $q_0 = 0.62$. The calculated turbulence rejection metrics for the trim condition described in the previous section and the optimized sensitivity function defined are shown in Table 4.10, along with relevant statistical parameters and

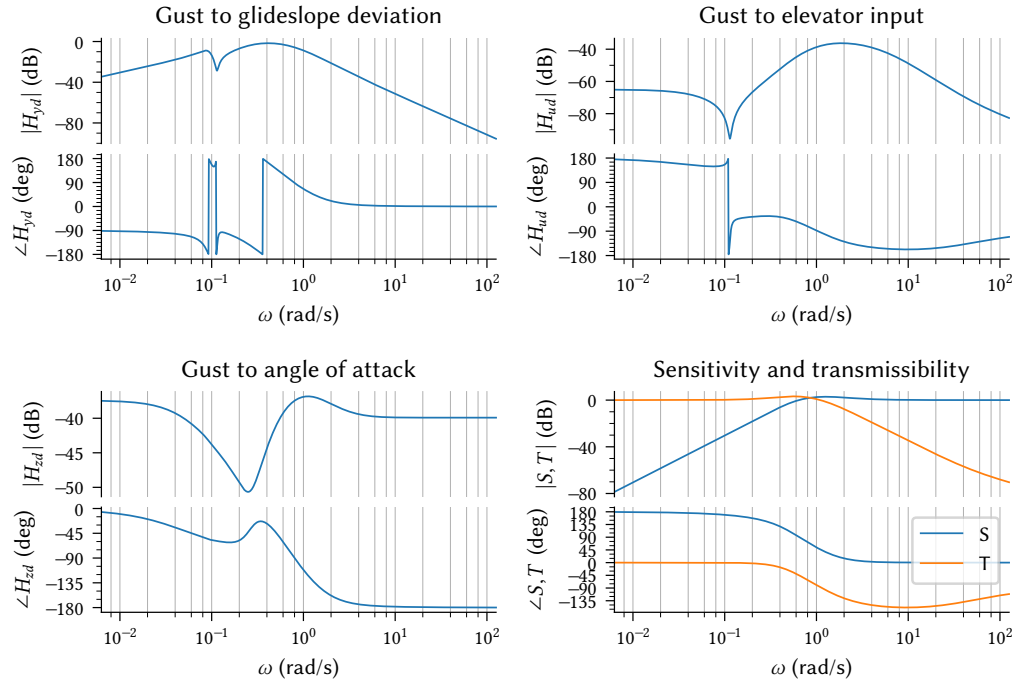
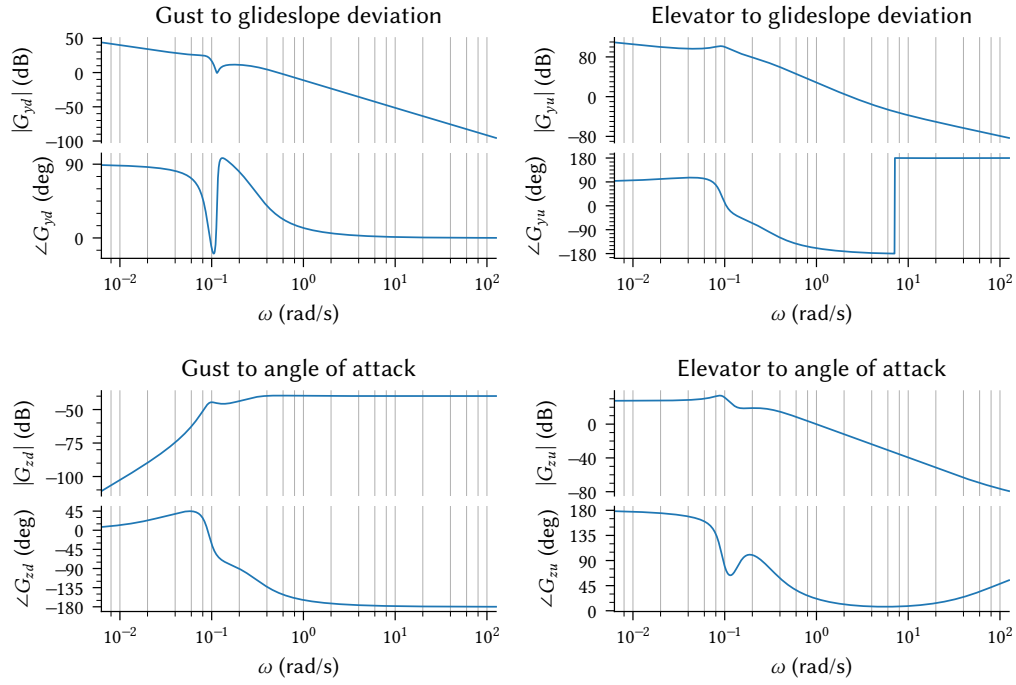


Figure 4.16 Bode plots for longitudinal aircraft model.

Table 4.10 Design metrics, design limits, mean and standard deviation for the quantities of interest

	Probability of exceedance (-)	Frequency of Exceedance (1/s)	Frequency of zero crossings (1/s)	Lower limit	Upper limit	Mean	Standard deviation	Units
Glideslope deviation	0.019	0.011	0.17	-18	18	0	7.7	(m)
Angle of attack	0.015	0.17	3.5	-10	40	20	9.1	(deg)
Elevator deflection	0.034	0.075	0.78	-30	30	-17	7.2	(deg)
Elevator rate	0.000 63	0.0078	2.7	-60	60	0	18	(deg/s)

the design limits for each quantity of interest.

The optimal design has low bandwidth if compared to the available bandwidth for this class of airplanes (usually over 60 rad/s). This is due to the elevator deflection needed for trimming the aircraft (-17 deg) being relatively close to the lower limit of -30 deg. Indeed, the probability of exceeding the elevator deflection limits (3.4%) is the higher of the probabilities being minimized, followed by the probability of exceeding the glideslope deviation limits (1.9%), and the probability of exceeding the angle of attack limits (1.5%). The probability of exceeding the elevator rate limit is very small for this design. The low bandwidth of the control loop reduces the control action but at the cost of larger tracking errors.

More insight into the calculated metrics can be obtained from analyzing the closed-loop Bode plots in Figure 4.16b, the probability density functions (PDF) in Figure 4.18, and the power spectral densities (PSD) in Figure 4.17. The chosen parameterization for the sensitivity function adds two (closed-loop) zeros at the origin of the gust to glideslope deviation transfer function, one that cancels the integrator from the deviation velocity and another one that removes the steady state error, as can be seen in its PSD, which is zero at zero frequency and has peaks at the phugoid and short-period modes. The angle of attack PSD shows very high spectral content in low frequencies, and another peak for the short-period mode, which contributes to the relatively high expected frequency of exceedance and mean-crossings for this quantity.

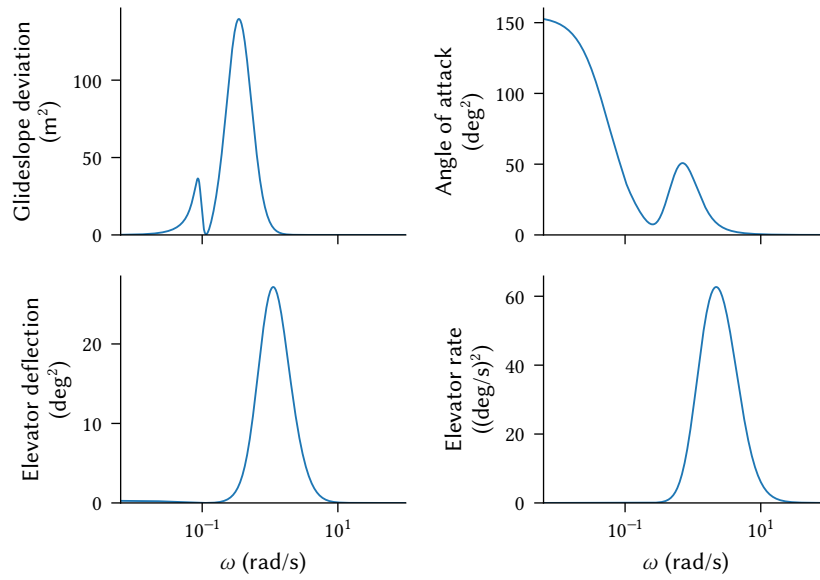


Figure 4.17 Power spectral densities for the quantities of interest.

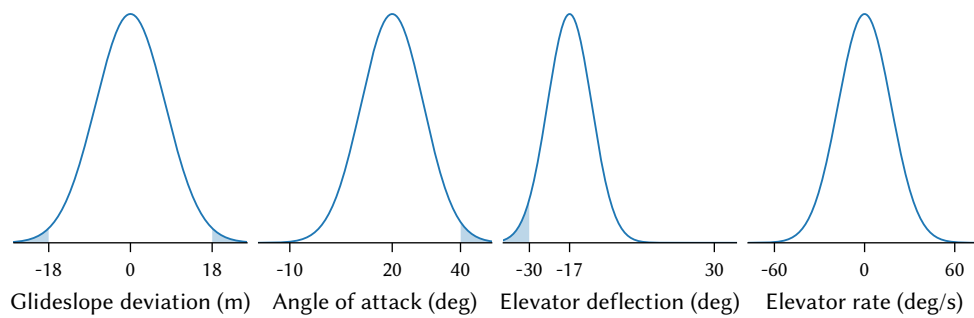


Figure 4.18 Probability density functions for the quantities of interest. The portion outside the design limits is highlighted, and the values shown correspond to the lower limit, trim value, and upper limit.

Table 4.11 Verification of the analytic derivatives against finite differences.

of	w.r.t.	Euclidean norm	Relative error
λ_y^0	vec A	$1.55 \times 10^{+5}$	1.76×10^{-7}
	vec B	$7.09 \times 10^{+4}$	1.55×10^{-11}
	vec C	$1.19 \times 10^{+2}$	6.58×10^{-9}
	vec D	$1.04 \times 10^{+2}$	1.66×10^{-9}
λ_y^2	vec A	$1.37 \times 10^{+4}$	2.27×10^{-8}
	vec B	$1.46 \times 10^{+4}$	1.34×10^{-11}
	vec C	$3.24 \times 10^{+1}$	5.06×10^{-9}
	vec D	$1.04 \times 10^{+2}$	7.09×10^{-10}
λ_z^0	vec A	$1.78 \times 10^{+1}$	2.15×10^{-9}
	vec B	$1.40 \times 10^{+1}$	3.93×10^{-11}
	vec C	4.67	6.66×10^{-11}
	vec D	6.57×10^{-1}	2.06×10^{-10}

4.2.6 Derivative verification

The analytical derivatives presented in Section 2.3.1 were verified against a finite differences approximation at the analysis point described in Section 4.2.4. The finite difference approximation was done using a central difference scheme and step size equal to 10^{-8} times the element being perturbed, but not smaller than an absolute step size of 10^{-8} , which was selected by performing a step-size study. The analytical and finite differences derivatives were in the very good agreement, as shown in Table 4.11 for the zeroeth order moment. The relative error between the methods was calculated as

$$\text{Relative error} = \frac{\|\text{FD derivatives} - \text{Analytical derivatives}\|}{\|\text{Analytical derivatives}\|} \quad (4.12)$$

where $\|\cdot\|$ denotes the Euclidean norm. The maximum relative error of the comparison was of 1.76×10^{-7} , achieved using double-precision floating numbers, which correspond to a machine precision of approximately 10^{-16} .

4.2.7 Sensitivity analysis

More insight into the configuration being analyzed can be gathered from the derivatives shown in Table 4.12. It is seen that moving the CG aft will improve every metric, possibly due to reducing the static elevator deflection necessary to trim the aircraft, as well as making it more responsive to control input (less stable). However, if the aircraft becomes open-loop unstable, the right-hand side of the Bode integral relation will be greater

Table 4.12 Selected design derivatives, color mapped by value.

(a) with respect to geometric parameters

	$\Delta CG/c$	bW2	bCAN	cCAN	bHT	cHT	dTLE1	dTLE2	dCAN
P, ΔGL	-0.015	-0.009	0.002	0.001	0.000	0.000	0.000	0.000	0.000
P, α	-0.015	-0.009	0.002	0.001	0.000	0.000	0.000	0.000	0.000
P, η	-2.723	0.228	-0.105	-0.010	-0.041	-0.034	0.009	0.010	-0.002
P, $\dot{\eta}$	-0.002	-0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
E[N], ΔGL	-0.008	-0.005	0.001	0.000	0.000	0.000	0.000	0.000	0.000
E[N], α	-0.158	-0.094	0.016	0.007	0.003	0.000	-0.002	-0.002	0.000
E[N], η	-4.843	0.405	-0.186	-0.018	-0.073	-0.060	0.016	0.018	-0.003
E[N], $\dot{\eta}$	-0.020	-0.012	0.002	0.001	0.000	0.000	0.000	0.000	0.000

(b) with respect to sensitivity function parameters

	ω_c	ω_0	g_0	q_0
P, ΔGL	-0.287	-0.020	0.043	-0.025
P, α	-0.038	-0.001	0.033	-0.008
P, η	0.181	0.078	0.090	-0.074
P, $\dot{\eta}$	0.016	0.004	-0.001	-0.005
E[N], ΔGL	-0.131	-0.010	0.022	-0.012
E[N], α	-0.308	-0.006	0.270	-0.069
E[N], η	0.378	0.139	0.106	-0.144
E[N], $\dot{\eta}$	0.278	0.004	-0.131	0.022

than zero, which will penalize performance. This effect cannot be captured by a derivative analysis of a stable design due to the derivative discontinuity in the Bode integral relation when a pole crosses the imaginary axis. Increasing wing area will benefit most metrics but penalize elevator deflection. Increasing the chord and span of the canard could counteract some of that penalty. Increasing the horizontal tail dimensions will help with that as well, this time because of increased control effectiveness. Introducing static deflections in the wing's trailing edge flaps or in the canard has small effects on the metrics because they are not very effective for trimming the vehicle. It is worth noting that changes in the geometric variables at this design point will affect the metrics related to elevator deflection most strongly.

We now turn our attention to the variables parameterizing the sensitivity function, starting with their effects on the probability of exceedance of design limits for the various signals of interest. Increasing ω_c and ω_0 is beneficial for the metrics related to the outputs (angle of attack and glideslope deviation) because it increases the bandwidth of the control loop but is detrimental to the metrics related to the control input (η) and control input

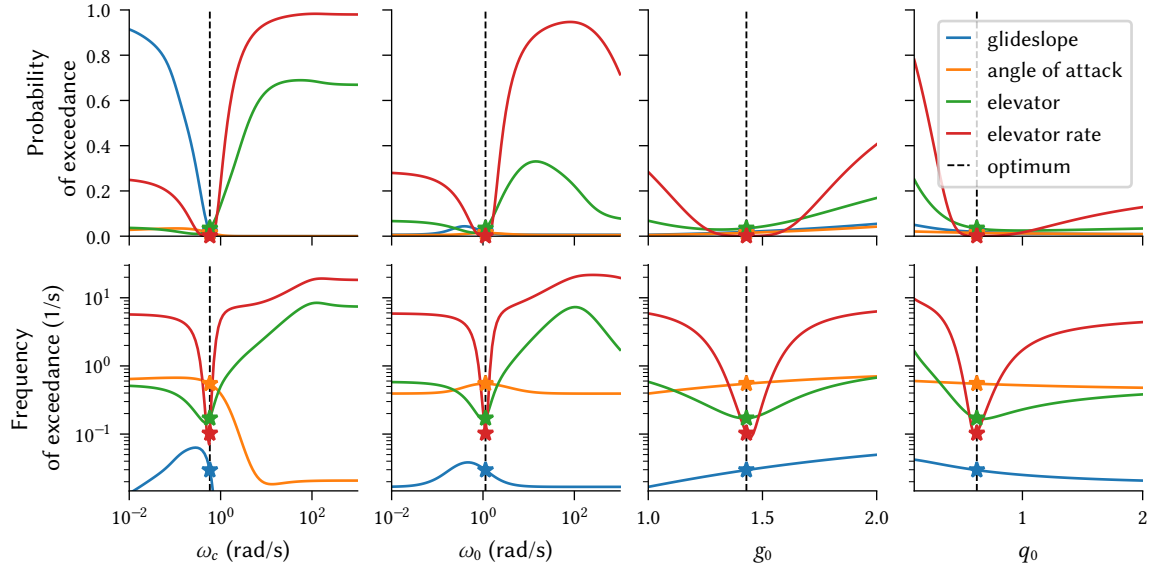
rate ($\dot{\eta}$). Increasing g_0 or decreasing q_0 , that is, making the peak in the sensitivity higher or wider is detrimental for all probability metrics and expected frequency of exceedance metrics, except for the ones relative to the rate of elevator deflection. This is further explored in Section 4.2.8.

4.2.8 Analysis of the optimal sensitivity function

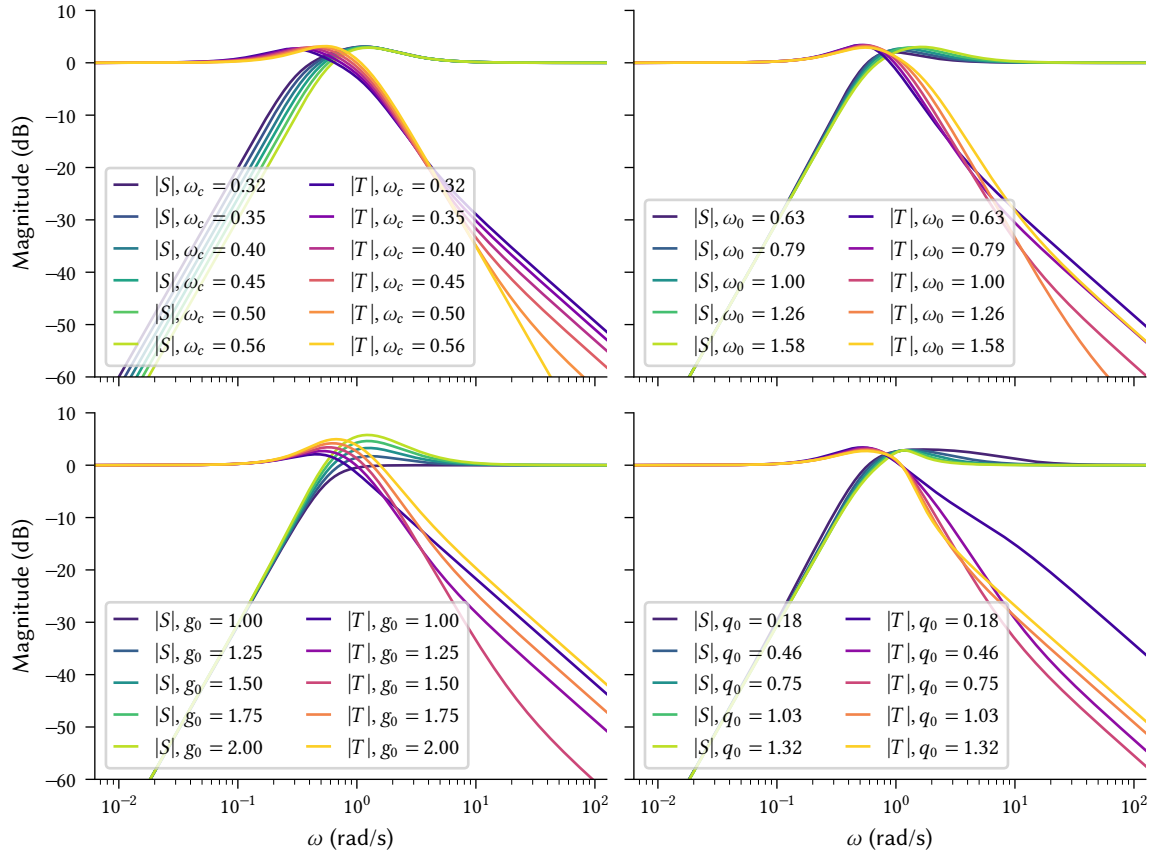
The optimal design of the sensitivity function may be explained by analyzing Figure 4.19a. This figure shows hyperplane cuts in the sensitivity design space, highlighting the effect of changing each design variable on the risk metrics. Since the objective function is the sum of the probabilities of exceedance of the four signals of interest, the *top left* plot shows that the value of the crossover frequency, ω_c , is set by the conflicting goals of keeping the glideslope deviation and elevator deflection within bounds. Suppose the crossover is moved to lower frequencies. In that case, the control is too loose and the glideslope cannot be well tracked, and if it is moved to higher frequencies saturation of the control surface begins to occur, followed by its rate saturation.

The *second plot from the left* shows that the value of the center frequency, ω_0 , lies in a narrow valley of reduced control effort. This valley exists because the magnitude of the transmissibility (which appears in the closed loop gust to elevator transfer function H_{ud} Equation (2.47b)) is not monotonic with respect to ω_0 , as can be seen in the *top right* plot of Figure 4.19b. There is a value of ω_0 which minimizes the value of the transmissibility for frequencies around the peak of the elevator deflection PSD (Figure 4.19a), and the optimizer converges to a value close to it, although slightly biased towards higher frequencies in order to satisfy the Bode integral relation. The same effect happens for g_0 and q_0 , which explains the observation made in Section 4.2.7 that making the peak of the sensitivity taller or wider is beneficial for some metrics of the elevator deflection rate (see Table 4.12b). Depending on which side of the minimum in the transmissibility function for high-frequencies the design value of these parameters is, it may be beneficial to either increase or decrease them to improve metrics that are strongly influenced by the high-frequency behavior of the transmissibility function. This is the case for the risk metrics on the elevator deflection rate, especially for the expected frequency of exceedance.

This suggests that the choice of sensitivity function parameterization could be modified to allow more control of the high-frequency behavior of the transmissibility function. In classical loop shaping, this is the role of a roll-off filter.



(a) on the control metrics



(b) on the sensitivity and transmissibility

Figure 4.19 Effect of varying one parameter of the sensitivity function at a time

	Probability of exceedance (-)		Frequency of exceedance (1/s)	
	second-order	first-order	second-order	first-order
Glideslope deviation	0.019	0.024	0.011	0.010
Angle of attack	0.015	0.0092	0.17	0.12
Elevator deflection	0.034	0.027	0.075	0.065
Elevator rate	0.000 63	0.000 66	0.0078	0.0073

Table 4.13 Comparison of control-related metrics for the first and second order high-pass filter sensitivity functions

4.2.9 Alternative parameterization of the sensitivity function

An alternative parameterization of the sensitivity function is considered, which differs from the one used in the results shown so far by the order of the high-pass filter. The current parameterization uses a second-order high-pass filter while the alternative uses a first-order high-pass filter:

$$\text{current, second-order: } S_2(s; \omega_c, \omega_0, g_0, q_0) = \frac{s^2}{s^2 + s\sqrt{2}\omega_c + \omega_c^2} \frac{s^2 + sg_0\omega_0/q_0 + \omega_0^2}{s^2 + s\omega_0/q_0 + \omega_0^2} \quad (2.27)$$

$$\text{alternative, first-order: } S_1(s; \omega_c, \omega_0, g_0, q_0) = \frac{s}{s + \omega_c} \frac{s^2 + sg_0\omega_0/q_0 + \omega_0^2}{s^2 + s\omega_0/q_0 + \omega_0^2} \quad (4.13)$$

This change is reflected in the optimum PSDs for each parameterization, shown in Figure 4.20. In particular, the first-order parameterization has non-zero steady-state error in the presence of constant disturbances since it has one fewer closed-loop zero at the origin compared to the second-order parameterization. This alternative parameterization has worse tracking at low frequencies, and better tracking at high frequencies because the shallower slope of the first-order filter allows for a greater cutoff frequency, ω_c , as will be shown in Figure 4.21. This is reflected in the control-related metrics shown in Table 4.13: the probability of exceedance of glideslope deviation is higher for the alternative parameterization, but the frequency of exceedance is almost the same because Rice's formula (Section 2.2.2) takes into account the second moment of the PSD. The alternative parameterization reduces control effort and the probability of exceedance of angle of attack, resulting in a small reduction in the sum of all probabilities, which is the value being optimized.

Figure 4.21 shows a design space exploration on the cutoff and peak frequencies (ω_c and ω_0 respectively). The remaining variables—namely g_0 , which defines the height of the peak, and q_0 , which controls the sharpness of the peak—are defined by solving the optimization problem of Section 4.2.3 for fixed values of ω_c and ω_0 . Figure 4.21 shows that

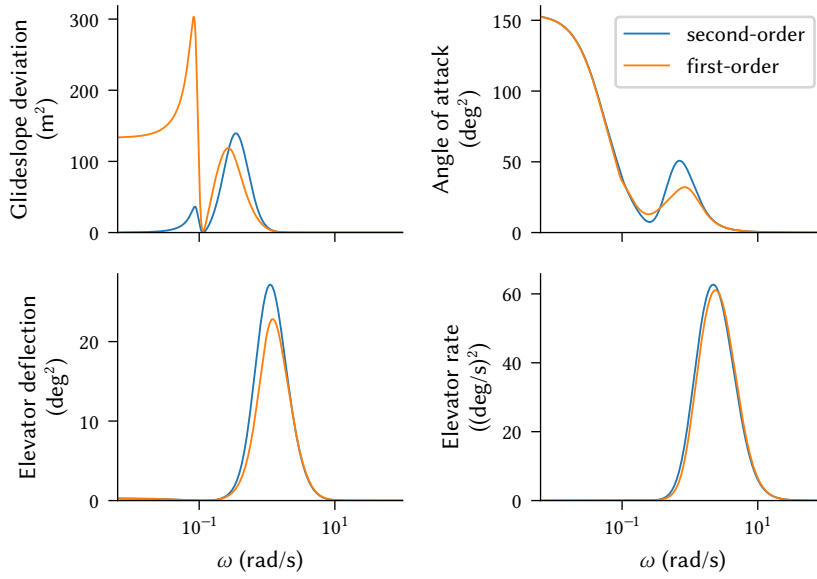


Figure 4.20 Comparison of optimal PSDs for first and second order high-pass filter sensitivity functions

while the first-order parameterization has a unique minimum, the second-order parameterization has a large flat region with two local minimizers. This highlights that choosing a parameterization for the sensitivity function is an important consideration that can affect the metrics' value.

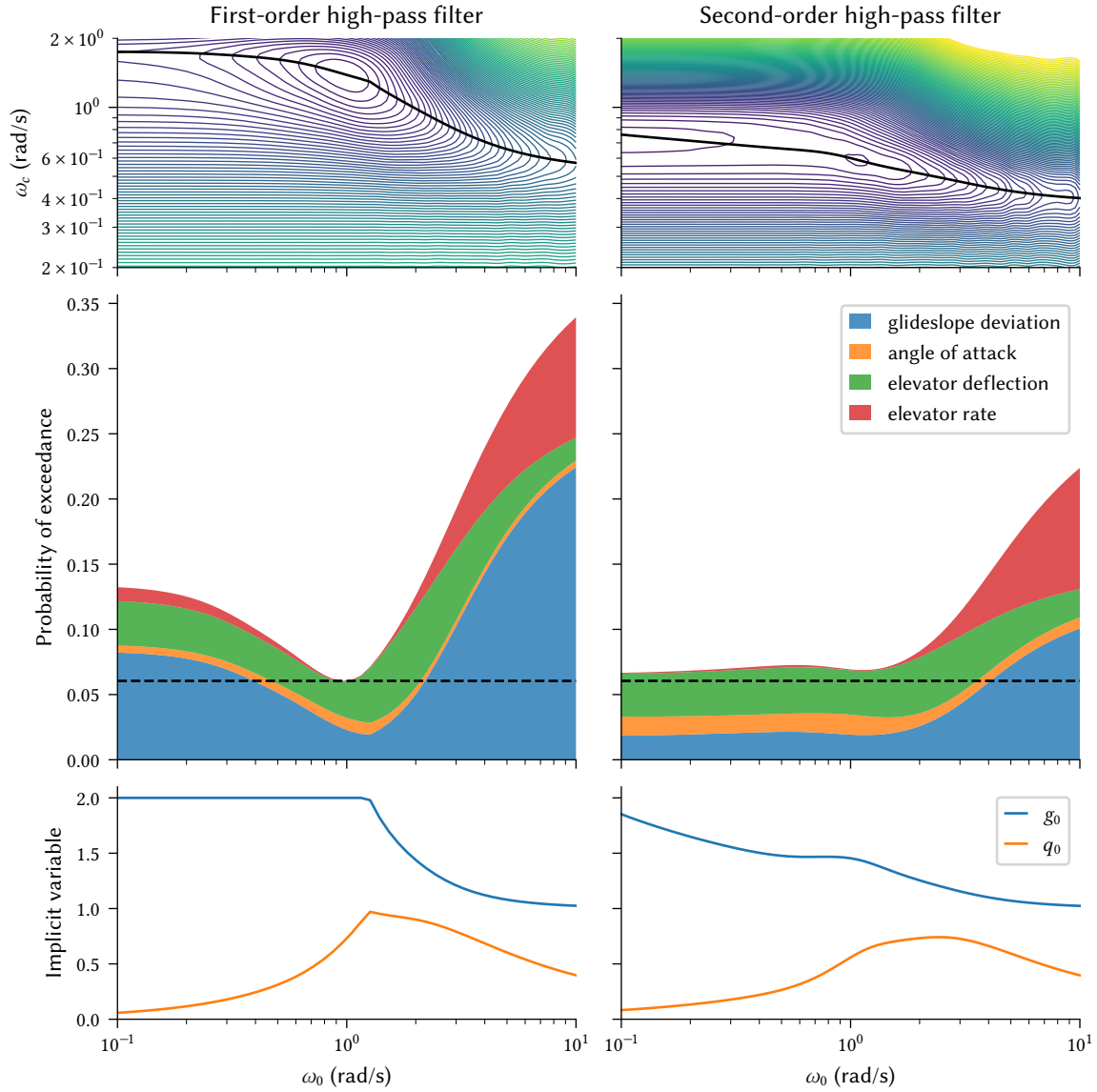


Figure 4.21 Comparison of different parameterizations of the sensitivity function. The first row shows contour plots of the minimum sum of the probabilities of exceedance, in 1% increments. In both figures, the lowest contour corresponds to 7%. The ridge lines are highlighted. In the middle row, the probabilities of exceedance along the ridge are shown. The bottom row shows the optimized parameters of the sensitivity function along the ridge. Multimodal behavior can be seen for the second-order high-pass filter parameterization.

4.3 Maneuver load alleviation

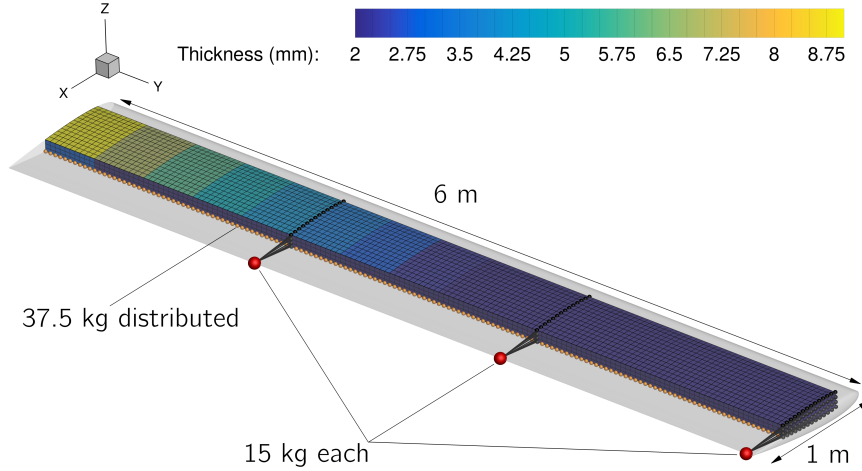
This section integrates a control agnostic MLA constraint similar to the one proposed by Xu and Kroo [10] into a mixed-complexity MDO problem. The optimization problem is a range maximization that builds on the one considered in [188], with the cruise range and cruise stresses calculated by detailed aerostructural analysis using a built-up shell finite element model (FEM) coupled to Reynolds-averaged Navier-Stokes (RANS) computer fluid dynamics (CFD). The MLA system is considered for load calculation in two corners of the V-g diagram, namely the maneuver and dive speeds (V_A, V_D), using a beam model that is obtained from a condensation of the built-up FEM. A bi-level optimization procedure is proposed to prevent the introduction of extra design variables in the outer, high-complexity, optimization problem. In this architecture, the inner optimizer uses a geometrically non-linear beam structural model [144] coupled with potential flow unsteady thin airfoil theory [205] to calculate optimal control deflections and trim condition for each static maneuver condition and returns the minimum stress level for a given maximum control deflection allocation. This formulation exploits the run-time difference between the detailed FEM coupled with RANS CFD analysis and the beam coupled with potential flow analysis, the latter being orders of magnitude faster.

To enable gradient-based optimization, the derivatives of the inner optimum solution with respect to the outer design variables are calculated analytically using the Lagrange multipliers of the optimal solution.

4.3.1 Model description

The baseline wing model used in the optimizations is shown in Figure 4.22 and is nearly identical to the one used in previous aerostructural optimization studies [148, 188]. The model consists of a rectangular, untwisted wing with a 1 m chord, an aspect ratio of 12, and a constant NACA 0012 aerodynamic cross-section shape.

The structural wingbox spans from 15% to 65% of the aerodynamic chord and is modeled using the properties of Aluminium 2024-T3 ($E = 73.1$ GPa, $\nu = 0.33$, $\sigma_Y = 345$ MPa, density = 2780 kg/m³). The baseline thickness distribution (shown in Figure 4.22) is determined from a preliminary structural optimization under representative 2.5-g aerodynamic loads, without flutter or MLA constraints, or geometric variables. The resulting optimized structural mass is 68.5 kg. In order to ensure that this wing is flutter critical within a reasonable flight-speed range, concentrated non-structural masses were added. As done in previous work, three 15 kg non-structural point masses were added along the trailing edge at each third of the semispan. Additionally, in this work, an evenly distributed mass of 37.5 kg over



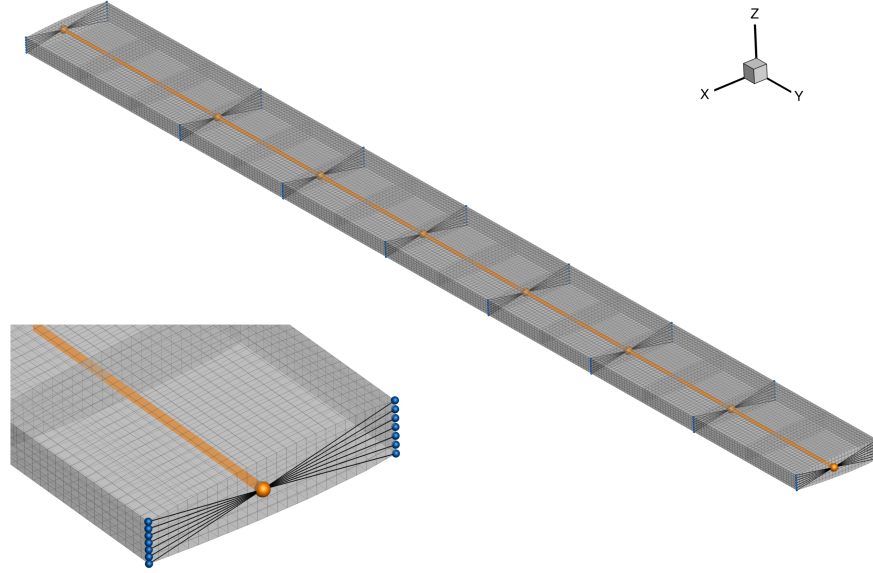
© Gray et al. [188], used with permission.

Figure 4.22 Baseline wing model

the lower skin of the wingbox was added to model the effect of fuel loading. This mass is modeled both as a non-structural mass for the flutter and failure constraints evaluation and as a static inertial load in the MACH analyses used to compute quantities such as cruise drag and maneuver stresses. The wingbox mesh consists of 5424 second-order MITC4 shell elements, giving a total of 31 296 degrees of freedom.

To evaluate the MLA-related stress constraints and the geometrically nonlinear flutter constraint, the wingbox built-up FEM is condensed to an equivalent beam model at each optimization step, using UM/EF2S [200, Appendix C] (see also [201–204]). The beam reference axis is at 41% of the aerodynamic chord and is discretized in 8 reference nodes, corresponding to 7 three-node beam finite elements in UM/NAST. For the stiffness condensation, each beam reference axis node is connected to the leading and trailing edges of its corresponding cross-section by TACS RBE3 element as illustrated in Figure 4.23. The equivalent beam model is coupled to an unsteady potential flow thin airfoil model [205]. The stresses are calculated by solving integrals Equations (2.96) to (2.98) analytically using the symbolic algebra system SymPy² and sampling the resulting closed-form solution for shear stress at the corners and the center of each wall, resulting in a total of 12 points for each cross-section. These points are chosen because they are at the extrema of the piecewise parabolic distribution of shear stress that results from this calculation. Eight different cross sections were considered, corresponding to the root of the beam elements used in the UM/NAST model.

²<https://www.sympy.org>, accessed April 14, 2024



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Figure 4.23 Baseline wing TACS and UM/NAST structural models

Due to the symmetries of the parameterization, the stress value at some of the points will be repeated, but the number of points was not changed to demonstrate the capacity of the approach to handle many sampling points for stress.

In addition, seven control surfaces are added to the trailing edge of the UM/NAST aerodynamic model, one for each element, each spanning 15% of the chord, as shown in Figure 4.24. It uses linear aerodynamics with constants calculated by the same semi-empirical formulas as in [204], reproduced here:

$$c_{l_\delta} = \frac{c_{l_\alpha}}{\pi} (\cos^{-1} c^* + \sqrt{1 - c^{*2}}) \quad c_{m_\delta} = -\frac{(1 + c^*)\sqrt{1 - c^{*2}}}{2} \quad (4.14)$$

where c_{l_α} is assumed to be 2π 1/rad, in line with the linearized aerodynamic model used for the NACA 0012 airfoil, and $c^* = 1 - 2c_{fr}$, where c_{fr} is the length of the control surface as a fraction of the chord.

A structured multi-block CFD mesh is used for aerostructural analyses. The wing's surface mesh contains 22 cells along the chord, 33 along the span, and 4 across the trailing edge. This mesh is then extruded, creating 40 layers that extend 100 chord lengths from the wing. The first layer height is 5.55×10^{-6} m, which was chosen so to achieve a y^+ value of approximately 2 at the maximum studied Reynolds number of approximately 10^7 .

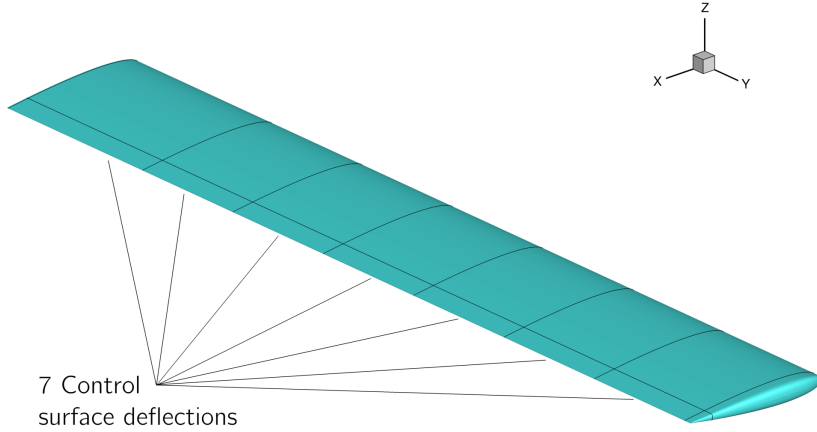


Figure 4.24 Aerodynamic control surface effects included in the UM/NAST aerostructural analysis

4.3.2 Optimization problem

The optimization problem considered in this work is based on [188], with the original stress at the 2.5-g load factor replaced by the MLA-related reduced order stress constraints at both the maneuvering speed, V_A , and dive speed, V_D , at 2.5-g load factor. The modified problem is shown in Table 4.14.

The flight conditions used in the optimization are described in Table 4.17. The cruise condition is used in the aerostructural analysis of the full-order model and has an operating Mach number of 0.5. The cruise altitude is 10 000 m while the maneuver is performed at sea level, with flow properties being computed based on standard atmospheric values. The reduced order flutter calculations are also performed using sea-level standard atmospheric conditions, using 20 flight speeds up to 180 m/s, a 15% margin on V_D . The maneuvering and dive speed conditions are used for the MLA-related reduced-order stress analyses. These analyses involve solving the sub-optimization problems shown in Tables 4.15 and 4.16.

The sub-optimization problems are used to both calculate the trim condition at V_D and V_A and the optimal static control surface deflections δ that minimize the critical stress due to bending on the wing upper and lower surfaces. A constraint on the vertical force F_Z is included to trim the wing at the 2.5-g condition. The trim is accomplished by varying the body angle of attack at a fixed speed for the dive condition and by varying the speed at a fixed body angle of attack at the maneuvering condition, i.e., V_A is effectively calculated as a by-product of the optimization. Bounds on the control surface deflections are included to limit the authority of the MLA system.

In addition to the stress constraints at the maneuvering and dive speeds, a stress con-

Table 4.14 Outer optimization problem.

	Function/variable	Description	Quantity
maximize	R	Aircraft range	
with respect to	t	Panel thicknesses of skins/spars/ribs	28
	x_{span}	Wing semispan scaling	1
	x_{sweep}	Wing sweep	1
	α_C	Body angle of attack at cruise	1
	Total		31
subject to	$KS_{\text{flutter}} \leq 0$	KS aggregate of damping	2
	$KS_{\text{failure}} _{\text{cruise}} \leq 1$	KS aggregates of 1.0 g stress values divided by allowable stress at V_C , accounting for the safety factor	4
	$\min_{\delta, \alpha_D, V_A} (KS_{\text{failure}}) _{\text{maneuver}} \leq 1$	KS aggregates of 2.5 g stress values divided by allowable stress at V_A and V_D , accounting for the safety factor	2
	$ t_{\text{skin}, i} - t_{\text{skin}, i+1} \leq 1 \text{ mm}$	Skin adjacency constraints	12
	$ t_{\text{spar}, i} - t_{\text{spar}, i+1} \leq 1.5 \text{ mm}$	Spar adjacency constraints	12
	$t_{\text{skin}, i} \leq 2 \text{ mm}$	Skin minimum thickness constraints	14
	$t_{\text{spar}, i} \leq 2 \text{ mm}$	Spar minimum thickness constraints	14
	$L = \text{MTOW}$	Trim constraint at cruise	1
	Total		47

Table 4.15 Inner optimization problem at V_D .

	Function/variable	Description	Quantity
minimize	KS_{failure}	KS aggregates of 2.5-g stress values divided by allowable stress multiplied by the safety factor	
with respect to	δ	Control surface deflections	7
	α_D	Root angle of attack at V_D	1
	Total		8
subject to	$F_Z = 2.5 \text{ MTOW}$	Trim constraint at constant speed	1
	$ \delta_i \leq \delta_{\text{max}}$	Bounds on control surface deflections	14
	Total		15

Table 4.16 Inner optimization problem at V_A .

	Function/variable	Description	Quantity
minimize	KS_{failure}	KS aggregates of 2.5-g stress values divided by allowable stress multiplied by the safety factor	
with respect to	δ	Control surface deflections	7
	V_A	Maneuvering speed	1
		Total	8
subject to	$F_Z = 2.5 \text{ MTOW}$	Trim constraint at constant body angle of attack	1
	$ \delta_i \leq \delta_{\text{max}}$	Bounds on control surface deflections	14
		Total	15

straint in the cruise condition (V_C , 1-g) is added to cover the case when the MLA system has enough authority to make the cruise condition the critical one. This constraint is calculated by the shell FEM solver and consists of the von Mises stress in the wingbox aggregated into four constraint values using KS aggregation, one each for the ribs, spars, upper skin, and lower skin panels. All stresses are constrained to lie below the material yield strength by a safety factor of 1.5.

Other maneuver cases, including cases with the MLA system failed, could be incorporated into the optimization to increase the realism of the design. In the failure cases, the safety factor could be reduced from 1.5 to, e.g., 1.3, as was done in [212].

Furthermore, linear adjacency constraints are included. The thickness of adjacent panels must be within 1 mm of one another, while for the spar panels, the limit is 1.5 mm.

The last constraint considered is the flutter constraint, which has been described in detail by Jonsson et al. [148]. It consists of linearizing the reduced-order aircraft model at sampled flight conditions in the envelope (20 points are used for this example), calculating its eigenvalues, and aggregating via KS functions the real part of the eigenvalues of each flight condition, which corresponds to the values of modal damping at particular speeds. A second KS aggregation is used to combine the different flight conditions. The final value is then constrained to be below zero to ensure that the aircraft is aeroelastically stable.

Figure 4.25 shows the structure of the OpenMDAO model used to compute the flutter and MLA stress constraints. The objective of the outer optimization problem is to maximize range as calculated by Breguet's equation:

$$R = \frac{V_C}{c_T} \frac{C_L}{C_D} \ln \left(\frac{M_{\text{init}}}{M_{\text{init}} - M_{\text{fuel}}} \right), \quad (4.15)$$

where R is the flight range, V_C the cruise speed, c_T the thrust-specific fuel consumption

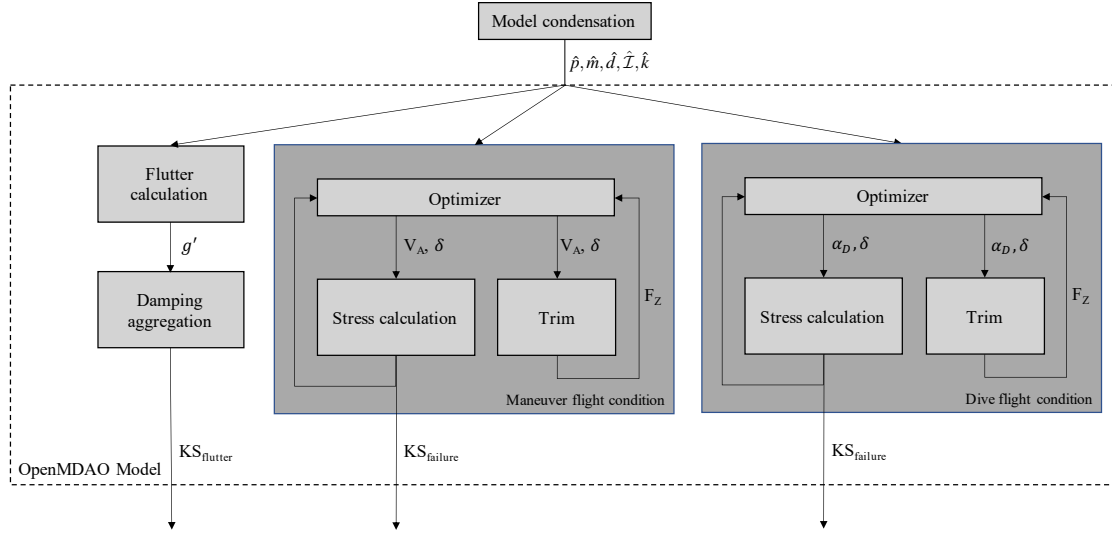


Figure 4.25 Calculation of MLA constraints and geometrically nonlinear flutter constraint on the beam model

Table 4.17 Flight conditions used in this numerical investigation.

Condition	Speed (m/s)	Body a.o.a. (deg)	Mach	Load Factor	Altitude (m)
Cruise (C)	149.7	–	0.5	1	10000
Maneuver (A)	–	10	–	2.5	0
Dive (D)	156.5	–	0.46	2.5	0
Flutter	10–180	2	0.029–0.529	–	0

ratio (assumed here to be 0.5 kg/h) and C_L , C_D are the cruise lift and drag coefficients, respectively, computed by the shell FEM solver coupled with RANS CFD. The initial cruise mass M_{init} consists of a fixed 400 kg plus the wingbox structural mass and the fuel mass of 37.5 kg. Note that these three masses are all half of the full aircraft values, as we only model one wing.

The design variables of the outer optimization problem are the wingbox panel thicknesses, span, sweep angle, and the body angle of attack in cruise to ensure the lift matches the load factor.

To investigate how much MLA can improve the efficiency of the wing, the optimization problem just described was solved with control surface deflection allocation limits ranging from 0 to 10 deg, where an allocation of 0 deg represents a baseline case with no MLA. These 11 optimizations were then repeated without the flutter constraint to investigate under which conditions the flutter constraint dominates the static stress constraints.

Furthermore, a constraint at V_A was included, and this constraint impacts the design even without the MLA system being active, as will be shown in the next section.

4.3.3 Verification of derivatives

The adjoint-based derivatives of the MLA stress constraint are verified against a finite-difference approximation. Table 4.18 compares the adjoint derivatives against a central difference approximation computed by perturbing the span and sweep design variables by 1×10^{-4} . The values of the stress constraint are computed with the baseline wing in the V_D condition, with a maximum control surface deflection of 3 deg. These step sizes were found to result in good matching between analytical and finite difference derivatives by a step-size study. The largest relative difference between adjoint and finite difference values is 3.1×10^{-6} , indicating the correct implementation of the derivative chain. The derivatives with respect to the thickness design variables were also verified.

Table 4.18 MLA failure constraint derivatives with respect to the geometric variables $dKS_{\text{failure}}/dx_{\text{geo}}$.

	$dKS_{\text{failure}}/dx_{\text{Span}}$	$dKS_{\text{failure}}/dx_{\text{Sweep}}$
Adjoint	0.4150052091137539	-0.0941349926526464
Finite-difference	0.4150052181406405	-0.0941352829342268
Relative error	2.2×10^{-8}	3.1×10^{-6}

4.3.4 Termination criteria for the inner and outer optimizer

The solver used for both the inner and outer optimization problems, SNOPT [211], implements the sequential quadratic programming (SQP) method. It divides the structure of the method into major and minor iterations, where each major iteration corresponds to a step in the nonlinear optimization problem calculated by solving a quadratic programming (QP) subproblem, which is obtained by linearizing the nonlinear problem. The minor iterations are the iterations taken to solve the QP subproblem.

To ensure the convergence of the inner optimization, which is needed for accurate calculations of the derivatives with the method described in Section 2.4.1, the tolerances of the inner optimization were defined as follows: The “major optimality tolerance”, which is imposed on the maximum norm of a scaled estimate of the gradient of the Lagrangian after each major iteration, was set to 1×10^{-10} and the “major feasibility tolerance”, which is imposed on the maximum constraint violation normalized by the norm of the solution, was set to 1×10^{-6} . The maximum number of major iterations was set to 40, and the initial guess used was the solution found in the previous iteration of the outer optimizer. For a detailed description of the convergence criteria informed by these tolerances, the reader should re-

fer to the SNOPT manual.³ It was checked that the inner optimizer met the convergence criteria at the last point evaluated by the outer optimizer.

The “major optimality tolerance” set for the outer optimizer was of 1×10^{-5} and the “major feasibility tolerance” was set to 1×10^{-6} . Early terminations that satisfied a “major optimality tolerance” of 1.5×10^{-3} and a “major feasibility tolerance” of 1.2×10^{-5} were considered successful. A search for other local minimizers was not attempted.

Figure 4.26 shows convergence plots for the cases with and without the flutter constraint for various levels of allowable control deflection for the MLA system ($\delta_{\max}$). When the flutter constraint is included in the problem, i.e., Figure 4.26b, the number of iterations necessary to satisfy the optimality criteria is increased in comparison to when the flutter constraint is not included, i.e., Figure 4.26a. In particular, the optimizations with the flutter constraint and with $\delta_{\max} = 5, 9, 10$ deg failed to converge on the first try and were restarted with the converged solutions with $\delta_{\max} = 4, 8, 7$ deg respectively.

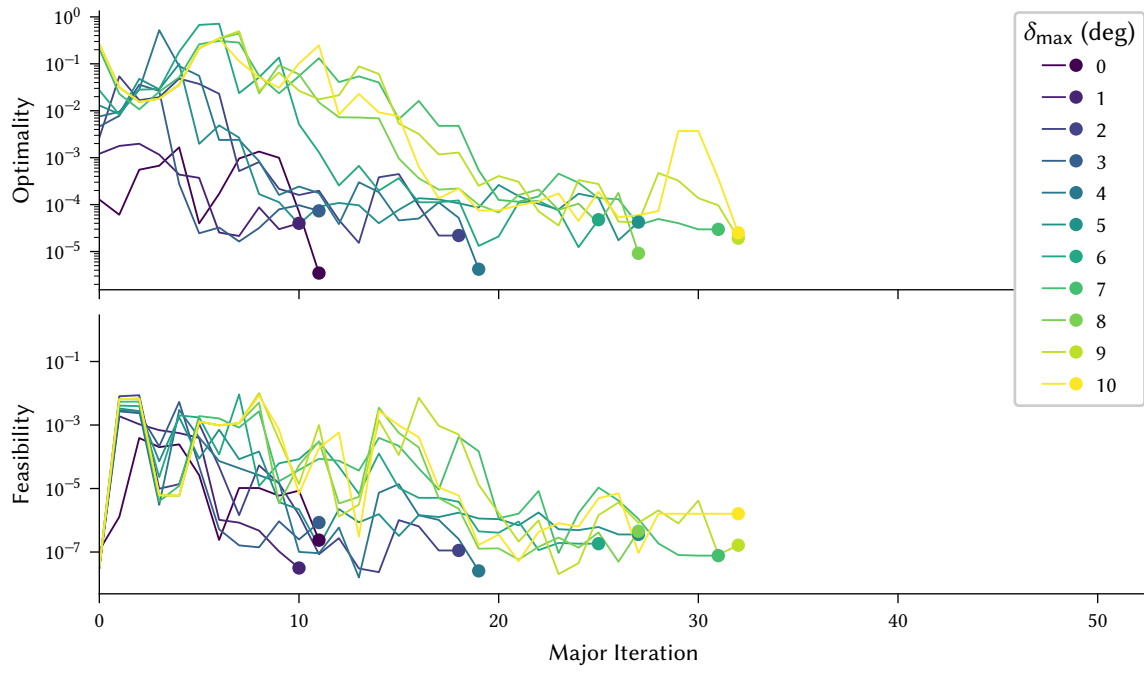
The optimizations shown in Figure 4.26a, that is, with the MLA constraint but without the flutter constraints were run again to obtain the run times, which are shown in Figure 4.27. Comparing the processing times with the number of function evaluations it is apparent that the function evaluation cost increases with the increase in the control deflection allocated for MLA, $\delta_{\max}$.

4.3.5 MLA stress constrained results

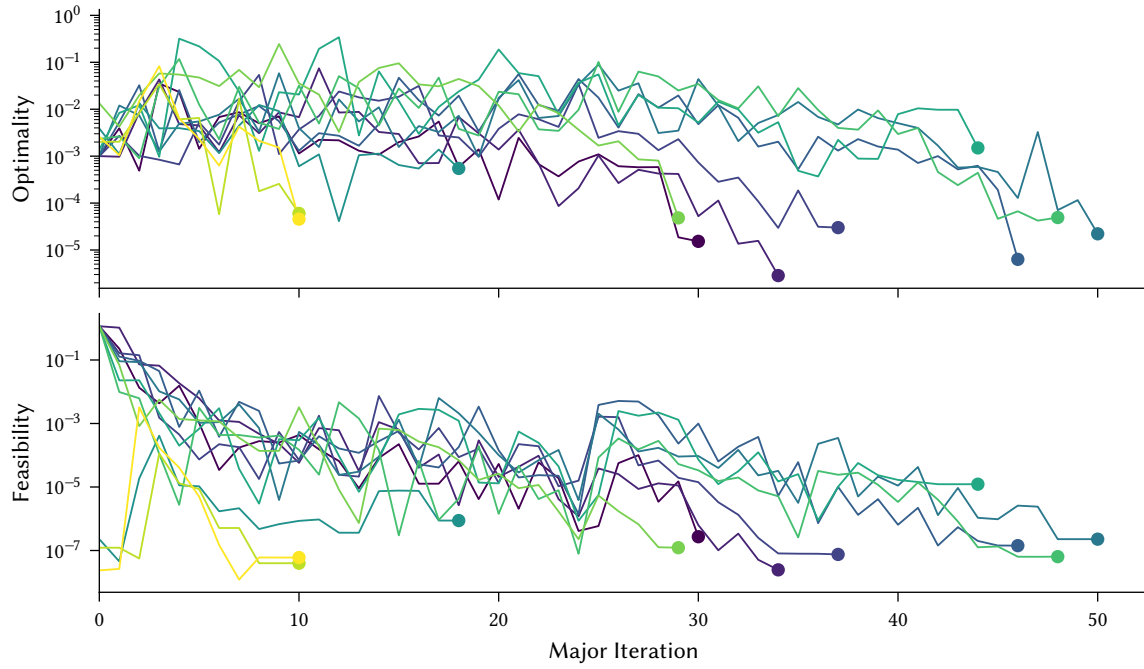
Figure 4.28a shows trends in a selection of values of interest across the wings optimized with differing MLA control surface allocations, represented by the maximum up or down control surface deflection, $\delta_{\max}$, and *without a flutter constraint*. In each plot, the dashed grey line denotes the value from the design optimized without MLA ($\delta_{\max} = 0$ deg), and in the plots of the failure (MLA stress) constraints, the red line denotes the upper limit for feasibility.

In the case with the highest MLA allocation ($\delta_{\max} = 10$ deg), the cruise range is 4.3% higher than the wing optimized without MLA. This range increase is due to a 16.6% reduction in structural mass and an increase of 1.9% in the lift-to-drag ratio (L/D). The increase in aerodynamic efficiency is attributed to an increase in span (5.7%) that outweighed the increase in sweep angle from 15 deg to 26 deg. It is worth noting that since only the span but not the chord is included as a design variable in the problem, the optimizer cannot increase the wing aspect ratio without simultaneously increasing the planform area. Furthermore, in these optimizations, the aircraft’s MTOW and, therefore, the lift required in

³Available at <https://web.stanford.edu/group/SOL/guides/sndoc7.pdf>, accessed on April 14, 2024



(a) without flutter constraint



(b) with flutter constraint

Figure 4.26 Convergence plots for the optimization problems with different control authority allocations for MLA

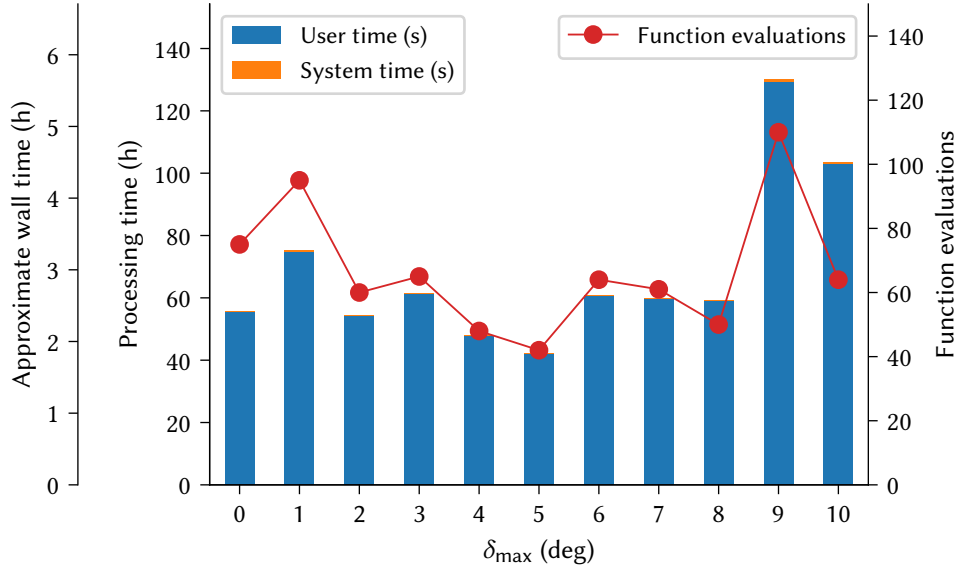


Figure 4.27 Time taken to run optimizations with MLA and without the flutter constraint, using 23 cores on a system with 2×12 cores of Intel® Xeon® CPU E5-2650 v4 @ 2.20GHz

cruise were held constant, so the induced drag was not changed due to a reduction in the structural weight. The only impact of reducing the structural weight is on the initial mass M_{init} used in Breguet's range equation (4.15), since that is calculated by summing the structural mass with a non-structural mass with a fixed value of 400 kg and the fixed fuel mass of 37.5 kg. Therefore decreasing the structural mass increases the fraction of the total mass corresponding to fuel, which slightly increases the range of the design.

For all designs, the maneuvering speed constraint is active, while the dive speed constraint is not. Two factors contribute to this effect. Firstly, at low angles of attack (high speeds), the normal force applied to each airfoil section is larger for the same lift distribution. This causes more passive load alleviation due to the kinematic coupling caused by the back sweep, even without MLA. Secondly, the increased dynamic pressure at V_D allows the control surfaces to produce more force than at V_A , which results in more active load alleviation. This is clearly illustrated by the observation that the peak stress in the dive speed condition decreases by 49.2% across the range of control surface deflection limits, while the stress in the maneuvering speed condition remains constant at the allowed limit. The stresses at V_C rise substantially with increased control allocation for MLA, as expected, but it is worth highlighting that this can lead to fatigue issues that would require further consideration in such an MDO problem.

The optimal designs for the cases of no MLA ($\delta_{\max} = 0$ deg) and MLA with maximum control surface allocation ($\delta_{\max} = 10$ deg) are shown in Figure 4.29a. The use of MLA allows for a 2 mm reduction in skin thickness in the region close to the wing root compared to the

optimal design without MLA. There was no change in thickness at the tip, which is at the lower bound defined for this problem of 2 mm.

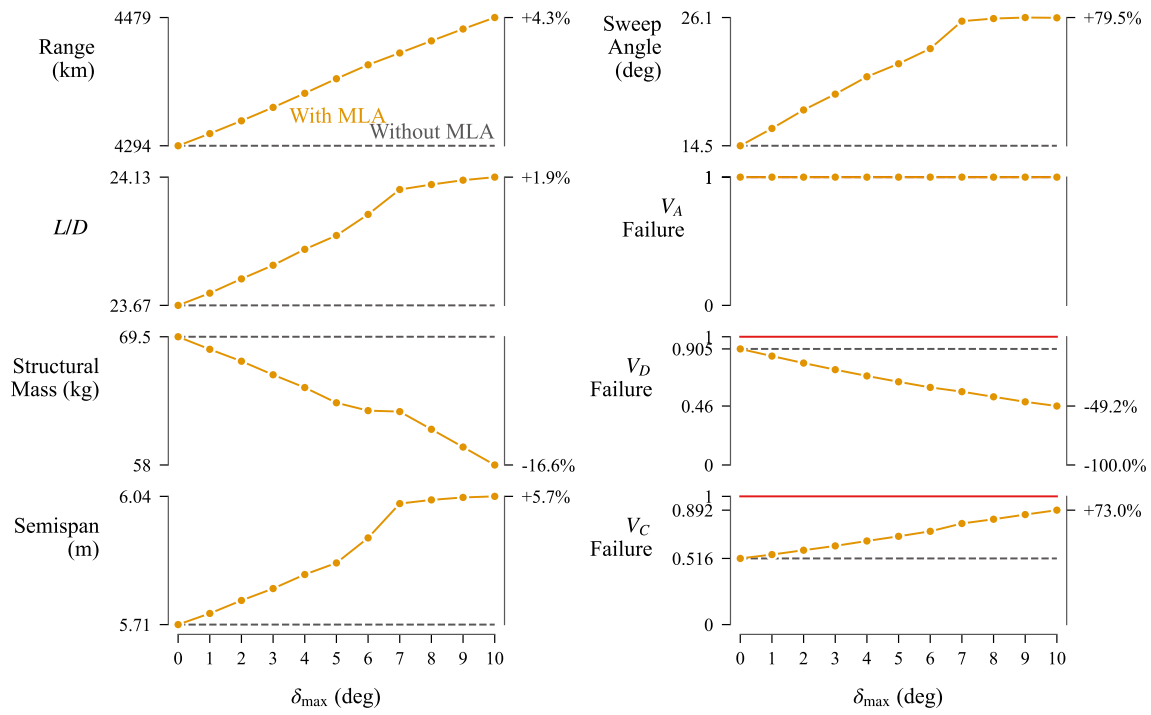
4.3.6 MLA stress and flutter constrained results

Figure 4.28b shows the trends of the optimal wing design for varying MLA control allocations for the case *including a flutter constraint*. In this case, the flutter constraint is active for all designs and this reduces the increase in range (0.7%) enabled by the MLA system. The failure constraint at V_A is active for all designs while the one at V_D is not, for the reasons already discussed in the previous section.

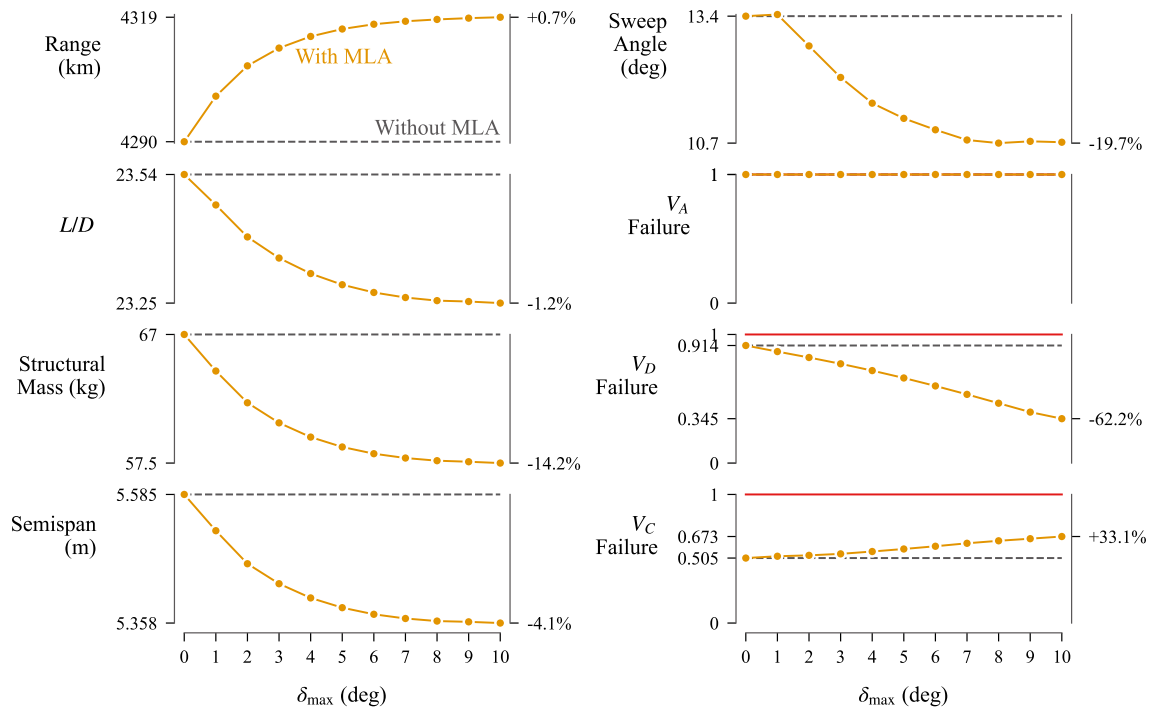
The trends for changes in span and sweep with increasing control authority allocation for MLA are reversed when the flutter constraint is considered. In this case, span decreases by 4.1% and sweep decreases from 13.4 deg to 10.7 deg. This causes a reduction of 1.2% in the lift-to-drag ratio, but the loss in aerodynamic efficiency is compensated by a decrease in structural mass of 14.2%, comparable to the one achieved without taking flutter into account.

The observations made for the case without the flutter constraint remain valid for this case, that is, the structural mass has a small impact in the range and no fuel mass is added from the savings in structural mass. There is a substantial increase in the cruise speed stresses of 33.1%, corresponding to 67% of the allowable stress times the 1.5 safety factor.

The optimal designs for the cases of no MLA ($\delta_{\max} = 0$ deg) and MLA with maximum control surface allocation ($\delta_{\max} = 10$ deg), and the initial design (baseline) for the optimization *with* the flutter constraint are shown in Figure 4.29b. Comparing just the two optimized cases, the result with MLA constraints shows a reduction in panel thickness for the first half of the wing span and of up to 2.5 mm in the wingbox skins close to the root. In that same region, there is an increase in the thickness of the spars to satisfy the flutter constraint since, for this particular model, the flutter is caused by the interaction between the in-plane bending and torsion modes (see [188] for a detailed discussion on the flutter mechanism).



(a) without flutter constraint



(b) with flutter constraint

Figure 4.28 Selected figures of merit for the optimal designs for varying MLA control authority

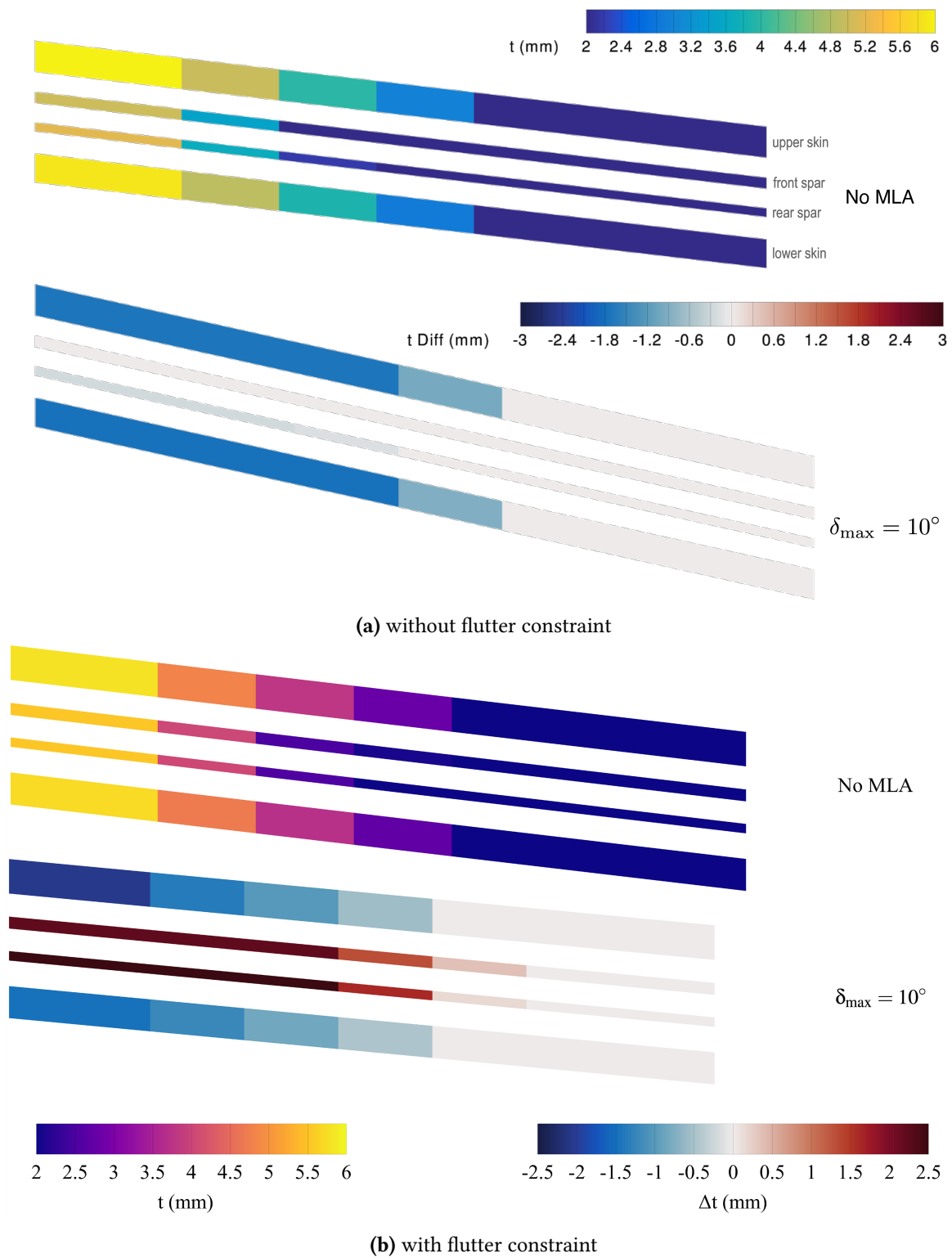


Figure 4.29 Optimal planform shape and thickness distribution for the cases without an MLA system and with 10 deg of MLA control authority, as a difference from the optimum without MLA.

CHAPTER 5

Concluding Remarks

This chapter summarizes the developments done on the methodology, its implementation in software, and its application in simplified multidisciplinary design optimization (MDO) problems. It highlights the key contributions of this dissertation and presents recommendations for future work based on gaps that were identified but not tackled by this work.

5.1 Summary and takeaways

The goal of this dissertation is to develop new methodologies for consideration of control-related metrics in aircraft MDO, without controller design. It emphasizes applications for maneuver load alleviation (MLA), gust load alleviation (GLA), and glideslope tracking during the approach to land under atmospheric turbulence. A literature review of the approaches for including control considerations in aircraft MDO was conducted, and these approaches were classified into four categories (Section 1.2.4), namely:

1. Combined plant and controller synthesis
2. Open-loop metrics augmented with heuristics
3. Closed-loop metrics based on fundamental limitations
4. Optimal control input co-design

The first approach involves designing the controller at the same time as the plant, while the other three are controller-agnostic ways of including control-related considerations in MDO: open-loop metrics can be used directly as a way to estimate the risk in developing a controller based on how far they deviate from the acceptable values; considerations based on the fundamental limitations of control can be included in order to estimate an upper bound for closed-loop performance; or the control input for performing a maneuver can be

designed at the same time as the aircraft. The two last approaches were pursued since they tackle the problem directly and are less dependent on heuristics.

For the problems that involve stochastic disturbance rejection, as is the case of GLA and glideslope tracking, an approach based on the fundamental limitations of control was developed because it integrates seamlessly with the powerful frequency domain methods already used for the computation of continuous gust loads. This approach is based on designing one of the closed-loop transfer functions, from which the sensitivity function was selected. A simple parameterization that reflects the objectives of feedback control and has enough degrees of freedom to satisfy the Bode sensitivity integral relation was developed and used. The Bode sensitivity integral relation is imposed by a constraint on the problem, which limits the bandwidth of the closed-loop system, reflecting its practical limitations.

Various stochastic risk metrics were selected to quantify the closed-loop performance of the system, including an estimation of the fatigue life of the aircraft which has not been used in MDO before. This estimation has the advantage of being differentiable, while the rainflow counting method it approximates is not. It was observed that all these metrics depend on moments of the power spectral density (PSD) of output signals of the stochastic system, therefore formulae for the calculation of these moments including the parameterized sensitivity function were developed. Matrix-free and adjoint methods were employed to develop efficient derivative computation for these PSD moments.

For the MLA problem, the idea of optimizing control surface deflections during the maneuver, already used in the literature, was adopted. The novelty introduced was the use of a bi-level optimization architecture with a model order reduction in the loop, which allows the objective function and failure criteria to be calculated at different levels of complexity. This has the potential to enable the computation of a greater number of load cases within MDO. A simple stress model was used, which can be improved in the future by using information from the model order reduction process.

These methodological developments were implemented using the UM/NAST, MACH, and OpenMDAO computer programs. The MDO capabilities of UM/NAST were enhanced by the development of a new interface for algorithmic differentiation. This interface abstracts the AD tool and allows the definition of new metrics with greater ease, because the AD code is reused. A new wrapper that exposes the UM/NAST design metrics (with AD derivatives) in the OpenMDAO environment was developed. Finally, the integration between UM/NAST and MACH was generalized to allow for arbitrary UM/NAST-based metrics to be used in a MACH MDO problem.¹

¹MACH is being gradually replaced by MPhys in newer works. The OpenMDAO wrapper for UM/NAST

These methodological and software developments were applied to three test cases:

1. The MDO of the wingbox from a blended-wing-body aircraft with GLA (Section 4.1)
2. The analysis of the glideslope tracking performance of a supersonic configuration aircraft in low-speed flight (Section 4.2)
3. The MDO design of a wingbox for a cantilevered wing including MLA (Section 4.3)

In each of these cases, the analytical derivatives were verified against finite differences approximations, demonstrating correct derivation and implementation.

In the GLA application (Section 4.1), the use of the parameterized sensitivity function was able to reduce the mass of the wing (objective), while still satisfying the other constraints. The need for having a working flutter constraint was highlighted, but there were problems with its derivatives which were investigated but not solved. The sensitivity of the optimum design with respect to changes in the design variables was calculated, and it was shown that the performance of the closed-loop design is more robust to changes in the structural variables.

In the application to glideslope tracking (Section 4.2), the design of an optimal sensitivity function was analyzed in detail, along with the effect of changing its parameterization. The bandwidth of the design in this case is driven by the tradeoff between tight control and the requirement of avoiding saturation, while the shape of the peak of the sensitivity function is driven by the constraint on the Bode integral. Depending on the choice of parameterization for the sensitivity function, the presence of multiple minima can be observed. The design sensitivity of the risk metrics with respect to the aircraft sizing parameters was analyzed, and a strong influence of the center of mass position was identified, along with wing and control surface sizing.

In the MLA application (Section 4.3) usage of MLA with up to 10 deg of control surface deflection allocation resulted in a 16.6% reduction in structural mass and a 4.3% increase in range without the flutter constraint. On the other hand, the benefit of increasing MLA control surface allocation is reduced when the flutter constraint is used. The optimal range improves by only 0.7% when the MLA control authority is changed from 0 to 10 deg, while the structural mass is reduced by 14%. In both cases, the inclusion of the MLA system allows for a significant reduction in the thickness of the upper and lower surfaces of the wingbox, while the sides remain virtually unchanged for the case without the flutter constraint. When considering the flutter constraint, the side wall thicknesses close to the root

makes integration with MPhys possible and should be considered for future work.

are substantially increased with the inclusion of MLA, in order to prevent flutter in a more flexible wing design.

5.2 Key contributions

The most relevant novel and citable contributions of this dissertation are:

1. *Developed a method based on the parameterization of the sensitivity function to compute closed-loop performance under stochastic disturbances, along with their derivatives for MDO.* Developed formulae for the calculation of moments of the PSDs of output signals of the closed-loop system, without designing the controller (Section 2.3). Identified relevant design metrics that can be computed from these moments (Section 2.2). Derived the equations for analytical calculation of the derivative of the PSD moments with respect to the open-loop system and the parameters of the sensitivity function, using matrix-free, direct, and adjoint methods for time and memory efficiency (Section 2.3). This approach was demonstrated for the design of the wingbox of a blended wing body aircraft with a GLA system flying in continuous turbulence (Section 4.1) as well as for the analysis of a supersonic configuration aircraft flying the approach to land under continuous turbulence (Section 4.2). These developments can also be used for computing these disturbance rejection metrics in open-loop designs.
2. *Included a frequency-domain fatigue constraint into gradient-based aircraft MDO* (Section 2.2.4). The method from Dirlik [77] for estimating the fatigue life along with the approach by Preumont and Piéfort [176] for calculating the von Mises stress PSD were incorporated into an aircraft MDO problem for estimating the fatigue life of the wingbox in continuous turbulence. The usual method of estimating the fatigue life in the time domain using rainflow counting is not differentiable, but this difficulty is avoided by using a frequency domain approach that makes this metric differentiable. The calculation of fatigue life can be done in open or closed-loop, using the sensitivity parameterization approach. The importance of a fatigue constraint is increased when considering closed-loop effects in MDO because the control action may increase loads at higher frequencies, where the gust intensity is diminished but the cycles increase more rapidly. This constraint was demonstrated in the MDO of the wingbox from a blended wing body aircraft equipped with a GLA system flying in continuous turbulence (Section 4.1).
3. *Developed a bi-level optimization approach for including MLA in aircraft MDO with mixed fidelity* (Section 2.4). The approach of accounting for an MLA system through

the design of the control surface deflections during the maneuver is used to implement an MLA-aware stress constraint into an MDO problem without additional design variables. This is accomplished via a bi-level optimization architecture. This structure enables performance improvements from the use of a model order reduction process performed within the optimization (first used in [148]), which will potentially allow the analysis of more maneuver cases in a single MDO. This approach was demonstrated for the design of a wingbox of a cantilevered wing (Section 4.3).

4. *Implemented an OpenMDAO interface for UM/NAST with AD derivatives, in which the metrics are general across different aircraft models and the AD tool is abstracted* (Chapter 3). This replaces the previous interface for accessing the derivative capabilities of UM/NAST [193, Chapter 11]. The new interface, named OpenMDAO-NAST, has the following advantages:

Abstraction of the AD tool (Section 3.1.3). New metrics have to be coded in C++, but the AD procedure is fully reused, avoiding the possibility of bugs.

Mix-and-match of metrics and models (Section 3.1.1). The procedure for updating the model from the new design proposed by the optimizer is common across all metrics. This means that any metric can be used with any model, changing only the relevant settings through the OpenMDAO interface (in Python) and that multiple metrics can be used with the same UM/NAST model in one MDO problem because all metrics share the common set of inputs that define the variable properties of the model.

5. *Classified the existing ways to include control-related considerations into aircraft MDO* (Section 1.2.4). This classification divides the existing literature on the topic into four categories depending on how the control-related considerations are made:

1. Combined plant and controller synthesis
2. Open-loop metrics augmented with heuristics
3. Closed-loop metrics based on fundamental limitations
4. Optimal control input co-design

The approaches presented in this dissertation can be classified into these categories, with the methods for stochastic disturbance rejection falling in category 3 and the method for bi-level optimization of the control surface deflections during the maneuver falling in category 4.

5.3 Recommendations for future research

Three main gaps were identified but not addressed in this work:

1. *There is no clear extension to the approach based on parameterization of the closed-loop transfer functions (Section 2.1) for the case of nonlinear plants:* These nonlinearities may arise from the geometrically nonlinear deformations of very flexible high-aspect-ratio wings from future low-emissions aircraft. There are extensions of the Bode integral relation to nonlinear systems [93, Chapter 13], and the Volterra series can be used for certain classes of nonlinear systems to compute PSDs [224, 225]. In this sense, the Volterra series can be thought of as an extension of the transfer function concept to nonlinear systems. However, the identification of the Volterra series is a challenge since the available methods depend on simulation (e.g., [226]). Furthermore, I could not find a link between the nonlinear extensions of the Bode integral relation and a Volterra series of the closed-loop system in the literature. Hence, the feasibility of this approach seems to depend on the closing of these theoretical gaps.

An alternative approach to deal with the problem of estimating the best closed-loop performance of a system excited by stochastic inputs is outlined in Section 5.3.1.

2. *Information from the built-up FEM is not used to inform the calculation of stress from the reduced order model,* which resulted in using a low-fidelity stress model in Section 4.3. Stress recovery procedures available in the literature (e.g., [227–229]) rely on the assumption that information regarding the geometry of the cross-section of the beam is readily available, which is then used to compute warp influence functions which are fundamental for the recovery of the stress field. This is not the case for a beam model generated from a built-up FEM, as was done in this work. Therefore there is a need to develop a stress recovery procedure that uses only information that can be extracted from the built-up FEM.
3. *There are problems with the derivatives of the flutter constraint that prevent the convergence of the optimization problem in some cases.* These problems have been observed not only in the results of Section 4.1 but also in more complex models that were tested internally in the scope of the Airbus-Michigan CASE-VFA. It remains unclear what is the cause of this problem.

A potential reason is the use of AD over an iterative process, namely the static analysis used to compute the in-flight deformed shape. In this case, there is no guarantee that a numerically stable process for computing the primal values is also stable in

the computation of the derivatives, and there is no guarantee that an adequate stop criterion for the primal values is also adequate for the derivatives [230]. One solution for this is using a Newton method to solve the static analysis (as done in [231], for UM/NAST) and to compute the derivatives of that step analytically.

Another idea worth trying is to separate the aggregation of the damping values by frequency bands. This will show whether the problem with the calculation of the derivatives occurs across all frequency bands or is restricted to some of them. Knowledge about the problem is required to select band separations that avoid the eigenvalues switching bands. Alternatively, a windowing process can be implemented to keep the metric differentiable in that case.

4. *The problem of proving the existence of a controller that achieves the estimated performance metric was not tackled.* Boyd and Barratt [42] show some cases where this problem can be solved, i.e., it is possible, given some design requirements, to either find a controller that meets the design specifications or prove that none exists.² One such case is when there are limits on the RMS values of the inputs and outputs of a plan with full-state feedback, and the solution is an LQR controller with weighting matrices that can be found by convex optimization [42]. This case can be used to validate the proposed methodology. Furthermore, controllers can be designed for the examples in Sections 4.1 and 4.2 and their performance compared with the performance estimated by the methods in this dissertation.

Next, a possible approach to tackle gap 2 is outlined, along with opportunities to enhance the examples presented in this dissertation.

5.3.1 An approach for computing risk metrics of nonlinear aircraft in continuous gust

The idea of using risk metrics based on PSD moments is also applicable to nonlinear systems. The PSD itself is a concept applicable to every time signal, linear or nonlinear, given that it is just the Fourier transform of the autocovariance function (Section 2.3). Therefore, the challenge is calculating the best achievable PSD-based metric for a nonlinear plant without controller design. One approach is to combine the ideas of the *stochastic discrete gust* (Section 1.2.3) with the use of an optimal control problem in MDO, which can be cast as a suboptimization problem as shown in [73]. This results in an adversarial problem, where

²Boyd and Barratt [42] refers to this concept as the “fundamental problem” of control co-design (see Section 1.2.1).

the gust parameters must be chosen to maximize the risk metric while the control input is chosen to minimize it. I could not find any readily available adversarial optimization solvers for optimal control, and even less so for MDO. However, due to the worst-case gust having few parameters, it could be added as another layer of suboptimization, for a hierarchical problem with three total layers. The suboptimization problem to be integrated into MDO would then be written as a minimax problem:

$$\min_{\text{control action}} \max_{\text{gust parameters}} \text{Risk Metrics(PSD moments(control action, gust parameters))}$$

The risk metrics would then be used to inform constraints in the top-level MDO problem. The hierarchical problem lends itself very well to the parallelization of the multiple flight points that the aircraft needs to be analyzed at when doing a realistic design.

A closely related development was done by Iannelli et al. [232], who developed a method to compute the worse case disturbance for a linear time varying (LTV) system, demonstrated in the example of a flexible aircraft subjected to gust.

5.3.2 Suggested improvements for the example problems

In the application to GLA (Section 4.1), the assumption of a uniform wingbox should be dropped to make it more realistic. This will add distributed design variables to the problem, which will highlight the performance advantage of calculating the derivatives in the adjoint mode. To aid convergence with the additional design variables, adjacency constraints on the thicknesses should be included. The calculation of gust loads with a failed GLA system should be incorporated, and it can be done through the same methodology by setting the sensitivity function to a constant value of one. The derivative of the optimum design with respect to changes in the values of the constraints (shadow costs) can be calculated using the Lagrange multiplier ideas presented in Section 2.4, as well as the sensitivity of the optimum design with respect to changes in the design variables but still respecting the physical constraints, namely the trim and the Bode integral relation.

In the application to glideslope tracking (Section 4.2), a more representative aerodynamic model should be used, for example, one based on a linearized UVLM [219]. The additional states from the unsteady aerodynamics will help demonstrate the scalability of the approach to larger systems. The inclusion of unsteady effects will also help in relaxing the assumptions on the integration limits.

Furthermore, the effects of nonminimum phase zeros due to actuator selection should be analyzed in more depth, including their implication for optimal actuator selection. Including actuator selection in the MDO is more complex since it changes the topology of the

problem, but it was done by Stanford [135]. The effect of these zeros can be included in the parameterization of the transmissibility function (Section 2.1), but the calculation of their derivatives with respect to changes in the open-loop system has to be developed.

In the MLA application (Section 4.3), the trends for the optimal span and sweep are reversed in the cases with and without the flutter constraint when looking at the optimal values parameterized by the maximum control deflection available for MLA. Without the flutter constraint, increasing the control authority available for the MLA system causes the optimal span and sweep to increase, which results in a net increase in aerodynamic efficiency. On the other hand, for the case with the flutter constraint, the optimal values span and sweep decrease with an increase in MLA control authority. This causes the decrease in structural mass for both cases to be similar, even though the increase in range is not. Further analysis is needed to verify whether this trend reversal can be attributed to the under-parameterization of the problem: The only design variables that modify the wing planform are span and sweep. Therefore the optimizer cannot choose wing area and aspect ratio independently.

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